

ACHIEVING CYBER RISK RESILIENCE

ANALYZING ALTERNATIVE POLICY APPROACHES

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Analyzing Alternative Policy Approaches

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Introduction: Governance Pathways for Cyber Resilience

Andrew Grotto

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For over two decades, digitalization has served as a major driver of economic growth and job creation in countries around the world. This is true for not only advanced economies, which could build on decades of previous investments in telecommunications infrastructure to take early advantage of digital innovations, but emerging economies as well. Although large gaps still exist between developed countries and many emerging economies in key digitalization metrics (e.g., broadband connectivity), countries that have succeeded in narrowing those gaps have reaped significant economic benefits, including faster economic growth, higher employment, rises in consumer welfare, declines in poverty, and productivity gains.¹ Artificial intelligence (AI) is poised to compound the potential of these gains from digitalization, and governments at every stage of economic development are considering how their societies can participate in the burgeoning AI economy.

All technologies have risks, and governments must make choices about how to manage those risks. For digital technologies (including AI), one of the most significant risks is cybersecurity. The more dependent a society is on digital technologies, the more exposed it is to cybersecurity risks from threat actors seeking to exploit the confidentiality, integrity, or availability of those technologies for economic, political, or strategic purposes. Cybersecurity is not a new problem, but the magnitude of cybersecurity risk grows as national economies digitalize. It is imperative for all governments to develop and implement a vision for addressing this challenge in a manner that both facilitates the economic and technological benefits of digitalization and builds resilient infrastructure.

Cybersecurity will be especially important for the diffusion of AI into emerging economies. For domains like critical infrastructure and government services, AI has the potential to improve service delivery, lower costs, and catalyze new business models that support economic growth and innovation. The owners and operators of these services, however, will be slow to adopt AI if they lack confidence in their ability to defend this new attack surface from cyberthreats on top of the risk and vulnerabilities in legacy systems. In addition, the diffusion of AI will force governments to adapt their resilience strategies to new forms of AI-enabled cyberattacks targeting their critical infrastructure and business communities. AI also has the potential to aid defense—not just against AI-enabled attacks but against all cyber hazards.²

The goal of this report, as well as a second forthcoming report from this project, is to serve as a resource for global policymakers as they consider different policy and regulatory strategies for supporting digitalization that is resilient to cybersecurity risks. We use the term “cybersecurity risks” to refer primarily to adversarial threats against the confidentiality, integrity, or availability of information or information systems, consistent with established definitions.³ In lay terms, confidentiality means that information meant to stay private remains private; integrity means that information is unadulterated and information systems work as intended; availability means that information and information systems are available for use by authorized parties. While this is the standard formulation, as Elaine Korzak elaborates in her chapter in this report, some governments believe the definition of confidentiality ought to extend into policies concerning censorship and

¹ Vivien Foster et al., “The Impact of Infrastructure on Development Outcomes: A Qualitative Review of Four Decades of Literature,” World Bank, Policy Research Working Paper, no. 10343, March 2023, <https://documents1.worldbank.org/curated/en/099529203062342252/pdf/IDU0e42ae32f0048304f74086d102b6d7a900223.pdf>.

² This topic will be explored in more detail in a subsequent report from this project.

³ See, for example, National Institute of Standards and Technology (NIST), “The NIST Cybersecurity Framework (CSF) 2.0,” February 26, 2024, <https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>.

information flows. When formulating cybersecurity policy, governments must make choices about desired end states and the policy means for achieving those ends. These choices inevitably involve trade-offs among competing values and interests, as well as hard decisions about where to spend scarce resources and hard realities about the practical limits of government capacity for administration in a technologically complex domain. This report aims to shed light on these trade-offs, decisions, and realities to help countries at different stages of their cyber policy journey—from early to mature—make more effective decisions about priorities and approaches.

As a policy domain, cybersecurity is still relatively new. As recently as 2018, countries with national cyber strategies were a distinct minority.⁴ The majority of countries today have national strategies, but a recurring theme among them is the persistent difficulty of building the human capital within government and the private sector that is necessary for continued maturation and refinement of cyber policies. This introduction presents an executive-level overview of cybersecurity risk and policy. It first surveys key concepts and features of the cybersecurity risk landscape, including reasons that cybersecurity risk presents an especially vexing risk management challenge. It then presents an analytic framework for describing and organizing cybersecurity regulations to make the abstract and technical concept of cybersecurity risk management more concrete and easier to understand for policymakers and executive leaders. The introduction concludes with an overview of the four chapters of this report.

Cybersecurity Risk: Key Concepts and Features

This section discusses key terms and provides a brief overview of cyberthreat actors and the cybersecurity risk landscape.

Cybersecurity Risk

At its simplest, risk (any risk) exists when there is a credible threat to something of value (an asset). Cybersecurity risk emerges whenever there is a possibility of cyberthreat actors identifying and exploiting a vulnerability in a digital asset that could put the confidentiality, integrity, or availability of information or an information system at risk of harm. The harm could be to the asset itself or to societal functions that depend on it. The magnitude of cyber risk depends on the probability of an adverse event happening and the consequences if the adverse event were to occur. Reducing either of these two factors—probability and consequences—reduces risk. Thus, at a high level, cybersecurity risk regulation aims to reduce the probability of an adverse event from happening and the consequences if the adverse event were to materialize. The latter goal is known as resilience.

It is practically impossible to eliminate cybersecurity risk without abandoning digital technologies altogether. Indeed, a defender could execute a risk management plan perfectly but still fall victim to a cyber incident—for example, if a hacker were to exploit a previously unknown vulnerability in software (such vulnerabilities are known as zero-days). Since immunity from cyberthreats is not a realistic goal for digital systems, the challenge for policymakers is defining

⁴ International Telecommunication Union (ITU), “National Strategies,” <https://www.itu.int/en/ITU-D/Cybersecurity/Pages/cybersecurity-national-strategies.aspx>.

what constitutes an acceptable non-zero level of residual risk and identifying how policy can support achievement of that level of resilience.

Cyberthreat Actors

There are two main categories of cyberthreat actors, plus a hybrid category. Nonstate actors such as criminals, hacktivists, and terrorists have primarily financial or ideological motives. State actors (i.e., governments) usually hack in support of military, intelligence, or law-enforcement objectives. A hybrid category exists in the form of proxies, or threat actors who blur or straddle the line between state and nonstate actors. They are often nonstate actors who may be enlisted by the state to carry out cyberoperations on its behalf or given tacit consent to operate. In many cases, a proxy actor may be formally employed by a government's security or military services but moonlight as a nonstate cyber criminal.

Cybersecurity Challenges

Four factors contribute to the difficulties that defenders of digital assets face. First, modern software is inherently complex, with tens of millions of lines of computer code or an AI model architecture whose behaviors humans cannot fully explain. While secure software development practices can reduce the prevalence of vulnerabilities, they cannot eliminate them entirely.

Second, the working environment within which software operates is both complex and broad. It is complex in that modern software can be difficult to maintain, and it is broad in that a typical enterprise IT environment could have at least as many devices connected to its network as it does employees, let alone customers, business partners, and others.

Third, human behavior further complicates defense: incentives do not always align in favor of security. For example, complex passwords help protect systems against brute force attacks, but people may resist using them because they are difficult to remember and burdensome to type. Security measures are seldom free; they typically require developers, system administrators, or end-users to invest scarce resources (time, money, and attention span) in implementing them. Governments face similar constraints when formulating policies to encourage cyber resilience among their citizens and private sectors: policy initiatives such as public-private partnerships, cyber rules and regulations, and international engagements require public administration resources. These resources are finite for any government but are especially precious for governments with limited institutional capacity in general. As a consequence, security often competes with other interests or values and may not prevail if the costs of security outweigh the benefits.

Finally, unlike natural disasters and accidents, cyberthreat actors are goal-oriented and adaptive. They are sensitive to the perceived costs and benefits of their actions, and they respond to defensive measures by adjusting their tradecraft.

Reducing Cyber Risk

A defender can reduce the probability of an attack by disrupting cyberthreat actors' operations (e.g., by arresting them) and encouraging stronger defenses. A defender can reduce the consequences of an attack by enhancing the resilience of its digital assets against cyberthreats. Stronger resilience can be achieved by taking actions to limit the harm that threat actors can cause in the event that they succeed in compromising a digital asset. (The focus of this study is on defenders' efforts to enhance the defenses and resilience of digital assets. It does not consider the

role of offensive cyberoperations or other policies and actions to disrupt cyberthreat actors. These are generally more complex operations that countries early in their cyber policy development process are unlikely to be able or eager to prioritize.)

Defenders' actions will ultimately be shaped by the legal and regulatory regime established at a national level, making effective cyber policymaking critical to cybersecurity outcomes. It is a truism of regulatory policy that laws and regulations should be as narrowly tailored and proportional as possible to the risks they are seeking to mitigate—and should be targeted toward a clear, defined set of risks as precisely as possible—to avoid becoming overly expansive. These can be challenging objectives, as discussed in chapter 4 of this report. Achieving them requires clearly defining what risks cyber policy is seeking to mitigate. It also requires identifying, implementing, and enforcing regulatory interventions that address those risks without causing excessive compliance burdens or unduly impeding other important policy objectives, such as the ability of companies to innovate. This additionally requires institutional capacity to carry out these actions. As discussed in chapter 2, many countries largely lack this capacity and must leverage their partnerships with countries that have developed and implemented more advanced cyber policy models.

In other words, governments have choices to make about what threats drive their assessments of risk and which assets to protect within the scope of their cybersecurity laws and policies. For example, some laws treat critical infrastructure differently from noncritical infrastructure and require that entities meeting that definition (or one of its comparable terms, such as critical entities or critical information infrastructure) implement more robust safeguards against cyber risks. Determining what entities are part of critical infrastructure is therefore an important scoping decision for policymakers. Most governments include in their definitions of critical infrastructure such core infrastructure as power generation and distribution, communications, financial services, government services, transportation, and healthcare, to name a few examples. Many jurisdictions, however, take a more expansive view of critical infrastructure. The European Union, for example, includes digital infrastructure such as cloud-computing and data center services within its definition of critical entities.⁵

As examined in chapter 1, governments also define the very concept of cybersecurity differently. There are two definitional paradigms for cybersecurity that recur in national and international debate. One of them is reflected in the Budapest Convention on Cybercrime, with 50 signatories, which describes its purpose as to “deter action directed against the confidentiality, integrity and availability of computer systems, networks and computer data as well as the misuse of such systems, networks and data.”⁶ This definition, with its emphasis on the confidentiality, integrity, and availability of information systems and their content, mirrors national-level definitions of the convention's signatories. In the United States, for example, the National Institute of Standards and Technology defines cybersecurity in similar terms, with added reference to authentication and

⁵ European Parliament and Council of the European Union, “Directive (EU) 2022/2557 on the Resilience of Critical Entities and Repealing Council Directive 2008/114/EC,” December 14, 2022, annex, <https://eur-lex.europa.eu/eli/dir/2022/2557/oj>.

⁶ Ibid., preamble.

nonrepudiation. Other signatories have broadly similar definitions in their national regulatory and policy regimes for cybersecurity.⁷ As noted earlier, this is the definition used in this report.

The other definitional paradigm extends the emphasis on confidentiality, integrity, and availability to include perceived threats to sovereignty and social stability perpetrated through the use of information and communications technologies. This paradigm, which often employs the phrase “information security” instead of “cybersecurity,” has its origins in Russia’s proposed resolution at the First Committee of the UN General Assembly in 1998 on “Developments in the field of information and telecommunications in the context of international security.”⁸ Referring to the risks of “information weapons,” the proposed resolution couched the risks in “the unprecedented level of information available to the public and, at the same time, the vulnerability of a society’s information structure.” As chapter 1 author Elaine Korzak explains in her previous work, this definitional paradigm goes “beyond concerns regarding the security of information and communication technologies systems and also includes the regulation of information or content flows.”⁹ As noted above, one of the primary challenges when this definition is used as the basis for policymaking is that it is inherently broader, making it more difficult to effectively implement narrow, tailored policies.

Sovereignty

All digital assets exist within the jurisdiction of some country in the world. States have an obvious interest in who owns and operates these digital assets; how those assets are used, including their security attributes; and what risks such uses pose to important political and economic interests. The concept of sovereignty is therefore an important contextual factor for cybersecurity regulations. All governments care about their sovereignty, and international law has a well-developed doctrine on what the concept means from a strictly legal standpoint. The concept also carries political-strategic weight that varies depending on a jurisdiction’s strategic culture. It informs how governments assess and respond to risk, including cyber risks.

Information Requirements for Mitigating Cyber Risks

Identifying appropriate cyber risk mitigations requires not only knowing the values of the two risk variables—threat and consequence—but also understanding the benefits and costs of proposed risk mitigations. Risk mitigation rules often have a primary regulatory goal that must be balanced against other interests. For example, cybersecurity regulations for critical infrastructure have a primary goal of improving the resilience of critical infrastructure against cybersecurity threats. But cybersecurity is only one of many policy interests that regulators must balance. For example, if a regulation is too expensive to implement, critical infrastructure providers might not be able to comply with it, which could lead to an unhealthy fatalism about cyber risks or a

⁷ For example, the EU’s Cybersecurity Act defines cybersecurity as “the activities necessary to protect network and information systems, the users of such systems, and other persons affected by cyber threats.” It defines “cyber threats” as “any potential circumstance, event or action that could damage, disrupt or otherwise adversely impact network and information systems, the users of such systems and other persons.” European Parliament and Council of the European Union, “Regulation (EU) 2019/881 on ENISA (the European Union Agency for Cybersecurity) and on Information and Communication Technology Cybersecurity Certification (Cybersecurity Act),” April 17, 2019, art. 2, <https://eur-lex.europa.eu/eli/reg/2019/881/oj>.

⁸ See the letter dated September 23, 1998, from the minister for foreign affairs of the Russian Federation to the UN Secretary-General, A/C.1/53/3, https://digitallibrary.un.org/record/261158/files/A_C.1_53_3-EN.pdf. See also Elaine Korzak, “Russia’s Cyber Policy Efforts in the United Nations,” NATO Cooperative Cyber Defence Centre of Excellence, Tallinn Paper, no. 11, 2021, https://ccdcoe.org/uploads/2021/06/Elaine_Korzak_Russia_UN.docx.pdf.

⁹ Korzak, “Russia’s Cyber Policy Efforts,” 15.

false sense of comfort that risk is being managed. Worse yet, an excessive focus on security could detract from the ability of providers to deliver reliable, cost-effective services to their customers—undermining their ability to support the digitalization and economic growth that effective cyber policies ultimately should facilitate. Rules that specify in detail the actions a provider must take to comply with a regulatory command can leave little room for innovation and adaptation to changing circumstances. On the other hand, highly general commands (e.g., “take reasonable measures” or “implement a risk management program”) could give excessive discretion to regulator and provider alike, potentially resulting in arbitrary enforcement by the regulator and lax performance by the provider.

When deliberating about trade-offs such as these, governments have significant information needs. They need information on threat actors and their capabilities and motivations. They need information on the consequences that could follow if a regulatory target suffers a cyber incident. They need information on the defensive abilities of their regulatory targets and the incentives that targets encounter when deciding whether to invest more in security compared with some other priority. They need information on security mitigations that work and those that fall short of expectations, including their costs and benefits. Gathering this information can be challenging for governments because the regulatory targets will often have direct, first-hand knowledge. This information asymmetry between regulators and their targets is a long-recognized challenge for regulators.¹⁰ How governments overcome it—and their degrees of success at overcoming it—is an important facet of how they regulate.

Cybersecurity Regulation: A Closer Look

All regulations consist of a rule with normative force that has some probability of being enforced. Research on regulatory policy has identified a range of design factors for regulations. Cary Coglianese identifies four dimensions: the entity that creates the regulation (“regulator”), the entity or entities that are covered by the regulation (“target”), the regulatory rule (“command”), and the consequences for upholding or violating the rule (“consequences”).¹¹ These dimensions, in turn, can take different forms. For example, the regulator could be a legislature that has enacted specific laws on cybersecurity, a regulatory agency issuing regulations on the basis of general or cybersecurity-specific regulatory powers granted to it by a legislative body or other authoritative source (e.g., the national constitution), or even a private or nongovernmental actor (e.g., a standard-setting body). The target could be a specific sector or numerous sectors (cross-sectoral). A command could prescriptively specify the actions a covered entity must take to comply or identify an outcome but leave it to covered entities to determine what actions to take to achieve the outcome. In addition to this means-ends optionality, the command could be more or less specific as to the means or ends. Consequences, finally, could be negative or positive.

Regulators

Jurisdictions have taken a broad range of different approaches to identifying which regulator or regulators are responsible for cyber risks. For example, the original regulator for the EU NIS2

¹⁰ Steven Shavell, “Liability for Harm versus Regulation of Safety,” *Journal of Legal Studies* 13, no. 2 (1984): 357–74.

¹¹ Cary Coglianese, “Management-Based Regulation: Implications for Public Policy,” in *Risk and Regulatory Policy: Improving the Governance of Risk*, ed. Gregory Bounds, Nikolai Malyshev, and Josef Konvitz (Paris: OECD Publishing, 2010), chap. 5.

Directive is the EU, which enacted the law, but individual member states must enact implementing regulations and are responsible for enforcement.¹² China's Cybersecurity Law directs "departments responsible for security protection work for critical information infrastructure" to develop and enforce implementing regulations.¹³ Many countries assign a lead role for cybersecurity regulation to their telecommunications regulators. In the United States, sector-specific regulatory agencies enact cybersecurity regulations within the span of their statutory jurisdiction.

Regulatory Targets

The target of regulations can vary as well. China's Cybersecurity Law, for example, covers "the construction, operation, maintenance, and use of networks"—an expansive scope of jurisdiction that cuts across all sectors of the economy and even includes ordinary, casual users of IT.¹⁴ The EU's NIS2 Directive is narrower in target scope, applying to organizations above a certain size as well as organizations that provide or support critical societal functions, with differentiated obligations depending on the type of organization.¹⁵ The United States, on the other hand, selectively regulates critical infrastructure, with individual regulatory agencies enacting and enforcing regulations within their span of jurisdiction. There is no federal, baseline cybersecurity law in the United States.

The ownership status of the targets is subject to further variation. Within critical infrastructure, owners and operators could be state-owned entities or private firms. For example, in most jurisdictions water utilities are typically owned and operated by governments; communications providers are state-owned or private, depending on the country; and financial services providers are usually privately owned.

Regulatory Commands

There is wide variation in the types of commands that cybersecurity regulations impose. Coglianese identifies four main categories,¹⁶ though cybersecurity laws often include more than one of the four types of regulatory commands. Each type is described below.

Design standards. Commands can take the form of design standards that specify the means that a regulated entity must employ to reach a cybersecurity goal. Design standards are relatively uncommon in cybersecurity regulations but are becoming more common as regulators push organizations to implement basic cybersecurity controls. For example, California's IoT Act (2018) effectively bans the use of hard-coded, non-unique passwords in consumer Internet of Things devices—a design requirement intended to advance the law's goal of requiring these devices to have a "reasonable security feature."¹⁷ Another example is multifactor authentication: regulators

¹² European Parliament and Council of the European Union, "Directive (EU) 2022/2555 on Measures for a High Common Level of Cybersecurity Across the Union (NIS2 Directive)," December 14, 2022, art. 31, <https://eur-lex.europa.eu/eli/dir/2022/2555>.

¹³ People's Republic of China (PRC), *Cybersecurity Law of the People's Republic of China*, adopted November 7, 2016, effective June 1, 2017, art. 32, trans. DigiChina, <https://digichina.stanford.edu/work/translation-cybersecurity-law-of-the-peoples-republic-of-china-effective-june-1-2017>.

¹⁴ *Ibid.*, art. 2.

¹⁵ European Parliament and Council of the European Union, "Directive (EU) 2022/2555 on Measures for a High Common Level of Cybersecurity Across the Union (NIS2 Directive)," arts. 2–3.

¹⁶ Coglianese, "Management-Based Regulation."

¹⁷ California Legislature, Senate Bill 327, chap. 886, signed October 4, 2018, https://leginfo.legislature.ca.gov/faces/billTextClient.xhtml?bill_id=201720180SB327.

are increasingly mandating its use, as in the example of the U.S. Transportation Security Administration's directive on pipeline security.¹⁸

Management-based regulation. A regulatory command can also take the form of management-based regulation, where the regulator requires the target to implement a management process aimed at mitigating risk. The EU's NIS2 Directive, for example, requires that member states ensure that "essential and important entities take appropriate and proportionate technical, operational and organisational measures to manage" cybersecurity risks. The directive goes on to specify the risk factors that organizations must address.¹⁹ China's Cybersecurity Law includes management-based regulation as well: Article 21(1) requires network operators to "formulate internal security management systems and operating rules, determine persons who are responsible for cybersecurity, and implement cybersecurity protection responsibility."²⁰

Performance requirements. Another approach to crafting a regulatory command is for the regulator to specify performance requirements or a desired end state but leave it to targets to determine how best to achieve those goals. Some regulatory regimes are aimed at achieving specific, measurable outcomes. An automobile regulator, for example, can set objectively measurable targets for fuel economy for an automobile manufacturer's fleet and let the manufacturer figure out how to meet them.

Performance requirements are rare in cybersecurity regulations, however, because it is difficult for regulators to specify what they might be in an objective, consistent manner. A standard of "no breaches" is unrealistic for most sectors of the economy because threat actors sometimes discover zero-day vulnerabilities in commonly used software. There is no defense for a zero-day vulnerability until the vendor of the software issues a fix, or patch, for it. Indeed, the fact of a breach is not always the best indicator of performance. Accidents happen, and the nature and degree of fault will often depend on the circumstances of the accident: Was it foreseeable? Could it have been prevented? If so, would prevention have come at a reasonable cost? Resilience in the face of adverse circumstances is important, but measuring it in the context of a cyber incident is difficult, especially when trying to apply a single standard across many different organizations or sectors.

Ex post liability. The first three categories—design standards, management-based regulation, and performance requirements—involve the regulator requiring targets to take actions *ex ante* that are aimed at preventing a harm or achieving some objective.²¹ A target could be liable under these commands if it fails to meet the regulatory requirements, regardless of whether the harm that the command was designed to avoid happens. In the context of cybersecurity, an organization subject to China's Cybersecurity Law that failed to "store network logs for at least six months" pursuant to Article 21 could be fined under Article 59, even if the lapse did not lead to a cybersecurity incident or other immediate harm.²²

In contrast, *ex post* liability emerges after an incident or some other harm-causing event. For this fourth category of command, an adverse event occurs that causes the regulator (or possibly

¹⁸ U.S. Transportation Security Administration, "Security Directive Pipeline-2021-02D: Pipeline Cybersecurity Mitigation Actions, Contingency Planning, and Testing," July 26, 2023, https://www.tsa.gov/sites/default/files/tsa-sd-pipeline-2021-02d-w-memo_07_27_2023.pdf.

¹⁹ European Parliament and Council of the European Union, "Directive (EU) 2022/2555 on Measures for a High Common Level of Cybersecurity Across the Union (NIS2 Directive)," art. 21.

²⁰ PRC, *Cybersecurity Law*, art. 21(1).

²¹ Shavell, "Liability for Harm versus Regulation of Safety."

²² PRC, *Cybersecurity Law*, arts. 21, 59.

a court) to examine whether the target complied with a general duty requirement (e.g., to act “reasonably”). An example of a regulator pursuing ex post liability in response to a harm is enforcement by the U.S. Federal Trade Commission (FTC) of its authority to police “unfair or deceptive acts or practices” under Section 5 of the FTC Act when an IT product vendor or other company has made claims about the security attributes of their products or data security practices and then suffers a breach. In such cases, the FTC moves to hold the company liable for failing to live up to its security claims.²³

Regulatory Enforcement

Enforcement is the final dimension of regulation. The tools of enforcement generally—investigatory authority, injunctions, fines, and criminal sanctions—are also found in cybersecurity laws. For example, the FTC enforces its Section 5 authority through investigatory actions and the pursuit of court-issued injunctions against targets that require the target to take remedial actions.²⁴ China’s Cybersecurity Law permits injunctions and fines, as well as criminal penalties for certain violations.²⁵ The EU’s General Data Protection Regulation links fines to an organization’s annual turnover.²⁶

Cybersecurity Policy: A Closer Look

The four chapters that follow build on this foundational background to illuminate important dimensions of cybersecurity policymaking as governments around the world seek to enhance their digital resilience. In chapter 1, Elaine Korzak maps the relationships between the positions on cybersecurity that different states take in multilateral international forums (such as the United Nations) and their approaches to cybersecurity resilience policies within their domestic economies. She describes a “cybersecurity approach” championed by the United States and other like-minded nations in their multilateral engagements focused narrowly on protecting the confidentiality, integrity, and availability of information and communications technology systems and contrasts it with a more expansive “information security approach” championed by China and Russia that emphasizes risks and controls over information flows that could destabilize domestic political order. These differing approaches are often associated with different positions on issues such as the relative desirability of new international treaties on information security and the role and adequacy of existing international law and the established UN framework for responsible state behavior in global cybersecurity.

At the national level, Korzak identifies four governance archetypes derived from an analysis of national cybersecurity strategies:

1. A market-driven approach (associated primarily with the United States) emphasizes innovation, private-sector leadership, and market incentives as foundational for cybersecurity resilience.

²³ See, for example, *Federal Trade Commission v. D-Link*, “[Proposed] Stipulated Order for Injunction and Judgment,” U.S. District Court, Northern District of California, 2017, https://www.ftc.gov/system/files/documents/cases/dlink_proposed_order_and_judgment_7-2-19.pdf.

²⁴ *Ibid.*

²⁵ PRC, *Cybersecurity Law*, chap. 6.

²⁶ European Parliament and Council of the European Union, “Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data (General Data Protection Regulation),” April 27, 2016, art. 83, <https://eur-lex.europa.eu/eli/reg/2016/679/oj>.

2. A digital rights approach (associated primarily with the EU) focuses on safeguarding the fundamental rights of individual citizens through regulatory interventions.
3. An information isolationism approach (associated primarily with Russia) focuses on controlling the domestic information environment by decoupling from foreign technology and information ecosystems.
4. An advanced information isolationism approach (associated primarily with China) mirrors the information isolation approach but draws on a competitive domestic technology sector to service the domestic market and support the government's censorship priorities.

Korzak concludes that these divergent approaches, rooted in distinct historical, cultural, legal, and political contexts, lead to different economic and security outcomes—a topic examined further in subsequent chapters.

In chapter 2, Anthony Adams explores how nations' cyber resilience maturity influences their willingness to engage in international collaboration and capacity-building initiatives, using Pacific Island nations as case studies. He applies qualitative measures to categorize nations into four cyber resilience maturity levels and describes how cyber maturity shapes countries' willingness and ability to productively make use of various forms of international cyber engagement, such as capacity building, information sharing, legislative assistance, and global norms development. He concludes that cybersecurity assistance and engagement should be calibrated to where a recipient nation actually sits on the maturity spectrum, rather than applying a one-size-fits-all approach. This marks a shift from donor-centric to recipient-centric framing of cybersecurity aid.

In chapter 3, Josephine Wolff examines why countries sometimes elect to restrict cross-border data flows and foreign digital services in the name of cybersecurity, using examples from a diverse range of jurisdictions, including India, the EU, Brazil, Somalia, and Indonesia. She finds that while such restrictive measures are rarely justified as broad-spectrum cybersecurity improvements, governments do view them as providing resilience against specific, high-probability threats—particularly cyberespionage from identified adversaries or politically destabilizing information flows—while also aiding in prevention and enforcement by providing law enforcement with increased access to communications. She also describes the difficult and potentially costly trade-offs that countries must navigate when considering in what instances to restrict data flows or access to foreign digital services in pursuit of cybersecurity resilience. The chapter concludes that governments' positions on internet openness are not static but shift as their threat perceptions evolve. Ostensibly security-motivated policies frequently intermingle with economic and political goals that are difficult to fully disentangle.

In chapter 4, Jaclyn Kerr argues that policymakers around the world face a common struggle: how to make decisions about cybersecurity policy that account for long-term consequences and trade-offs. While there is no perfect antidote to this struggle, she points out that many trade-offs are known and indeed recurring: blocking foreign services or requiring data localization can improve resilience against specific threats like espionage but reduce resilience against outages and undermine the economic benefits of openness; restricting encrypted communications aids law enforcement but exposes users to the very espionage threats that encryption guards against; vendor consolidation improves short-term security posture but risks lock-in and dangerous monocultures over time. Even though factors like cognitive bias, limited state capacity, fragmented decision authority (across municipal, national, and supranational levels), and geopolitical pressure from major powers can distort cybersecurity policymaking away from rational, comprehensive analysis

of trade-offs, Kerr concludes that policymakers have a menu of options for improving their ability to identify and weigh these trade-offs.

Kerr identifies one element as being of particular importance to understanding these trade-offs: proactively soliciting and listening to diverse stakeholder perspectives—including private sector and civil society—to disclose hidden costs and unintended consequences before policies are locked in. While no cybersecurity policy can eliminate all risks, decision-makers who build feedback mechanisms, maintain stakeholder trust, and keep long-term trade-offs front of mind are far better positioned to adapt as the threat environment inevitably evolves.

The conclusion extends the arguments from the preceding chapters and provides policy implications for governments and other stakeholders. It highlights the enhanced importance of effective cybersecurity policymaking in an era in which AI is rapidly transforming the digital domain—and in which countries’ abilities to secure their digital ecosystems and capture the benefits of AI deployment are increasingly interdependent. It notes the difficult trade-offs inherent to the design and implementation of cybersecurity policy, particularly in countries that are early in their digital development, and the range of legitimate policy factors that pull countries toward the various models outlined throughout this report. It ultimately emphasizes the importance of adequately weighting the long-term economic, technological, and practical costs associated with different approaches to cybersecurity policy—in addition to near-term concerns about specific risks. Because of these costs, the conclusion argues that what is described throughout this report as an “open” model of cybersecurity governance—one that emphasizes broad stakeholder engagement in cyber policymaking and remains broadly open to international markets—more effectively mitigates against these long-term risks to growth and innovation than the “information isolationist” model.

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Mapping Divergent National Approaches to Cybersecurity Policy

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EXECUTIVE SUMMARY

This chapter analyzes and typologizes diverging national- and international-level approaches to cybersecurity policy in key states influencing the development of global norms, standards, and policies in cyberspace.

MAIN ARGUMENT

States around the world are developing national cybersecurity strategies and policies to address emerging risks in cyberspace and are implementing these measures at the domestic level through strategies, laws, and regulations and at the international level through discussions in the United Nations and other forums. The development of distinct approaches to cybersecurity policy among several key actors—namely the U.S., the European Union, Russia, and China—is shaping the global landscape of cybersecurity policy by driving international negotiations on the rules that will govern cyberspace, as well as the strategic and policy directions of other states. Each of the four key players takes a unique domestic approach, which, while largely aligning on the protection of critical infrastructure and promotion of economic interests in cybersecurity policy, differ significantly on the role of the state in cyberspace, the protection of the individual versus the political system, and the adequacy of current international rules governing cyberspace. These national policies align broadly with the countries' positions in international debates, with the U.S. and EU leading a cluster of states advocating for a narrow cybersecurity approach, while Russia and China lead a cluster of states advocating for a more expansive information security approach. These varying approaches to national- and international-level policy offer key insights into the factors driving the development of different cybersecurity policies globally.

POLICY IMPLICATIONS

- National-level cybersecurity policies reflect the economic reality and goals of each country. The U.S. and EU seek to maintain strong digital markets and technology industries, while China and Russia seek to compete economically in technology and cybersecurity.
- States pursuing the information security approach, including both Russia and China as well as countries whose cyber policy models are shaped by these countries, will likely continue to advocate for substantial changes to international law governing cyberspace in international negotiations.
- The information revolution and its implications for the stability of political regimes is a key factor influencing the approaches to cybersecurity policies and the involvement of nongovernmental stakeholders in policy development.

Alongside the rapid proliferation of digital technologies, states at all stages of development and of varying governance types are designing and implementing policies to address growing cybersecurity concerns. Governments have attempted to address these cyber risks both internationally through diplomatic forums such as those convened under the auspices of the United Nations and nationally through regulatory and other policy efforts within their domestic jurisdictions, resulting in the emergence of a diverse array of governance approaches. However, a comprehensive or systematic mapping of divergent national- and international-level approaches to cybersecurity policies and governance is lacking.

This chapter aims to fill this gap and is guided by the following research question: What are the main points of divergence among competing approaches to cybersecurity policies and governance? The chapter examines how key states—that is, those at the forefront of shaping the development of global norms, standards, and agreements that will affect a wider range of countries, particularly developing ones—approach cybersecurity policies, analyzing both international- and national-level discussions as well as how these two layers relate to each other. The chapter presents an initial mapping of these approaches to cybersecurity. As these approaches evolve and new ones emerge, future research could expand and update the mapping.

With regard to the international level, this chapter focuses on discussions in the United Nations, where two distinct diplomatic positions have evolved. One, led by the United States and European Union member states, advocates an international framework of responsible state behavior and is based on a conceptual understanding of cybersecurity that focuses on risks to cyber resilience, namely the confidentiality, integrity, and availability of information and communications technology (ICT) systems. The other position is anchored in the concept of information security and is rooted in a broader definition of cybersecurity that includes risks associated with information and information flows. Both Russia and China have been proponents of this approach and have, among other things, emphasized the sovereignty and prerogative of states in cyberspace. This chapter refers to these distinct approaches as the “cybersecurity approach” and the “information security approach,” respectively.

With regard to the national level, this chapter focuses on four major states or groups of states—the United States, the EU, Russia, and China—and examines how they approach governance and regulation of cyber risks within their domestic jurisdictions. The proposed methodology utilizes national strategies as shorthand to identify signature elements of these national-level approaches to cybersecurity governance. The result is an initial mapping that identifies four distinct national approaches: (1) a “digital rights approach” (EU), (2) a “market-driven approach” (United States), (3) an “information isolationism approach” (Russia), and (4) an “advanced information isolationism approach” (China). As explained below, the sharpest divergence in approach is between the digital rights and market-driven approaches, on the one hand, and the information isolationism and advanced information isolationism approaches, on the other.

These national-level governance models are consistent with the respective states’ positions in international-level discussions. However, the analysis reveals notable similarities and differences both across and within the cybersecurity and information security camps. While all four national-level approaches attempt to achieve both cyber resilience and the advancement of economic interests in cyberspace, significant divergences emerge over the role of the state in governance and regulation, the protection of individual citizens versus the political system as a whole, and the

adequacy of current international rules for cyberspace. As will be examined in chapter 4 of this report, these differing approaches lead to divergent economic and security outcomes.

The first part of this chapter surveys the current understanding of national-level approaches to cybersecurity governance and identifies important gaps. The second part lays out a proposed methodology to begin to address those gaps by examining national cybersecurity strategies of major states. The third and fourth parts identify divergent governance approaches in international-level discussions and national-level strategies, respectively. The chapter concludes with observations about similarities and differences among the divergent national-level approaches and how national-level positions fit in with dominant international-level approaches.

Mapping Divergent Approaches: Current Gaps

Although cybersecurity has become a burgeoning field of academic and policy research,¹ a critical gap is the absence of analysis studying relationships between the regulatory approaches that states take within their domestic jurisdictions to frame and address cybersecurity risks and the positions that states take in international forums on cyber issues. This is surprising. Researchers have analyzed in detail the cybersecurity efforts and policies for individual countries, as well as different aspects of cybersecurity and resilience, such as critical infrastructure protection or public-private partnerships.² Yet, despite evident differences among states in terms of their economic, political, and regulatory environments, the range of approaches that states take to cybersecurity policy has not been comprehensively or systematically surveyed worldwide. Further, divergent approaches to cybersecurity at a national level have not been mapped to or linked back to policy positions that states have advanced in international-level governance conversations, despite the latter being a topic of long-standing scholarly interest and focus. Regional and international organizations, including the United Nations, have actively discussed cybersecurity governance, and distinct governance approaches have crystallized in these negotiations.

On the other hand, considerable work has gone into assessing the maturity or capacity of states worldwide. These types of assessments typically seek to review a state's cyber resilience capacity or posture. One prominent example is the Cybersecurity Capacity Maturity Model, an analytical framework developed by the University of Oxford that has been employed in over 90 states.³ Another example is the Cyber Readiness Index developed by the Potomac Institute for Policy Studies.⁴ Capacity or maturity assessments are important for benchmarking different countries' cyber resilience postures. However, they do not analyze and categorize the underlying motivations and risk decision frameworks driving divergent national-level approaches to cybersecurity policies overall. Instead, these assessments typically map a state's capacities against one standard that has been set within the context of a given assessment tool's methodology.

¹ For an overview introduction, see Paul Cornish, ed., *The Oxford Handbook of Cyber Security* (Oxford: Oxford University Press, 2021); Eneken Tikk and Mika Kerttunen, eds., *Routledge Handbook of International Cybersecurity* (New York: Routledge: New York, 2020); and P.W. Singer and Allan Friedman, *Cybersecurity and Cyberwar: What Everyone Needs to Know* (Oxford: Oxford University Press, 2014).

² See, for example, the body of literature in the *Journal of Cyber Policy*, available at <https://www.tandfonline.com/journals/rcyb20>, and the *Journal of Cybersecurity*, available at <https://academic.oup.com/cybersecurity>.

³ For more information on the Cybersecurity Capacity Maturity Model, see Global Cyber Security Capacity Centre, University of Oxford, <https://gcscc.ox.ac.uk/the-cmm>.

⁴ For more information on the Cyber Readiness Index, see the Potomac Institute for Policy Studies, <https://www.potomac institute.org/index.php/centers/past-centers/cyber-readiness-index>.

This chapter begins to address existing gaps by mapping alternative approaches to cybersecurity, focusing on a select number of states that have been prominent and active in international-level discussions driving cybersecurity policy and regulation. It does not provide a comprehensive overview of all national-level and international-level approaches to cybersecurity. Rather, it provides a blueprint for future analysis, including the examinations later in this report of how states taking different approaches to cybersecurity implement cyber resilience policies and what their relative efficacy in achieving both security and economic objectives is. As more states and their policy approaches are analyzed over time and mapped onto the spectrum of approaches identified herein, findings from this chapter will continue to evolve as they are supplemented and supplanted by future analysis.

Mapping Divergent Approaches: Methodology

To systematically map and analyze divergent national-level approaches to cybersecurity, this chapter first examines different approaches that states have advanced in international-level cybersecurity governance debates, which often compel states to be explicit about their underlying visions for effective cybersecurity policies and marshal the support of like-minded states. This analysis yields broad categories of divergent approaches that are shared by a number of states.

Second, within these broad categories of states' approaches to cybersecurity, the chapter then maps the national-level policy approaches of select states, identifying the spectrum of national-level approaches within international-level categories. It does so by examining countries' published national cybersecurity strategies as a main proxy to distill characteristics of divergent national-level approaches to cybersecurity. National-level strategies, frameworks, or doctrines provide a high-level, yet fundamental, overview of a state's understanding of cybersecurity and its approach to governing it. National strategic documents are highly contextual and can even change with new iterations or updated versions. As one analysis by Microsoft notes, "priorities for national cybersecurity strategies will vary country by country. In some countries, the focus may be on protecting critical infrastructure risk, while other countries may focus on protecting intellectual property, and still others may focus on improving the cybersecurity awareness of newly connected citizens."⁵

At their core, national strategies identify a state's current cybersecurity objectives, a desired future state of cybersecurity, and the policies and measures needed to get there, given various resource and other constraints.⁶ All these aspects provide important insights for an analysis of divergent national-level approaches.

This chapter's examination of key states and their national cybersecurity strategies will focus on three components in particular: (1) the perceived nature of threats in cyberspace, (2) the main entities (governmental versus nongovernmental) responsible for cybersecurity, and (3) the prioritized equities (or interests) shaping these components.

This methodology enables a mapping of divergent approaches to cybersecurity governance at both the international and national levels. However, important limitations need to be recognized.

⁵ Cristin Flynn Goodwin and J. Paul Nicholas, "Developing a National Strategy for Cybersecurity: Foundations for Security, Growth, and Innovation," Microsoft, October 2013, 4, https://download.microsoft.com/download/b/f/0/bf05da49-7127-4c05-bfe8-0063dab88f72/developing_a_national_strategy_for_cybersecurity.pdf.

⁶ Ibid.

National cybersecurity strategies might not reflect the totality of influential stakeholders' views on cybersecurity policy. Independent legislatures, disaffected agencies, and private actors may be among the potentially influential actors whose views are not necessarily captured in national strategies. In addition, these strategies provide high-level, aspirational directions for a state's cybersecurity policies that can change over time, as reflected in various iterations of national strategies. These changes could reflect shifts in threat perceptions, resource constraints, and political factors, among other potential drivers of change. Any given national cybersecurity strategy provides a snapshot of states' approaches at the time of publication.

A final limitation is that while national cybersecurity strategies have proliferated over the past two decades, with some countries on their second or third iteration, not all member states of the United Nations have one.⁷ This limits the number of national approaches that can be analyzed using this methodology. A state could have many reasons for not pursuing a strategy. In particular, states without national strategies tend to be earlier in their cyber policy development trajectory, suggesting that additional types—perhaps belonging to countries with the most severe resource constraints—may not be captured. However, as discussed in chapter 2, these states are generally very early in their development of cyber resilience capacity, meaning that they are far more likely to be highly reliant on policy and capacity-building support from those with more established models rather than be active drivers of those models themselves.

Despite these limitations, the examination of national cybersecurity strategies provides an initial way to map and categorize contrasting national-level governance approaches.

Mapping Divergent Approaches: The International Dimension

For more than two decades, states have discussed cybersecurity issues in diplomatic forums—most notably at the United Nations, where discussions have focused on state conduct in cyberspace as well as cybercrime.⁸ Processes in the United Nations provide an essential starting point for how states approach national-level cybersecurity policies, given that consensus often represents the lowest common denominator of agreement among states.⁹ In other words, these processes provide an international baseline that all participating states have been able to agree to, which, in turn, sets a minimum standard for the design of national approaches to cybersecurity policies.

So far in UN negotiations, states have agreed to cooperate to promote “an open, secure, stable, accessible and peaceful ICT environment.”¹⁰ Their agreement provides a set of high-level elements for the governance of cyberspace. The international framework for responsible state behavior in

⁷ The UN Institute for Disarmament Research's Cyber Policy Portal collates national cybersecurity documents and regulations and is a great resource to find national policies worldwide. It provides an up-to-date overview of the number of UN member states with cybersecurity strategies and documents. More information is available at <https://cyberpolicyportal.org>.

⁸ For overview information, see UN Office for Disarmament Affairs, “Developments in the Field of Information and Telecommunications in the Context of International Security,” <https://disarmament.unoda.org/ict-security>; and UN Office on Drugs and Crime, “Cybercrime,” https://www.unodc.org/unodc/en/cybercrime/index_new.html.

⁹ See, for example, Alexandra Kulikova, “Cyber Norms: Technical Extensions and Technological Challenges,” *Journal of Cyber Policy* 6, no. 3 (2021): 341.

¹⁰ See, for example, the UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, “Report of the Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security,” A/70/174, July 2015, 6, available at <https://documents.un.org/doc/undoc/gen/n15/228/35/pdf/n1522835.pdf>.

cyberspace (the “UN framework”),¹¹ which is meant to enhance cyber resilience by addressing the malicious use of ICT, provides further considerations for states. The framework for responsible state behavior includes, among other things, the application of existing international law and a set of eleven voluntary nonbinding norms, both of which are relevant for how states approach and implement their national cybersecurity policies. Most notably, the voluntary norms include a commitment to respect human rights online and the need to protect critical infrastructure.¹²

However, beyond these considerations, international-level commitments leave states with considerable discretion in their national-level approaches to cybersecurity policies. Unsurprisingly, distinct approaches to cybersecurity policies have emerged. Within the discussions at the United Nations and beyond, participants have advanced two main (and competing) conceptual approaches: information security and cybersecurity. On the one hand, several states, most notably Russia and China, have advanced the concept of information security. On the other hand, states, including most members of the Organisation for Economic Co-operation and Development (OECD), have approached national and international policies based on a narrow concept of what this chapter will refer to as the cybersecurity approach.¹³

The main difference in these governance approaches can be explained by differing views of threats, responsible stakeholders, and prioritized equities in providing cybersecurity. Proponents of the cybersecurity approach are principally concerned with threats to individuals and organizations within the state and take a narrow approach in securing information and communications technologies from malicious uses and enhancing cyber resilience.¹⁴ Cybersecurity policy is predominantly oriented toward securing ICT systems, networks, and the data they contain from threats and risks. A particular focus is placed on the confidentiality, integrity, and availability of ICT systems and their data.¹⁵

In contrast, proponents of information security take a broader approach to cybersecurity that goes beyond the confidentiality, integrity, and availability of ICT systems to also include concerns related to information or content flows that political elites consider to be harmful or threatening. These concerns focus on the potential of information or information flows to destabilize societies and states.¹⁶ Thus, the information security approach is principally concerned with threats to the overall social and political order, whether these involve the targeting of ICT systems or the use of information flows.

¹¹ The framework for responsible state behavior consists of successive UN documents: UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, “Report of the Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security,” A/68/98, July 2013, https://dig.watch/wp-content/uploads/A_68_98_E.pdf; UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, “Report of the Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security,” 2015; Group of Governmental Experts on Advancing Responsible State Behaviour in Cyberspace in the Context of International Security, “Report of the Group of Governmental Experts on Advancing Responsible State Behaviour in Cyberspace in the Context of International Security,” A/76/135, July 2021, <https://digitallibrary.un.org/record/3934214?ln=en&v=pdf>; and Open-Ended Working Group on Developments in the Field of Information and Telecommunications in the Context of International Security, “Final Substantive Report,” A/AC.290/2021/CRP.2, March 2021, <https://front.un-arm.org/wp-content/uploads/2021/03/Final-report-A-AC.290-2021-CRP.2.pdf>.

¹² See, for example, provisions in the 2015 report of the Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security.

¹³ See Elaine Korzak, “Russia’s Cyber Policy Efforts in the United Nations,” NATO Cooperative Cyber Defence Centre of Excellence, Tallinn Paper, no. 11, 2021, https://ccdcoe.org/uploads/2021/06/Elaine_Korzak_Russia_UN.docx.pdf.

¹⁴ *Ibid.*, 5–16.

¹⁵ See, for example, the national view of the United Kingdom submitted to UN proceedings in the “Report of the Secretary-General,” A/68/156, July 2013, 15, <https://documents.un.org/doc/undoc/gen/n13/397/35/pdf/n1339735.pdf>.

¹⁶ See the discussion in Korzak, “Russia’s Cyber Policy Efforts,” 15–16.

The divergent governance approaches of cybersecurity and information security result in two key policy positions in international negotiations. First, states' positions differ as to the adequacy of the regulatory status quo at the international level, particularly whether new international legal agreements are needed. Second, states advance differing views regarding the role of the state and governmental entities in the regulation of cybersecurity.

First, proponents of information security have advocated for the negotiation and adoption of new international legal agreements to regulate information security.¹⁷ Russia, along with China and a handful of others, have persistently called for a treaty on “international information security” that would codify the broad approach of information security.¹⁸ In contrast, proponents of the cybersecurity approach have opposed negotiations of new international legal agreements, preferring instead to apply existing international law as well as the international framework on responsible state behavior.¹⁹

Second, for proponents of information security, governments are the primary actors responsible for the protection of the ICT environment. This view acknowledges a role for nongovernmental stakeholders—business, academia, and civil society—but sees them as subordinate or even subservient to governmental efforts and leadership.²⁰ Information security advocates emphasize international law principles, such as state sovereignty and noninterference in the internal affairs of states.²¹ In contrast, states advancing the cybersecurity approach have stressed the need for close cooperation with nongovernmental stakeholders, among others, through public-private partnerships. As will be discussed later in this report, this fundamental difference in allocation of responsibility for cybersecurity has major impacts on the implementation of cybersecurity policies and on the effects these policies have on cyber resilience, security, economic, and innovational outcomes.

Mapping Divergent Approaches: The National Dimension

Having explained key differences in international approaches, the following part examines national-level cybersecurity strategies to develop four archetypes of national-level approaches to cybersecurity. It distinguishes the four archetypes according to their depictions of (1) the threats identified by states, (2) the main stakeholders responsible for ensuring cybersecurity, and (3) the equities that each state prioritizes in its respective approach.

Market-Driven Approach: The United States

The United States has been a critical voice in international-level cybersecurity discussions since their inception, advancing and often leading the diplomatic efforts of like-minded states and promoting the cybersecurity approach. The United States has also been an early adopter of

¹⁷ Korzak, “Russia’s Cyber Policy Efforts,” 13–15.

¹⁸ Over the years Russia has submitted several draft conventions. The most recent proposal—“Updated Concept of the Convention of the United Nations on Ensuring International Information Security”—is available at [https://docs-library.unoda.org/Open-Ended_Working_Group_on_Information_and_Communication_Technologies_-\(2021\)/ENG_Concept_of_UN_Convention_on_International_Information_Security_Proposal_of_the_Russian_Federation.pdf](https://docs-library.unoda.org/Open-Ended_Working_Group_on_Information_and_Communication_Technologies_-(2021)/ENG_Concept_of_UN_Convention_on_International_Information_Security_Proposal_of_the_Russian_Federation.pdf).

¹⁹ See the national view of the United States submitted to UN proceedings in the Report of the Secretary-General, A/59/116/Add. 1, December 2004, 3, <https://documents.un.org/doc/undoc/gen/n04/475/11/pdf/n0447511.pdf>.

²⁰ Korzak, “Russia’s Cyber Policy Efforts,” 18–19.

²¹ Ibid., 17–18.

national-level cybersecurity strategies and plans, having gone through several iterations over the years.²² The latest update to its national-level approach to cyber resilience is the 2026 Cyber Strategy for America.²³ In addition, different parts of the U.S. government have issued strategy documents focused on other dimensions of cyber policy, including the 2024 International Cyberspace and Digital Strategy.²⁴

Perceived threats: Targets and threat actors. The current Cyber Strategy for America posits that digital interconnectivity is only deepening and that while this trend will entail numerous opportunities for development, it will also introduce new risks and exacerbate existing ones. The strategy highlights risks to critical infrastructure and identifies its protection and resilience as one of the key pillars for action in the national strategy. It describes critical infrastructure as vital but increasingly vulnerable as a result of next-generation connectivity, which exposes essential systems to disruption, including data centers, the energy grid, and water utilities.²⁵ As a result, the strategy points to the need to safeguard critical infrastructure from threats and ensure the availability and resilience of the essential services they provide.

In addition, the U.S. approach is concerned with cybercrime activities. In particular, the Cyber Strategy for America highlights the threat from criminal syndicates to national security, public safety, and economic prosperity.²⁶ With regard to individual citizens, the strategy points to both direct cyberthreats and the importance of the “right to privacy for Americans and American data.”²⁷

Lastly, the greatest threats in cyberspace emanate from malicious military, intelligence, and criminal adversaries that need to be confronted and defeated.²⁸ In particular, adversary activities challenge freedom and safety in cyberspace.²⁹

Responsible stakeholders. The U.S. approach to national-level cybersecurity policy recognizes the fundamental and critical role of nonstate actors, in particular from the private sector, in enhancing cyber resilience. The latest national strategy affirms that robust public-private collaboration is essential to securing cyberspace.³⁰ The U.S. national-level approach to cybersecurity governance has long emphasized the roles and responsibilities of nonstate actors in cybersecurity, deferring to market forces and incentives that drive the behavior of different nongovernmental stakeholders in lieu of government regulatory activity.

Concomitant with the view that the private sector should play a leading role in cybersecurity implementation, U.S. government regulation has relied on shaping market dynamics and creating incentives, resulting in a relatively limited regulatory footprint. As highlighted in the Cyber Strategy for America, government interventions are geared toward shaping the market rather than

²² For a succinct overview of U.S. strategies and documents, see Jason Healey, “Twenty-Five Years of White House Cyber Policies,” Lawfare, June 2, 2023, <https://www.lawfaremedia.org/article/twenty-five-years-of-white-house-cyber-policies>.

²³ White House, *President Trump’s Cyber Strategy for America* (Washington, D.C., March 2026), <https://www.whitehouse.gov/wp-content/uploads/2026/03/president-trumps-cyber-strategy-for-america.pdf>.

²⁴ U.S. Department of State, *United States International Cyberspace and Digital Policy Strategy: Towards an Innovative, Secure, and Rights-Respecting Digital Future* (Washington, D.C., May 2024), https://www.state.gov/wp-content/uploads/2024/07/United-States-International-Cyberspace-and-Digital-Strategy-FINAL-2024-05-15_508v03-Section-508-Accessible-7.18.2024.pdf.

²⁵ White House, *President Trump’s Cyber Strategy for America*, 5.

²⁶ *Ibid.*, 4.

²⁷ *Ibid.*, 5.

²⁸ *Ibid.*, 4–5.

²⁹ *Ibid.*, 3.

³⁰ *Ibid.*, 4.

replacing or commanding it, under the premise that the “private sector has the agility necessary to keep pace with rapidly evolving threats.”³¹

Prioritized equities. Examining the U.S. approach to national-level cybersecurity policies reveals that among the many equities to be balanced, three stand out. First, protecting (national) security in cyberspace and beyond features prominently in U.S. policy. This includes an emphasis on the protection of critical infrastructure from malicious cyberoperations as well as the policy prerogative to disrupt and dismantle threat actors ranging from states to nonstate actors.³² Due to critical infrastructure services being predominantly owned or operated by private-sector stakeholders in the U.S. market, the U.S. approach pays particular attention to securing businesses.

Second, advancing economic prosperity and the United States’ leading role in innovation and technology, particularly AI, are major concerns.³³ While the market is seen as the most effective propellant of innovation (as well as a preferred tool of regulation itself), the U.S. approach seeks to intervene in order to maintain U.S. dominance in innovation and its accompanying economic benefits. To that end, policies include strategic investments in research and development, technology development, education, and workforce development.

Third, building on its engagement internationally, the U.S. approach emphasizes the need to maintain and defend international norms and standards that reflect U.S. views and priorities.³⁴

Digital Rights Approach: The European Union

As a consistent proponent of cybersecurity and opponent of the information security approach in international-level policy negotiations, the EU has emerged as a prolific regulator setting overarching national-level policies for its member states that have at times served as blueprints for regulation in other parts of the world.³⁵ Following several iterations, a key overarching strategy was published in 2020 as the EU’s Cybersecurity Strategy for the Digital Decade.³⁶ Perhaps even more impactfully, the EU has adopted several high-profile regulations and policies governing cybersecurity and broader technology issues.³⁷ A few examples include the revised NIS2 Directive and the Directive on the Resilience of Critical Entities (CER).³⁸

Perceived threats: Targets and threat actors. The EU’s 2020 Cybersecurity Strategy for the Digital Decade firmly places citizens at the center of its analysis of threats in cyberspace. The document begins by stating the following: “Cybersecurity is an integral part of Europeans’ security. Whether it is connected devices, electricity grids, or banks, aircraft, public administrations, or hospitals

³¹ White House, *President Trump’s Cyber Strategy for America*, 5.

³² *Ibid.*, 4.

³³ *Ibid.*, 6.

³⁴ *Ibid.*, 4.

³⁵ See Anu Bradford, *The Brussels Effect: How the European Union Rules the World* (New York: Oxford University Press, 2020).

³⁶ European Commission and the High Representative of the Union for Foreign Affairs and Security Policy, “Joint Communication to the European Parliament and the Council: The EU’s Cybersecurity Strategy for the Digital Decade,” December 16, 2020, <https://digital-strategy.ec.europa.eu/en/library/eus-cybersecurity-strategy-digital-decade-0>.

³⁷ For an excellent overview of EU policies and regulations, see Christina Rupp, “Navigating the EU Cybersecurity Policy Ecosystem: A Comprehensive Overview of Legislation, Policies and Actors,” *Interface*, June 27, 2024, available at <https://www.stiftung-nv.de/publications/navigating-the-eu-cybersecurity-policy-ecosystem#policy-area-1-overarching-policies>.

³⁸ “Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on Measures for a High Common Level of Cybersecurity across the Union, Amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and Repealing Directive (EU) 2016/1148 (NIS 2 Directive),” *Official Journal of the European Union*, December 27, 2022, available at <https://eur-lex.europa.eu/eli/dir/2022/2555/oj/eng>; and “Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the Resilience of Critical Entities and Repealing Council Directive 2008/114/EC,” *Official Journal of the European Union*, December 27, 2022, available at <https://eur-lex.europa.eu/eli/dir/2022/2557/oj>.

they use or frequent, people deserve to do so within the assurance that they will be shielded from cyber threats.”³⁹ Consequently, the EU’s approach emphasizes that cybersecurity is essential for citizens to trust, use, and benefit from the digital revolution as well as for safeguarding fundamental human rights, including the rights to privacy and to the protection of personal data and the freedom of expression and information.⁴⁰

In light of this understanding, the EU—like the United States—highlights malicious attacks on critical infrastructure as a major global threat. Digital services, the finance sector, manufacturing, and the public sector are frequently targeted while cyber readiness and awareness among both businesses and individuals remain low.⁴¹ Closely related, cybercrime and its economic cost are mentioned as a threat in cyberspace.⁴²

Further, the EU strategy posits that these threats in cyberspace are exacerbated by rising geopolitical tensions over the global and open internet and over the control of technologies across the supply chain.⁴³ Deepening digital restrictions are undermining the global and open nature of cyberspace as well as the rule of law, human rights, freedom, and democracy. Since these constitute core values of the EU, these developments are considered as serious threats to its interests in cyberspace.

Responsible stakeholders. Broadly speaking, the EU’s approach to cybersecurity governance seeks to utilize its regulatory powers to enhance resilience by creating and elevating common standards and norms. While it relies on investments and policy initiatives to influence market dynamics, this approach is marked by proactive regulatory action that sets minimum standards across the EU—a markedly more assertive stance than that adopted under the U.S. approach.

The review of the original NIS Directive, mandated in the 2020 Cybersecurity Strategy and completed and entered into force in 2023 as the NIS2 Directive, perhaps best illustrates this approach.⁴⁴ The revised directive on measures for a high common level of cybersecurity across the EU establishes a comprehensive set of rules for public and private entities, with a focus on protecting critical infrastructure.⁴⁵ These rules impose cyber risk management requirements and incident reporting and information-sharing obligations in a range of sectors, along with national supervision and enforcement requirements.⁴⁶

Beyond government regulation, however, an active role for nongovernmental stakeholders is acknowledged and explicitly encouraged, as “governments, businesses and citizens all share responsibility in [a] cyber-secure digital transformation.”⁴⁷

Prioritized equities. In its approach to cybersecurity governance, the EU places a firm emphasis on the defense of its core values in cyberspace: rule of law, human rights, and democracy. EU citizens and their fundamental rights are placed at the center of cybersecurity efforts. As the European Commission lays out, the 2020 Cybersecurity Strategy “aims to ensure a global and

³⁹ European Commission and the High Representative of the Union for Foreign Affairs and Security Policy, “The EU’s Cybersecurity Strategy for the Digital Decade,” 1.

⁴⁰ *Ibid.*, 4.

⁴¹ *Ibid.*, 3.

⁴² *Ibid.*

⁴³ *Ibid.*, 1–2.

⁴⁴ “Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022.”

⁴⁵ *Ibid.*

⁴⁶ *Ibid.*

⁴⁷ European Commission, “Cybersecurity Strategy,” <https://digital-strategy.ec.europa.eu/en/policies/cybersecurity-strategy>.

open Internet with strong safeguards where there are risks to security and the fundamental rights of people in Europe.”⁴⁸ To that end, the EU strives to cooperate with others to defend and maintain the like-minded vision of a global, open, stable, and secure cyberspace.⁴⁹

Other equities balanced in the EU approach to cybersecurity reflect its authorities and competencies. With the single market as the focal point and source of EU authority, the EU strategy focuses on economic aspects and business opportunities. It openly states that “the upcoming decade is the EU’s opportunity to lead in the development of secure technologies across the whole supply chain,” which would include cybersecurity by design for industrial processes, operations, and devices.⁵⁰ As part of this emphasis on economic equities, the EU strives to lead in defining and promoting norms and standards.⁵¹ In addition, with its limited competence in security and defense matters, the EU approach focuses on supporting member states in their capacity to defend their citizens, businesses, and institutions from cyberthreats by requiring them to adopt minimum standards and build capacity for preventing, detecting, and responding to threats.⁵²

Overall, the EU builds on its status as a strong economic and trading bloc to emphasize economic aspects alongside its core values of the rule of law, human rights, and democracy. Coupled with its perspective on individuals, these equities enable the EU to see itself as a leading voice in setting governance nationally and internationally.

*Information Isolationism Approach: Russia*⁵³

Russia has been a leading and vocal proponent of the information security approach at the international level since the late 1990s, and its national-level policies have evolved especially over the past decade to fully reflect that view as well. At present, Russia’s national-level cybersecurity policy is characterized by an overarching effort to decouple from international information infrastructure, replace foreign technology providers with domestic suppliers, and address a fundamental mistrust in unregulated cross-border information flows.⁵⁴ This approach is characterized by efforts to isolate and control the national information space.⁵⁵

Russia’s “Doctrine of Information Security” of 2016, supported by the National Security Concept of 2021, forms the main national strategy document.⁵⁶ These materials are supplemented by the *Strategy for the Development of the Information Society in the Russian Federation for 2017–2030* and a document titled “Main Directions of the State Policy in the Field of Automated Control

⁴⁸ European Commission, “Cybersecurity Strategy.”

⁴⁹ European Commission and the High Representative of the Union for Foreign Affairs and Security Policy, “The EU’s Cybersecurity Strategy for the Digital Decade,” 19.

⁵⁰ *Ibid.*, 5.

⁵¹ *Ibid.*, 19.

⁵² *Ibid.*, 13–19.

⁵³ For further discussion of the term “information isolationism,” see Sergey Sukhankin, “Russia Adopts New Strategy for Development of Information Society,” Jamestown Foundation, Eurasia Daily Monitor, May 16, 2017, <https://jamestown.org/program/russia-adopts-new-strategy-development-information-society>.

⁵⁴ For a recent analysis, see Justin Sherman, “Russia’s Digital Tech Isolationism: Domestic Innovation, Digital Fragmentation, and the Kremlin’s Push to Replace Western Digital Technology,” Atlantic Council, Issue Brief, July 2024, https://dfirlab.org/wp-content/uploads/sites/3/2024/07/AC_CSI_Russias_Digital_Tech_Isolationism.pdf.

⁵⁵ Sukhankin, “Russia Adopts New Strategy for Development of Information Society.”

⁵⁶ President of the Russian Federation, “Decree of the President of the Russian Federation dated 5 December 2016 No. 646 Approval of the Doctrine of Information Security of the Russian Federation,” available at <http://static.kremlin.ru/media/acts/files/0001201612060002.pdf>; and President of the Russian Federation, *National Security Concept of the Russian Federation* (Moscow, July 2021), <http://www.scrf.gov.ru/media/files/file/l4wGRPqJvETSkUTYmhepzRochb1j1jqh.pdf>.

Systems of Production and Technological Processes of Critical Infrastructure” from 2012.⁵⁷ All these policy documents are promulgated and approved at the highest levels of government by the president, the Security Council, or both.

Perceived threats: Targets and threat actors. Official documents mention a number of threats and risks in cyberspace that are common concerns for states. These include computer crimes, the use of ICT tools by terrorist and extremist organizations, the use of ICT tools by states’ militaries, and the increased complexity and scope of attacks on critical information infrastructure.⁵⁸ Beyond these, perceived threats also include the use of “transboundary information circulation” and “information and psychological tools” to destabilize the political and social situation in Russia as well as in various regions around the world.⁵⁹ Emphasis is placed on information as a threat to social and political stability.

Lastly, shortages in the cybersecurity workforce and a lack of competitive domestic information technologies (including hardware and software products) are cited as major concerns for Russia’s national approach to information security policy.⁶⁰ The resulting dependence on foreign information technologies is seen as an impediment to the socioeconomic development of the country as competitor states can leverage their dominant tech sectors in the pursuit of geopolitical interests. Further, dominance of foreign technology affects the global management of the internet, which is currently seen as neither fair nor trust-based.⁶¹

Responsible stakeholders. To address the threats underpinning the Russian approach to cybersecurity policy, sole responsibility lies with government actors, in particular the central government at the highest levels. Information security is seen as a part of the national security system of Russia, and governance is directed by state bodies.⁶² This includes activities to ensure information security as well as activities to develop and improve information security.⁶³

The government’s primacy to regulate information space or cyberspace comes with a plethora of state bodies whose composition is determined by the president of Russia. This central, top-down governance approach leaves a marginal role for nongovernmental stakeholders. Critical infrastructure providers, information technology providers and operators, educational organizations, citizens, and others are termed “participants” in the information security system who “participate in solving problems of ensuring information security” in accordance with the national laws of Russia—and not primarily as partners in the development of cyber resilience.⁶⁴

Prioritized equities. Russia’s approach prioritizes the protection of its sovereignty as a major strategic goal in ensuring information security and emphasizes the need to create new international legal treaties to regulate information technologies and prevent interstate conflict in cyberspace.⁶⁵ This goes hand in hand with an emphasis on maintaining political and social

⁵⁷ President of the Russian Federation, *Strategy for the Development of the Information Society in the Russian Federation for 2017–2030* (Moscow, May 2017), <http://static.kremlin.ru/media/acts/files/0001201705100002.pdf>; and President of the Russian Federation, “Main Directions of the State Policy in the Field of Automated Control Systems of Production and Technological Processes of Critical Infrastructure,” February 3, 2012, <http://www.scrf.gov.ru/security/information/document113>.

⁵⁸ President of the Russian Federation, “Doctrine of Information Security,” 6–7.

⁵⁹ *Ibid.*, 5–6.

⁶⁰ *Ibid.*, 7.

⁶¹ *Ibid.*, 7–8.

⁶² *Ibid.*, 12–13.

⁶³ *Ibid.*, 14–15.

⁶⁴ *Ibid.*, 13–14.

⁶⁵ *Ibid.*, 12.

stability and creating “a safe environment for the circulation of reliable information.”⁶⁶ National defense and national security figure prominently as well, with concerns for “the use of information technologies for military-political purposes...as well as for terrorist, extremist, criminal and other illegal purposes.”⁶⁷

Economic concerns are driven by an ambition to achieve greater independence and autonomy. Russia’s information technology and electronics industry are to be developed with the proclaimed goal to decrease Russia’s dependence on foreign technology vendors.⁶⁸ This reflects Russia’s relatively small international market share in information and communications technologies across the supply chain.

Advanced Information Isolationism: The People’s Republic of China

Following China’s rapid economic and technological development, cybersecurity became a priority, expressing the country’s ambition in 2014 to become a “cyber power.”⁶⁹ This also resulted in an ever-expanding regulatory regime with China’s first National Cybersecurity Strategy in 2016, an international cyber cooperation strategy in 2017, and a national cybersecurity law in 2016.⁷⁰ While Russia relies on “information security” as its preferred terminology, the Chinese usage has changed over time from “information security” to “network security.” The Chinese term for “network security,” however, is often translated as “cybersecurity,”⁷¹ eliding the nuance of shifting Chinese perspectives.

China’s approach shares core elements with Russia’s information isolationism but builds on it in important ways. The Chinese approach equally seeks to isolate and control the national information space and places paramount importance on the regulation of cross-border information flows. But, in a key difference to Russia, China has been able to decouple from international information infrastructure and decrease its dependence on foreign technology providers much more successfully and at scale. In stark contrast with Russia, major Chinese technology companies, ranging from ICT infrastructure providers to social media companies, have emerged. While some of these companies and their products are focused on the Chinese market, others have become global competitors in information and telecommunication technology. The success of China’s information ecosystem has spurred efforts to export it internationally, promoting Chinese technology products and services through the Digital Silk Road Initiative.⁷²

Perceived threats. Targets and threat actors. Chinese perceptions of threats and risks in cyberspace encompass concerns ranging from political stability and national defense to the culture, society, and economy of China. These include concerns over attacks on critical infrastructure that

⁶⁶ President of the Russian Federation, “Doctrine of Information Security,” 4, 9.

⁶⁷ *Ibid.*, 12.

⁶⁸ *Ibid.*, 11.

⁶⁹ Rogier Creemers, “The Chinese Conception of Cybersecurity: A Conceptual, Institutional, and Regulatory Genealogy,” *Journal of Contemporary China* 33, no. 146 (2024): 173.

⁷⁰ Cyberspace Administration of the People’s Republic of China (PRC), *National Cyberspace Security Strategy* (Beijing, December 2016), available in an unofficial translation at <https://chinacopyrightandmedia.wordpress.com/2016/12/27/national-cyberspace-security-strategy>; Ministry of Foreign Affairs and the Cyberspace Administration of China (PRC), *International Strategy of Cooperation on Cyberspace* (Beijing, March 2017), available in an unofficial translation at <https://chinacopyrightandmedia.wordpress.com/2017/03/01/international-strategy-of-cooperation-on-cyberspace>; and National People’s Congress (PRC), “Cybersecurity Law of the People’s Republic of China,” November 6, 2016, available in an unofficial translation at <https://www.newamerica.org/cybersecurity-initiative/digichina/blog/translation-cybersecurity-law-peoples-republic-china>.

⁷¹ Creemers, “Chinese Conception of Cybersecurity,” 175.

⁷² Charles Mok, “How China Governs Its Internet and Plans to Export Its Model,” Tech Policy Press, July 24, 2025, <https://www.techpolicy.press/how-china-governs-its-internet-and-plans-to-export-its-model>.

could impair critical sectors such as energy, transportation, and finance and that could ultimately affect China's national economic security. Further, online terrorism, separatism, extremism, and crime, as defined by Chinese authorities, are cited in the National Cybersecurity Strategy.⁷³

Emphasis is also placed on the negative effects of the information revolution on political and cultural security. In line with China's concerns in the UN discussions over the flow of information and its potentially destabilizing effects, the strategy specifically calls out "harmful information violating the Socialist core value view" that is "corroding the physical and mental health of minors, undermining the social atmosphere, misleading value orientations and endangering cultural security."⁷⁴ China's approach directly ties cybersecurity to the stability of the political regime. Cyber activities harm political security, which is the "basic precondition for national development and the happiness of the people." As a result, threats in cyberspace include the use of ICT to "interfere in the internal political affairs of other countries, to attack other countries' political systems, incite social unrest, subvert other countries' regimes, as well as large-scale cyber surveillance, cyber espionage and other such activities."⁷⁵

Lastly, China's strategy is concerned with an increasing international competition in cyberspace, which results in competition to set international norms, control strategic resources in cyberspace, and win an arms race in cyberspace.⁷⁶

Responsible stakeholders. China's approach to national-level cybersecurity governance is characterized by earlier and more impactful regulatory efforts, compared with Russia, and by the comprehensive nature of extending government control to cyberspace⁷⁷—a feat Russia has not attempted to fully replicate. As China's national strategy and accompanying regulations indicate, the state is the main actor in national cybersecurity regulation. The Cybersecurity Law, for example, starts with a list of general provisions that are all centered on government authority. Among others, it provides that "the state formulates and continuously improves cybersecurity strategy, clarifies the fundamental requirements and primary goals of cybersecurity, and puts forward cybersecurity policies, work tasks, and procedures for key sectors."⁷⁸

While nongovernmental stakeholders, in particular the private sector, play a role in cybersecurity policy development and implementation,⁷⁹ China's national-level approach relies on the primacy of government regulation rather than on market forces, reflecting a degree of similarity with the digital rights approach. China's advanced information isolationism approach is complemented by the creation of a comprehensive bureaucracy that further promulgates and implements China's cybersecurity policies at various levels of government, from central to local authorities. These policies reflect the priority role of the central government in areas ranging from combating cybercrime and protecting infrastructure to managing information flows, often giving government authorities extensive access to information and systems.⁸⁰

⁷³ Cyberspace Administration (PRC), *National Cyberspace Security Strategy*, section I (2).

⁷⁴ Ibid.

⁷⁵ Ibid.

⁷⁶ Ibid.

⁷⁷ Creemers, "Chinese Conception of Cybersecurity," 175.

⁷⁸ Cyberspace Administration (PRC), *National Cyberspace Security Strategy*, chap. I, article (4), trans. Rogier Creemers, Paul Triolo, and Graham Webster, New America, <https://www.newamerica.org/cybersecurity-initiative/digichina/blog/translation-cybersecurity-law-peoples-republic-china>.

⁷⁹ Creemers, "Chinese Conception of Cybersecurity," 173.

⁸⁰ "China's Digital Data Sovereignty Laws and Regulations," InCountry, August 20, 2024, <https://incountry.com/blog/chinas-digital-data-sovereignty-laws-and-regulations>.

Prioritized equities. As the National Cybersecurity Strategy explicitly stated, China strives to “safeguard the interests of national sovereignty, security and development, and [to] realize the strategic objective of building a strong cyber power.”⁸¹ An examination of equities prioritized in the Chinese national strategy confirms these objectives.

Major emphasis is placed on state sovereignty and its full extension into cyberspace. As the strategy plainly states, “no infringement of sovereignty in cyberspace will be tolerated...and the people of all countries are to decide on cyber affairs within the scope of sovereignty.”⁸² This emphasis on sovereignty goes hand in hand with a focus on order and political stability. Again, the strategy plainly states that sovereignty includes the opposition to “all actions to subvert [China’s] national regime or destroy [China’s] sovereignty through the network.”⁸³ Accordingly, sovereignty empowers the state to “prevent, curb and punish the online dissemination of harmful information endangering national security and interests, and to safeguard order in cyberspace.”⁸⁴ The goal is a network environment where the “free flow of information is organically unified with safeguarding national security and the public interest.”⁸⁵ The “great wall” of China as well as superapps like Tencent’s WeChat illustrate the extent to which China has been able to create and control a separate, Chinese information ecosystem, in stark contrast with Russian efforts. Economic aspects are also highlighted. Continuous “informatization” is taken as a prerequisite for modernization, and China strives to persist in its current economic position with regard to technology, continuing its leaps in innovation-driven development. This applies to software security, cybersecure products, and network infrastructure, as well as content.⁸⁶ Coupled with China’s broader technology and innovation policies, this prioritization goes further than Russia’s focus on simply decreasing dependence on foreign technology providers; it creates domestic Chinese providers that can compete and dominate globally.

In addition, security concerns, including the protection of critical infrastructure, are paramount. Cybersecurity is essential for national security, and thus China seeks to effectively control cybersecurity risks.⁸⁷ Notably, national security concerns include the need to safeguard regime stability.⁸⁸

Lastly, the Chinese approach to cybersecurity governance highlights the need to change international internet governance mechanisms to become “multilateral, democratic and transparent.”⁸⁹ As a result, China seeks to participate in regional and international processes to change the status quo of internet and cyberspace governance.

⁸¹ Cyberspace Administration (PRC), *National Cyberspace Security Strategy*, section II.

⁸² *Ibid.*, section III (1).

⁸³ *Ibid.*, section IV (1).

⁸⁴ *Ibid.*, section III (1).

⁸⁵ *Ibid.*, section II.

⁸⁶ *Ibid.*, section IV (6).

⁸⁷ *Ibid.*, sections II and III. (4).

⁸⁸ *Ibid.*, section IV (1).

⁸⁹ *Ibid.*, section II.

Conclusion

As states grapple with the security challenges of the rapidly increasing digitization of the world, divergent approaches to cybersecurity policies have become apparent at both the national and international levels. Every country's approach to cybersecurity governance will be unique, shaped by particular historical, cultural, legal, economic, and political contexts. Nevertheless, both internationally as well as nationally, emerging approaches or archetypes can be identified.

At the international level, underlying conceptual differences between cybersecurity and information security translate to corresponding differences in international approaches. Clusters of states have firmly converged around these two major concepts, which have come to form the spectrum on which international discussions have progressed. On one side, states led by the United States and EU member states emphasize the role and responsibilities of nongovernmental stakeholders in cybersecurity governance and seek to preserve the vision of a free, open, and secure cyberspace. On the other side, information security proponents assert governmental control over cyberspace and cybersecurity governance, with little to no room for nongovernmental actors. Concomitantly, information security proponents, led by Russia and joined by China, push for changes to the current regulatory regime, including to internet governance and the regulation of state conduct. While these two clusters have solidified, the majority of states have remained flexible in their international approach, embracing different elements of both approaches at different times.

For national-level policies, the preceding examination of countries' strategy documents has revealed four divergent approaches to cybersecurity governance. First, the U.S. approach can be characterized as market-driven, with the government relying principally on market forces or market self-regulation. Government interventions are still geared toward shaping market forces with a minimalist footprint rather than replacing them. Second, the EU approach emphasizes individual citizens and their digital rights in cyberspace, alongside the EU's core values of the rule of law, human rights, and democracy. Through this digital rights approach, the EU has emerged as an engaged regulator setting standards and norms for cybersecurity that have even reverberated beyond its borders. Third, Russia's approach of information isolationism couples a maximalist view of the government's responsibility for cybersecurity and its prerogative to regulate with a concern over the regulation and isolation of cross-border information flows. Whereas Russia is still heavily reliant on foreign technology providers in this endeavor, China, with its competitive private sector, is attempting to pursue a fourth approach, namely advanced information isolationism.

As individual states approach cybersecurity governance on national and international levels, interesting similarities and key differences emerge both between and within the camps of cybersecurity versus information security. First, Russia's and China's national-level strategies perceive the current international regime as problematic and its continuation as a threat to their interests in cyberspace. Conversely, both the U.S. and EU approaches emphasize defending the current international order, and advances by other states to change it are perceived as fundamental threats. These positions neatly map onto states' diplomatic positions in international discussions, where they are mirrored in the efforts of Russia and China to advance a new international treaty, and in the efforts of like-minded states to defend existing norms of state behavior.

Second, the potential implications of the information revolution for the political stability of states are a key concern uniting the national-level approaches taken by Russia and China. As a result, both approaches emphasize the extension and exercise of national sovereignty over

cyberspace. This view maps onto the positions held in UN discussions. Whereas Russia and China assert the state's prerogative for cybersecurity policies and governance, leaving little to no room for nongovernmental stakeholders, the EU and U.S. approaches reveal an alternative model for the role of governments and nongovernmental stakeholders. The EU approach, in particular, places individual citizens, rather than the political regime, at the center of its efforts to secure cyberspace. Although the EU and its national member states assume the main responsibility for setting cybersecurity regulations (and the EU advances its notion of digital sovereignty), this digital rights approach is decidedly geared toward the protection of individuals and their rights in cyberspace. In a key difference among proponents of cybersecurity, the United States has traditionally emphasized the role and responsibilities of the market in providing cyber resilience. For both the United States and the EU, however, the state plays a facilitative role focused on safeguarding private behavior.

Third, despite their differences, all four national approaches to cybersecurity are concerned with protecting and enhancing the resilience of critical infrastructure and preventing the economic loss and national security harm associated with any disruption of critical services.

Fourth, while economic aspects loom large in all four national approaches to cybersecurity governance, the specific approaches are firmly embedded in each country's individual circumstances. As a major economic and trading bloc, the EU seeks to extend its role into the digital market. The United States has been at the leading edge of innovation with a globally dominant tech sector and seeks to maintain this position. China has emerged as the main competitor to the United States for technology market share and global influence over digitalization and is striving to elevate its position. Russia also seeks to compete economically in cybersecurity, though it is doing so from a position of relative weakness with regard to its private sector—a situation it sees as imperiling its national security in cyberspace.

In summary, important cross-cutting differences and similarities can be identified when mapping and comparing divergent approaches internationally and nationally. These findings can be expanded, supplemented, and supplanted by future research examining and comparing a larger set of countries.

Cyber Resilience Maturity in the Indo-Pacific: Impacts on International Partnerships

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EXECUTIVE SUMMARY

This chapter examines how differences in cyber resilience maturity among Pacific Island nations shape their receptivity to international engagement and evaluates what this means for the design of effective cybersecurity assistance in the region.

MAIN ARGUMENT

Cyber resilience development in the Pacific Islands follows a dynamic, nonlinear trajectory in which a nation's willingness to engage internationally evolves alongside its domestic cybersecurity maturity. A four-category maturity model ranging from minimal to advanced cyber resilience demonstrates that Pacific Island governments adopt distinct collaboration strategies depending on their institutional capacity, resource constraints, and strategic priorities. States with minimal or emerging capabilities tend to engage internationally to secure targeted assistance for foundational capacity building, while more mature states increasingly seek collaboration to extend existing capabilities or project regional influence. The analysis further shows that nations oscillate between comprehensive and targeted engagement approaches as their maturity shifts rather than following a linear path. These dynamics have direct implications for cyber diplomacy: mismatches between donor expectations and recipient maturity levels risk reducing the effectiveness and sustainability of assistance. A calibrated approach, grounded in an understanding of maturity-driven behavior, enables development partners to better anticipate partner responses, align investments with domestic priorities, and support durable cyber resilience outcomes across the region.

POLICY IMPLICATIONS

- *Align assistance with maturity levels.* Regional cyber capacity-building initiatives should be explicitly tailored to a partner nation's cyber resilience maturity, prioritizing foundational capabilities for less mature states and capability extension for more advanced partners.
- *Sequence engagement strategies.* Policy engagement should recognize that receptivity to norms, treaties, and multilateral frameworks evolves as domestic cyber resilience matures rather than treating these as entry-point requirements.
- *Leverage trusted regional actors.* Sustained partnerships with regional epistemic communities and multistakeholder institutions can amplify donor assistance by embedding capacity building within trusted local networks.
- *Integrate cyber diplomacy and development policy.* Cybersecurity assistance should be coordinated with broader U.S. development and security objectives in the Pacific to ensure coherence, legitimacy, and long-term resilience beyond the period of direct support.

When examining how governments and their stakeholders improve their capabilities for cyber resilience, it is important to explore different approaches adopted by nations and other stakeholders in formulating and implementing their cybersecurity policies. This includes examining contrasting approaches to international collaboration, engagement, and cyber resilience capacity building.

The first chapter of this report examined diverging approaches to cybersecurity policy and governance at both the national and international levels, focusing on the four main archetypes of the United States, the European Union, Russia, and China. Chapter 2 explores how the readiness of nations to collaborate and engage in bilateral and multilateral initiatives relating to implementing cyber resilience evolves alongside their domestic cyber resilience maturity. Through a series of case studies on nations in the Pacific Islands region, this chapter illustrates how countries with differing levels of cyber maturity—at both the cybersecurity policy and cyber resilience levels—may choose to engage with international partners to uplift their domestic capabilities. The focus is on identifying both commonalities and disparities among approaches to cyber resilience development, opening up two lines of inquiries:

- What is the relationship between a country's level of cyber maturity and its receptivity to capacity-building assistance and other forms of international collaboration and engagement?
- How does a country's approach to engagement and collaboration on cybersecurity policy matters affect these relationships?

To address these questions, the chapter uses an analytical framework to consider cyber resilience maturity among nations within the Pacific Islands region that respond to different risk conditions and face varying economic and other capacity constraints. The first section discusses the concept of cyber resilience from the perspective of nations with differing levels of relative maturity. The second section then explores opportunities for developing cyber resilience capabilities and considers how nations with different levels of relative maturity choose to engage and collaborate. After outlining this framework and the characteristic behaviors of Pacific Island nations, the third section explores how these nations' approaches to engaging with partners on cyber resilience implementation evolve over time, influenced by their capability maturity. Additionally, it examines how their willingness to engage and collaborate with neighbors and partners affects this approach.

This analysis highlights significant impacts for international diplomacy and cybersecurity development assistance across the Pacific Islands region. The outcomes provide the basis for assessing in the fourth section how governments with different levels of comparative cyber resilience capability maturity are likely to engage with each other on security and development matters. By understanding the nature of these relationships, policymakers can target development assistance and investments in specific areas of mutual interest and better anticipate how the recipient national governments are expected to respond. Such understanding is crucial because it shifts the conversation away from how development partners determine the basis for cybersecurity capability assistance toward how the dynamic needs of the recipient determine the basis for targeted policy and investment interventions. By focusing on these needs, this chapter highlights the importance of aligning development assistance with the specific cybersecurity maturity levels and strategic priorities of the recipient nation. In doing so, development partners ensure that investments are not only more effective but also more relevant to the unique contexts of

the recipient nations, leading to a more balanced and sustainable development of cybersecurity capabilities that ultimately enhances the resilience of the recipient after assistance concludes.

Cyber Resilience Maturity

As a Pacific Island nation's cyber resilience maturity evolves relative to its neighbors and establishes increasingly mature practices and processes, its willingness to receive capability development assistance and to collaborate with partners will change at different rates.¹ Using a combination of qualitative measures of a nation's cyber resilience maturity (see **Table 1**)—specifically, Budapest Convention status, national cyber strategy status, domestic cyber legislation status, completion of a Cybersecurity Capacity Maturity Model for Nations (CMM) review, and establishment of a national cybersecurity emergency response team (CERT)—I have proposed the following four general categories of nations with increasing levels of cyber resilience maturity to explain this correlation:²

- *Category One.* Nations with *minimal* cyber resilience maturity show a willingness to base regional or global collaboration on the receipt of targeted support to develop foundational cyber resilience capabilities.
- *Category Two.* Nations with *emerging* cyber resilience maturity show a willingness to collaborate, based on a desire to rapidly expand their foundational capabilities.
- *Category Three.* Nations with *established* cyber resilience maturity show a willingness to collaborate based on a desire to extend their existing capabilities.
- *Category Four.* Nations with *advanced* cyber resilience maturity show a significant increase in willingness to collaborate, with a focus on extending their regional and global influence.

The implication of this correlation is that by understanding where an organization is on its cyber resilience maturity journey, policy engagement and support can be targeted toward expected behaviors and domestic priorities. The following discussion demonstrates these categories through brief examples of Pacific Island nations.

Category One: Minimal Cyber Resilience Maturity

Category One nations demonstrate minimal cyber resilience capabilities compared with their neighbors and may seek collaboration based on receiving targeted cyber capability development support. Examples of Category One nations include the Federated States of Micronesia (FSM), Tuvalu, and Nauru. Each faces unique challenges that hinder its efforts to develop capabilities without external support. Within the Pacific Islands Forum, the Smaller Island States (SIS) forum acknowledges the entrenched development challenges of FSM, Tuvalu, and Nauru, attributed to their small size, lack of natural resources, and geographic remoteness. These factors collectively limit their capacity to achieve sustainable cyber resilience outcomes.

¹ For further discussion of this argument, see Anthony M. Adams, "Cybersecurity Threat Detection and Response: Building a Pacific Islands CERT Framework" (PhD thesis, Monash University, 2024).

² Ibid.

TABLE 1 Four categories of cyber resilience maturity (December 2025)

	FSM, Nauru, and Tuvalu	Samoa	Fiji and Kiribati	Australia
Measure of maturity	Category One: Minimal	Category Two: Emerging	Category Three: Established	Category Four: Advanced
Collaboration approach	Targeted	Broad	Targeted	Broad
CMM review	Yes/No ^a	Complete	Complete ^b	No ^c
National CERT	No	Agreement to prepare	Operational	Operational
National cybersecurity strategy	No	No ^d	Complete ^e	Complete
Budapest Convention status	No	No ^f	Invited/partner	Partner
Domestic cybercrime legislation	No/Foundation	Foundation/ uplifted legislation	Uplifted legislation	Multilayered suite of legislation

NOTE:

- a:* Category One nations have either not completed a CMM review or have done so without the review progressing to policy development. In these case studies, all three countries have completed CMM reviews (FSM 2020, Tuvalu 2021, Nauru 2022/2023).
- b:* Fiji has completed its second CMM review, conducted by the Oceania Cyber Security Centre (OCSC) and funded by the UK and Australian governments.
- c:* Australia has not completed a CMM review but has assumed a partner-style role and has provided targeted support to help Categories One, Two, and Three nations complete reviews. For example, it provided support to the OCSC to complete a second CMM review for Fiji in alignment with the Fiji-Australia Memorandum of Understanding on Cyber Security Matters under the Vuvale Partnership (February 2024).
- d:* Samoa's National Cybersecurity Strategy (2016–21) has not been renewed.
- e:* Fiji's national strategy is currently being developed with support from the Australian government under the Vuvale Partnership MOU.
- f:* Samoa has made substantial progress in developing cybersecurity capabilities but has also prioritized resources toward other policy areas in recent years; these include responses to natural disasters and the decision to host the 2024 Commonwealth Heads of Government Meeting. This means that during 2024 the nation was reclassified from Category Three to Category Two.

Moreover, the World Health Organization classifies Tuvalu as a least-developed nation and FSM and Nauru as developing nations.³ These nations are characterized by small populations, remote locations, high telecommunications and transportation costs, and inadequate infrastructure. A significant portion of their population engages in subsistence agriculture, while an increasing

³ World Health Organization, "Country Cooperation Strategy at a Glance," May 2013, https://apps.who.int/iris/bitstream/handle/10665/136831/ccsbrief_pci_en.pdf.

number migrate to other nations for employment, with remittances now constituting a notable percentage of their GDP.

While the Oceania Cyber Security Centre (OCSC) has completed national cybersecurity capacity reviews at the invitation of the respective national governments, the findings have not been used to inform national cyber resilience strategies or related domestic legislation. The FSM government published a national cybersecurity strategy over a decade ago and appointed the Department of Transportation, Communication, and Infrastructure to lead its implementation.⁴ However, the strategy has not been updated in the twelve years since publication. FSM does not currently have domestic legislation in place that implements either substantive or procedural powers related to cybercrime and electronic evidence. Neither the Crimes Act (2014) nor the Criminal Procedure Act (2014) references cybercrime offenses. The Copyright, Patents, and Trademarks (Title 35) provides limited coverage for cyber-enabled crimes and general search and seizure powers, but the extension to electronic evidence is unclear.

Nauru does not have a published national cybersecurity strategy, although the Department of ICT (information and communications technology) is responsible for developing and implementing a strategy, together with supporting government policies and a cybersecurity roadmap. In addition, Nauru is currently reviewing its foundational national legislation pertaining to cybercrime, with key provisions related to the Budapest Convention in the Cybercrime Act (2015), Communications and Broadcasting Act (2018), and Police Force (Amendment) Act (2015).

Tuvalu does not have a national cyber resilience strategy or supporting domestic legislation but has drafted a cybercrime bill (approval remains pending).⁵ The national government administers cybersecurity initiatives through the Ministry of Justice, Communication, and Foreign Affairs. In addition, it does not have any recognized national (and sector-specific) cybersecurity frameworks for implementing internationally recognized cybersecurity standards. The government also does not have dedicated cybercrime legislation in place, with some offenses incorporated in different legislation, primarily the Tuvalu Telecommunications Corporation Act, which is not by its nature an act of substantive criminal law per se.⁶ Seizure of electronic evidence has limited coverage in the Criminal Procedure Code, while there are no legislative provisions to criminalize online child abuse.

The lack of progress in either publishing (Nauru and Tuvalu) or updating (FSM) a national cybersecurity strategy, as well as in uplifting domestic legislation (Tuvalu and FSM), suggests a potential reluctance by these nations to allocate limited resources away from addressing long-standing development issues toward enhancing cyber resilience capabilities. This reluctance could persist without targeted collaboration and assistance from development partners and neighboring nations.

Category Two: Emerging Cyber Resilience Maturity

Category Two nations display emerging cyber resilience capabilities in comparison with their neighbors and demonstrate a general readiness to collaborate to swiftly enhance their foundational capabilities. Samoa provides an example of a Category Two nation that targets the development of

⁴ Department of Transportation, Communications and Infrastructure (Federated States of Micronesia), “FSM National ICT and Telecommunications Policy 2012,” September 2012, <https://fsm-data.sprep.org/dataset/fsm-national-ict-and-telecommunications-policy-2012>.

⁵ For more information, see “Tuvalu,” Council of Europe, Octopus Cybercrime Community, <https://www.coe.int/en/web/octopus/-/tuvalu>.

⁶ The Tuvalu Telecommunications Corporation Act is available at <https://www.wipo.int/edocs/lexdocs/laws/en/tv/tv010en.pdf>.

emerging cyber resilience capabilities through collaboration with global cybersecurity partners, while remaining subject to ongoing development constraints.

The government of Samoa adopted the National Cybersecurity Strategy (2016–21).⁷ Its goals include strengthening existing cyber systems and protecting critical infrastructure. The strategy underpinned both completion of the OCSC CMM review in April 2018 and establishment of the Samoa national CERT in June 2019.⁸ However, it was not updated on expiration and is no longer being used to guide government investment priorities. The failure to replace the National Cybersecurity Strategy may have resulted from several external environmental factors that have led to changes in investment priorities, including national responses to the country's 2023 measles outbreak and the Covid-19 pandemic and the hosting of the Commonwealth Heads of Government Meeting in October 2024. Additionally, the government commenced working with the Council of Europe in 2018 to review and align its domestic legislation on cybercrime and electronic evidence with the provisions of the Budapest Convention, with a focus on uplifting the current Crimes Act (2013) rather than enacting new legislation.

Category Three: Established Cyber Resilience Maturity

Category Three nations exhibit established cyber resilience capabilities, with collaboration focused on maintaining and extending those capabilities. Samoa and Kiribati provide examples of Category Three nations, demonstrating established cyber resilience maturity and engaging in collaboration to extend their current capabilities.

As a leading economic partner in the Pacific Islands region, Fiji boasts a comparatively mature cybersecurity architecture, based on the National Security Strategy (2016), the 5-Year and 20-Year National Development Plan (2017), its membership in the Pacific Islands Law Officers Network (PILON) Cybercrime Working Group, its participation in the UN Open-ended Working Group on Security of and in the Use of Information and Communications Technologies 2021–2025, and its status as a party to the Budapest Convention. However, neither the National Cybersecurity Strategy nor the 5-Year and 20-Year National Development Plan include specific national cybersecurity initiatives for the subsequent five-year investment horizon. Progress in this regard will be driven by the signing of the Fiji-Australia Memorandum of Understanding (MOU) on Cyber Security Matters under the Vuale Partnership in February 2024, which will facilitate an updated national cybersecurity strategy, including the operationalization of a Fiji CERT.

In its efforts to expand its regional engagement, the government partnered with the OCSC in 2015 to complete one of the world's first cybersecurity capability assessments using the CMM, which was developed by the Global Cyber Security Capacity Centre. The outcomes of this CMM review informed the country's National Cybersecurity Strategy. In coordination with Australia, the United Kingdom funded the OCSC to undertake a second CMM review with Fiji in alignment with the new MOU, which informed the updated national cybersecurity strategy, making Fiji the only Pacific Island nation to complete two reviews.

Kiribati has entrenched development challenges. The Pacific Islands Forum recognizes its Smaller Island States status and the World Health Organization classifies it as a least-developed nation, with characteristic constraints similar to FSM, Nauru, and Tuvalu. In the case of Kiribati,

⁷ Ministry of Communication and Information Technology (Samoa), *Samoa National Cybersecurity Strategy 2016–2021* (Apia, 2016), <http://www.samoagovt.ws/wp-content/uploads/2017/02/MCIT-Samoa-National-Cybersecurity-Strategy-2016-2021.pdf>.

⁸ For more information about Samoa's national CERT, see <https://www.samcert.gov.ws/about-us>.

however, the national government has prioritized the development of its emerging cyber resilience capabilities through active engagement and collaboration with cybersecurity partners, including the PILON, the OCSC, the Council of Europe, and the Australian government.

In its National Cybersecurity Strategy (2020), Kiribati maps a path to establish an independent national cyber resilience posture within the Pacific Islands region.⁹ The strategy calls for using Kiribati's scarce cybersecurity resources to build a foundation on which citizens and the private sector can develop their own cyber resilience capabilities, with a focus on building critical infrastructure and strengthening existing public- and private-sector partnerships. As a key implementation action arising from the National Cybersecurity Strategy, the Digital Government Act 2023 provides for the establishment of a national CERT, to be overseen by the Digital Transformation Office within the Ministry of Information, Communication and Transport.

In its efforts to build on its emerging cyber resilience capabilities, Kiribati, working with the OCSC, conducted a national cybersecurity capacity review in July 2019. The government then used the findings of the review to inform the development of the National Cybersecurity Strategy. Additionally, the government signed the Nurturing the Partnership MOU with Australia in February 2023. Under the MOU, Australia agreed to provide a range of commitments to upgrade cybersecurity and ICT infrastructure across Kiribati.

Category Four: Advanced Cyber Resilience Maturity

Category Four nations exhibit relatively advanced cyber resilience capabilities coupled with a significant increase in willingness to collaborate in order to extend their sphere of influence. Australia is an example of a Category Four nation. The country possesses comparatively mature cyber resilience capabilities and has demonstrated a strong willingness to engage deeply across the Pacific Islands region, including by projecting its values and priorities as preconditions for delivering development aid to and collaborating with regional countries.

The Australian government's strategic cyber resilience roadmap is documented in the 2023–2030 Australian Cyber Security Strategy and outlines the country's strategic objective of protecting its national policy interests through developing interconnected layers of security across the Australian cyberthreat landscape.¹⁰ The government will use the strategy to project Australia as a leading partner within the region through two initiatives. First, Australia will promote enduring cyber resilience and technology security across the Pacific Islands region, with development initiatives prioritizing social and cultural values such as gender equality and social inclusion. Second, the government will build a multiagency regional cyber crisis response team. This team will use Australian industry and government expertise to provide targeted support across the region.

Comprehensive and Targeted Approaches to Cyber Resilience Maturity

When considering a nation's relative cyber resilience maturity and its willingness to engage in cyber resilience with international partners, it is important to also look at the nation's approach

⁹ Ministry of Information, Communication, Transport and Tourism Development (Kiribati), *Kiribati National Cybersecurity Strategy 2020* (Tarawa, 2020), https://www.mict.gov.ki/sites/default/files/publications/National%20Cybersecurity%20Policy_0.pdf.

¹⁰ Australian Government, *2023–2030 Australian Cyber Security Strategy* (Canberra, 2023), <https://www.homeaffairs.gov.au/cyber-security-subsite/files/2023-cyber-security-strategy.pdf>.

to engaging with its cybersecurity partners and stakeholders. There are two important drivers of expected behavior.¹¹ First, nations can be expected to exhibit different approaches to cyber resilience engagement based on whether they adopt a characteristically comprehensive or targeted approach. Second, nations can be expected to move between these approaches to engagement as their relative cyber resilience maturity changes. In general, there is no consistent or deterministic relationship between a country's approach to cyber resilience engagement (comprehensive or targeted) and the cybersecurity policy models outlined in the previous chapter. Nations adopting either approach may, in principle, engage with Category Four partners advocating any of the four models discussed in chapter 1.

However, an important qualification is relevant for policymakers. Countries that adopt a targeted approach to engagement tend to prioritize the rapid development or consolidation of core cyber resilience capabilities. In such cases, cybersecurity policy models that emphasize information control or security as an end in itself are likely to be less attractive, particularly where recipient governments perceive limited direct linkage between those models and tangible resilience outcomes. As a result, alignment between engagement approach and policy model becomes more consequential where resources are constrained and capability development is the dominant objective.

Category One and Three nations typically adopt a targeted approach to cyber resilience engagement. They prioritize the development and maintenance of domestic cyber resilience capabilities over regional engagement and collaboration. These nations are less inclined to identify and collaborate with development partners and other stakeholders on shared interests. Instead, they prioritize cyber resilience policies that promote domestic self-reliance ahead of engaging with international behavioral norms, including standards, treaties, and conventions.

On the other hand, Category Two and Four nations generally adopt a more comprehensive approach to cyber resilience engagement. They demonstrate a willingness to engage with regional and development partners and are more likely to collaborate on shared interests. These nations prioritize cyber resilience policies that promote domestic self-reliance while aligning with international standards, treaties, and conventions.

As a nation transitions between comprehensive and targeted approaches, we can expect to see changes in the degree to which it prioritizes its cyber resilience policy objectives over regional engagement and collaboration. These transitions correspond to movements between Categories One to Four, reflecting the evolution of a country's cyber resilience maturity and the effects of government decision-making in response to external environmental factors.

Cyber Resilience Maturity and Receptivity to International Engagement

Forms of International Engagement

A notable development over recent years across a range of countries is the growing recognition of the contributions made by nongovernment participants, including international organizations, academia, NGOs, and epistemic communities of practice, in supporting the development of cyber resilience capabilities through international engagement and collaboration. Notably, this aligns with chapter 1's findings that countries that adopt a cybersecurity approach—and are therefore

¹¹ The following discussion draws on Adams, "Cybersecurity Threat Detection and Response."

generally more focused on core cyber resilience outcomes rather than on information controls—are more willing to adopt strategies and policies that facilitate multistakeholder participation. Understanding how nations with different levels of relative cyber resilience maturity engage with these stakeholders requires examining different opportunities for international engagement and collaboration.

Capacity Building

Effective cybersecurity capacity building that is able to sustain its impact on cyber resilience outcomes over an extended period of time requires collaboration and coordination among multiple state and nonstate stakeholders. When considering the disparate economic and geographic differences across the Pacific Islands region, we observe that capacity building proves most successful when stakeholders offer sustained, long-term local presence. This presence sits on a foundation of trust relationships with the national government, local community, and global cybersecurity practitioner community. Prominent capability-building partners, including the Asia Pacific Network Information Centre (APNIC) and OCSC, use their established local presence and relationships to facilitate ongoing engagement between state and nonstate stakeholders from across the region and globally.

Operating across the wider Asia-Pacific region, APNIC serves as an epistemic community where cybersecurity practitioners, particularly those from low-income Pacific Island nations, can develop specific technical skills and engage with other members to share knowledge and experiences. In practice, APNIC actively participates in Pacific Island regional forums and conferences, delivering keynote sessions, training sessions, collaborative workshops, and individual consultations to engage with local practitioners and facilitate knowledge exchange.

The OCSC provides in-country, multi-stakeholder assessments of cyber resilience readiness in Pacific Island nations that request them, utilizing the internationally recognized CMM. Although the CMM review process is not legally binding, it allows participating governments to benchmark their cyber resilience capabilities relative to globally recognized nonstate norms and highlights areas for further investment and policy prioritization.

Assistance with Drafting Cybercrime Legislation

International lawmaking collaboration is generally focused on empowering nations to implement the provisions of the Budapest Convention, established by the Council of Europe in 2001 and supplemented by two additional protocols (CETS 189 and CETS 224). The Budapest Convention is the first international treaty on crimes committed via the internet and other computer networks, dealing particularly with infringements of copyright, computer-related fraud, child pornography, and violations of network security. Open for accession by any country, the convention requires that a country seeking accession to have implemented or be planning to implement domestic laws that enact its provisions.

To consider the effects of the Budapest Convention on different Pacific Island nations' cyber resilience outcomes, we need to explore their approaches to collaborating on the domestic lawmaking initiatives that enact the convention's provisions. Resources exist to help these nations develop the domestic legislative frameworks necessary for accession. PILON, for example, brings together senior law officers from Pacific Island nations within the PILON Cybercrime Working Group, which is an epistemic community that helps develop regionally supported law and justice

policy initiatives, with a special interest in targeting initiatives to support member nations preparing for accession to the Budapest Convention. Significantly, nations that are parties to the Budapest Convention or that have been invited to accede may become priority recipients of cyber capacity-building programs provided by the Council of Europe. Nations that have been invited to accession have five years to implement the necessary domestic legislation prior to their accession being ratified by all current parties to the convention. At the time of writing, Australia, Tonga, Kiribati, New Zealand, Vanuatu, and Fiji are the only Pacific Island nations that have become parties to the convention.

Information Exchange

Information is generally exchanged between cyber resilience practitioners through a combination of state-driven, top-down frameworks that act as guardrails for collaboration and nonstate-driven bottom-up information sharing among epistemic communities of practice that operate within these guardrails.

Top-down frameworks can take the form of either multilateral or bilateral agreements, with recent examples from the Pacific Islands showcasing both approaches. The Pacific Islands 2050 Strategy for the Blue Pacific Continent, the 2023 Lagatoi Declaration on Digital Transformation of the Pacific, and the 2018 Boe Declaration on Regional Security serve as multilateral frameworks wherein member nations have adopted strategic objectives aimed at bolstering regional cyber resilience, including safeguarding national critical infrastructure. These documents represent a consensus-driven approach among regional leaders to create frameworks for nations to work on matters of shared regional interest while respecting the different levels of cyber resilience policy maturity, capability, and overall prioritization.

Bilateral agreements within the region tend to be initiated by Australia (Category Four, comprehensive approach) as the dominant partner in the region and provide frameworks for exchanging information and collaborating with smaller nations. For example, the Fiji-Australia Vuvale Partnership establishes a foundation for information sharing and collaboration regarding specific, targeted cyber resilience initiatives with Fiji (Category Two). Similarly, the Nurturing the Partnership MOU facilitates collaboration and information sharing between Australia and Kiribati (Category Two). Australia also provides support for the Pacific Cyber Security Operational Network, which seeks to advance regional cyber resilience through information sharing, collaboration, and promotion of shared technical cyber capabilities between member nations.

Epistemic communities of practice consist of nonstate actors united by shared technical expertise and best practices that frequently transcend national borders. These communities rely on personal contacts and relationships between participants, fostering bottom-up information sharing and collaboration. Through sustained, trust-based interactions, these communities enable collective advances in cybersecurity capabilities and practices, which are particularly valuable for nations with varying levels of capacity and resources. PILON is an example of an epistemic community that promotes bottom-up transnational collaboration and information sharing. It provides a network of senior law officers from nineteen Pacific Island nations, including Australia and New Zealand, who collaborate on advancing key regional law and justice issues and developing regionally supported policies for consideration by lawmakers in the respective national governments.

Participation in the Development of Cyber Norms

When considering how nations participate in the development of cyber behavioral norms, a distinction needs to be made between top-down (state-driven, institutional) and bottom-up (nonstate-driven, organic) interactions among international, domestic, government, commercial, academic, NGO, and other actors. Multilateral frameworks such as the UN Group of Governmental Experts (UNGGE) can foster behavioral norms that filter down through domestic communities, while norms may also develop organically at the domestic level (through vendor specifications, intra-team cooperation, precedent practices) and then diffuse upward through the international system.¹²

The UNGGE framework of global cybersecurity norms (i.e., A/65/201, A/68/98, and A/70/174) is outlined in the 2010, 2013, 2015, and 2021 UNGGE reports.¹³ The norms are not legally enforceable and instead assign specific nonlegal duties and obligations for state cybersecurity actors. Significantly, UNGGE norm no. 7 calls on states to protect their critical infrastructure from cyberthreats: “States should take appropriate measures to protect their critical infrastructure from ICT threats, taking into account General Assembly resolution 58/199 on the creation of a global culture of cybersecurity and the protection of critical information infrastructures, and other relevant resolutions.”

A country wishing to uphold the UNGGE norms, and in particular UNGGE norm no. 7, must act in accordance with them. Some nations might already meet the standards set by the norms, and might even have played a leadership role in driving their development and endorsement as bottom-up change agents. Others, however, might need to revise or adapt their top-down national cyber-related initiatives to meet those standards. To assess the top-down impact of the UNGGE norms on cyber resilience outcomes, it is important to examine how different nations have approached the task of revising or adapting their incumbent national cyber-related initiatives. By examining a mix of high- and low-income Pacific Island nations, we can observe the effect of these policies on the implementation of UNGGE norm no. 7.

For Pacific Island nation governments that have issued national cybersecurity strategies (typically those in Category Two, Three, or Four), the strategies shape their relationships with the nongovernmental sector, including market participants, academic institutions, development partners, and agencies. For those nations without national strategies (Category One), governments tend to rely on development aid to either establish cybersecurity strategies that leverage shared regional outcomes or align with regional practices, such as the Pacific Islands 2050 Strategy for the Blue Pacific Continent, the 2018 Boe Declaration on Regional Security, and the 2023 Lagatoui Declaration on Digital Transformation of the Pacific.

Pacific Island nations adopt similar approaches to the role of the international private sector in funding national critical infrastructure.¹⁴ Whereas low-income nations explicitly acknowledge their relative cyber resilience disadvantage and seek international private-sector investment and

¹² Nazli Choucri, Stuart E. Madnick, and Jeremy Ferwerda, “Institutions for Cyber Security: International Responses and Global Imperatives,” *Information Technology for Development* 20, no. 2 (2014): 96–121; Ilona Stadnick, “What Is an International Cybersecurity Regime and How We Can Achieve It?” *Masaryk University Journal of Law and Technology* 11, no. 1 (2017): 129–54; and UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, Report, A/76/135, July 14, 2021, https://front.un-arm.org/wp-content/uploads/2021/08/A_76_135-2104030E-1.pdf.

¹³ UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, Report, A/70/174, July 22, 2015, <https://digitallibrary.un.org/record/799853?ln=en>; and UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, Report, A/76/135.

¹⁴ For further discussion, see Adams, “Cybersecurity Threat Detection and Response.”

development assistance to reduce their comparative disadvantage, high-income nations leverage their relative cyber resilience advantage and invite domestic and international private-sector participation as a means of sharing sovereign financial or borrowing risk.

Patterns of International Engagement

Category One: Minimal Cyber Resilience Maturity

Category One nations (minimal cyber resilience maturity, targeted collaborative approach) are likely to engage with the regional and global cybersecurity community, based on receiving targeted support to initiate collaboration on cyber resilience capability development. The focus for these nations is on overcoming environmental, domestic capability, and economic development challenges that inhibit their ability to devote significant resources in developing cyber resilience capabilities. FSM, Nauru, and Tuvalu again provide useful examples of how Category One nations can be expected to engage and collaborate internationally.

While Pacific Island nations have agreed by consensus to endorse regional declarations that commit to shared undertakings on cyber resilience matters, including the 2050 Strategy for the Blue Pacific Continent, the 2023 Lagatoi Declaration on Digital Transformation of the Pacific, and the 2018 Boe Declaration on Regional Security, these declarations are not legally binding on member nations. In this regard, implementation for Category One Pacific Island nations remains patchy, as might be expected, given their tendency to focus on foundational capacity building for cyber resilience.

Within the Pacific Islands region, Category One nations can be expected to prioritize investment in existential national threats, such as the impacts of climate change, ahead of developing domestic cyber resilience capabilities.¹⁵ In this case, they might choose to partner with nations with higher cyber resilience maturity and a broader approach to collaboration (Category Four) to leverage additional investment and will likely remain focused on core cyber resilience capabilities.¹⁶ In response, development support for Category One nations is likely to target establishing initial cyber resilience foundations, particularly through adoption of a national cybersecurity strategy and a basic domestic legislative framework for cybersecurity resilience.

Category Two: Emerging Cyber Resilience Maturity

Category Two nations (emerging cyber resilience maturity, comprehensive collaborative approach) can be expected to engage with the regional and global cybersecurity community based on receiving targeted support to nurture emerging cyber resilience capabilities. The focus for these nations is on establishing and reinforcing their emerging cyber resilience capabilities, typically while being hampered by environmental constraints and a lack of institutional capacity and skilled local resources.

Within the Pacific Islands region, Category Two nations can be expected to continue the Category One focus on investing in existential national threat responses. At the same time, they will also seek to cooperate with global and regional partners to expand their foundational domestic cyber resilience capabilities.

¹⁵ Adams, "Cybersecurity Threat Detection and Response."

¹⁶ Joe Burton, "Small States and Cyber Security: The Case of New Zealand," *Political Science* 65, no. 2 (2013): 216–38.

For example, the government of Samoa has prioritized resources toward responses to natural disasters (2022 earthquake and 2023 measles outbreak), with cybersecurity capability maturity uplift focusing on strengthening existing cyber systems and protecting critical infrastructure. In particular, the government has completed a CMM review in 2018 and established the Samoa national CERT in 2019. In addition, the government commenced working with the Council of Europe in 2018 to review and align its domestic legislation on cybercrime and electronic evidence with the provisions of the Budapest Convention, with a focus on uplifting the current Crimes Act (2013) rather than enacting new legislation. Development support for Category Two nations is likely to help reinforce the initial cyber resilience foundations, with a focus on nurturing self-reliance. As these countries seek to develop more self-reliant cyber resilience ecosystems, it is important that they engage with partners whose cyber models are aligned with these objectives so as to ensure that their scarce resources are targeted effectively.

Category Three: Established Cyber Resilience Maturity

Category Three nations (established cyber resilience maturity, targeted collaborative approach) are likely to engage with their regional and global cybersecurity community in order to seek support to expand their existing cyber resilience capabilities ahead of developing new capabilities or legislation. These nations are likely to seek collaboration on areas of shared interest that align with their existing, established cyber resilience capabilities, reflecting the importance of engagement at the Category Two stage.

The Kiribati government, in recognizing its structural limitations, has used its National Cybersecurity Strategy (2020) to declare its intention to expand bilateral and multilateral engagements in the region and globally to foster development of its national ICT and cyber resilience priorities. This includes engaging other regional nations and partners to help develop foundational domestic capabilities, including by leveraging regional epistemic communities such as the PILON, OCSC, and APNIC. Kiribati also needs to pursue opportunities for assistance from the Council of Europe through its Global Action on Cybercrime Extended project.

For example, Kiribati commenced working with the Council of Europe and the Australian government in 2019 to reform the Communication Act (2012), which covered a range of general offenses related to misuse of telecommunications systems, services, and related matters in preparation for the country's invitation to accede to the Budapest Convention. This reform process culminated in the adoption of the Cybercrime Act in 2021, which was later amended to include substantive articles on offenses and penalties in Part III (Article 720) and procedural provisions for procedural law in Part IV (Article 2330). Following this reform process, Kiribati became a party to the Budapest Convention in July 2024.

Fiji leverages bilateral agreements with neighboring nations to develop additional cyber resilience capabilities. The Vuale Partnership that Fiji signed with Australia in 2019 provides a framework for cooperation to increase Fiji's cyber resilience capability maturity. The two countries subsequently signed a five-year MOU in April 2024 to support the establishment of a national CERT in Fiji and the completion of the second CMM review. In addition, Fiji participates in the PILON Cybercrime Working Group and commenced working with the Council of Europe in 2019 to reform its domestic framework on cybercrime and electronic evidence. As a result of these efforts, the 2021 Cybercrime Act partially went into force in November 2022. Subsequently, Fiji became a party to the Budapest Convention in July 2024.

Development support for Category Three nations can be expected to acknowledge the prior establishment of cyber resilience capabilities for these countries to become more self-reliant. Additional targeted support is aimed at encouraging these nations to further strengthen and expand their cyber resilience capabilities.

Category Four: Advanced Cyber Resilience Maturity

Category Four nations (advanced cyber resilience maturity, comprehensive collaborative approach) can be expected to actively engage with their cybersecurity community in order to project their strategic policy interests beyond the region, guided by their overall cybersecurity model, as discussed in chapter 1.

Australia, for example, is a party to the Budapest Convention and has adopted domestic cyber resilience legislation grounded through three significant bills. First, the Security of Critical Infrastructure Act 2018 addresses national security risks of sabotage, espionage, and coercion posed by foreign involvement in Australia's critical infrastructure. Second, the Telecommunication and Other Legislation Amendment Act 2017, known as the telecommunication sector security reforms, amends the Telecommunications Act 1997 to establish a regulatory framework to better manage the national security threats of espionage, sabotage, and foreign interference to Australia's telecommunications networks and facilities. Finally, the Trusted Information Sharing Network for Critical Infrastructure Resilience was established by the Australian government in 2003. It is Australia's primary national engagement mechanism for business and government information-sharing and resilience-building initiatives on critical infrastructure resilience.

Through its 2023–30 Australian Cyber Security Strategy, Australia has a clear and unambiguous goal of projecting its international cyber policy priorities across the global cybersecurity landscape by 2030. For example, Shield 6, Action 19, of the strategy calls for the government to link the provision of economic and other development assistance for neighboring nations to whether their cyber policy initiatives are consistent with Australia's.

Development support from Category Four nations is likely to facilitate the delivery of in-kind aid to neighboring nations so that the Category Four nation is better able to project its priority values and outcomes. For example, the Fiji-Australia Vuvale Partnership provides a framework within which Fiji can receive cyber resilience support from Australian agencies and partners through the Cyber and Critical Technology Cooperation Program.

Conclusion

This chapter provides two key insights. First, it illustrates that among Pacific Island nations the relationship between national security and economic development is dynamic, evolving continuously as their cybersecurity capabilities mature. Second, it highlights that as governments' cybersecurity capabilities evolve, so too does the nature of their regional engagement and collaboration on matters of shared interest. Moreover, the priorities of their development partners, as well as those partners' ability to support shared priorities and shape beneficial outcomes, evolve commensurately as well.

The chapter outlines a positive, nonlinear relationship between a Pacific Island government's cybersecurity capability maturity and the nature of its approach to regional engagement and collaboration. This relationship is dynamic in nature, enabling nations to refine their strategies

for engagement with partners and neighboring states in response to changes in cybersecurity capability maturity.

This finding holds substantial implications for cyber diplomacy, especially for effectively directing diplomatic efforts and support for cyber capability development in the region. As highlighted above, Category One countries are likely naturally aligned with the cybersecurity approach outlined in chapter 1, given their pronounced resource constraints. As countries graduate into Category Two, continued support from like-minded partners is critical to ensuring that the country's own capabilities are directed toward core cyber resilience outcomes. Recognizing these relational dynamics enables policymakers to strategically direct development assistance and investments, focusing on areas that match the recipient nation's specific level of cybersecurity maturity and strategic priorities. By tailoring support to a country's needs, policymakers can ensure that investments are not only effective but also resonate with each nation's unique developmental context, leading to more balanced and sustainable development of cybersecurity capabilities above and beyond the initial provided support.

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Explaining Divergence: Stakeholder Views

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EXECUTIVE SUMMARY

This chapter discusses security, economic, and geopolitical factors motivating countries to adopt different cybersecurity policies with more or less open approaches to the global internet.

MAIN ARGUMENT

Where policymakers choose to adopt policies that limit global online openness, it is often because of a specific concern about the risks posed by particular international actors—for instance, tech firms based in foreign countries—rather than a belief that doing so will broadly improve the resilience of their cyber infrastructure in the face of most cyberattacks, outages, and other disruptions. This helps explain why countries' positions on global openness and the supporting policies sometimes shift as a result of changing cyberthreat models and perceptions of whether outside (and particularly international) online influences will be positive or negative from a national security standpoint as well as an economic one.

POLICY IMPLICATIONS

- Attempts to change policymakers' approaches to cybersecurity policy depend on understanding the specific threats and risks those policymakers most wish to address.
- Information isolationist approaches to cybersecurity policy are increasingly attractive to a wide range of countries as geopolitical tensions rise internationally.
- Information isolationist approaches to cybersecurity policy often involve trade-offs that make users more susceptible to certain types of risks, even as they may protect those users' governments from other threats.

As more countries have developed and implemented cybersecurity policies and regulations, it has become clear that there are a variety of different principles and priorities underlying their approaches. This chapter analyzes the competing priorities at play in the development of cybersecurity policies in order to better understand how and why these approaches have diverged from each other and identify different factors that may motivate governments to implement more or less “information isolationist” approaches, as discussed in chapter 1. More information isolationist approaches, in which countries attempt to lock down the internet within their borders and carefully control what outside information and services can be accessed, are sometimes referred to as “closed” internet policies, while the reverse type of policy, which encourages the free flow of global data and services, is sometimes referred to as policy that facilitates a more “open” internet. This chapter considers the underlying motivations for these different approaches to cybersecurity policymaking, examining whether each country’s respective policies are implemented to achieve cyber risk resilience or whether in some cases other goals and priorities are pursued under the guise of cybersecurity. For the purposes of this analysis, policies that promote more open online environments and generally align with the market-driven approach are those that encourage and facilitate the free flow of data and online expression across international borders; enable internet users to access foreign services and information, as well as store their data anywhere they wish; and enable companies to develop new online platforms and services that can operate across global markets.

Importantly, the notion of “openness” is used to refer to many other characteristics in internet design and policy, including ideas of open access, open source software, interoperability, and transparency. As one article notes in its analysis of how the idea of openness is applied to the internet:

“Openness” evokes various connotations in socio-technical contexts, including but not limited to interoperability between computer systems in a network as well as the freedom of users to access, modify, and (re)distribute source code, expressive works, or datasets. It also suggests a participatory and interdependent mode of (co)production; sharing tangible and intangible resources; greater civic engagement as well as greater accountability of the duty-bearers; alternative governance models; a socio-cultural movement against enclosure and monopoly; an organisational paradigm characterised by dissolving boundaries or reducing barriers to facilitate innovation, amongst others.¹

Many of these connotations are associated with specific policy efforts as well—for instance, debates about net neutrality and multi-stakeholder internet governance. Attempting to trace all the different policy initiatives that relate to these different meanings of openness is no small task. This chapter considers only one very narrow notion of openness: how open a country’s internet is to online services, websites, and data from other countries, or how globally open it is.

Many countries face competing priorities when enacting data protection and cybersecurity policies that lead them to implement sometimes seemingly contradictory policies relating to global openness. Most fundamentally, many countries would like their own domestic internet companies to be able to operate internationally because of the economic growth benefits. Yet they are sometimes wary of letting foreign online services operate within their borders for security reasons, as well as to enable the emergence of domestic service providers. For instance, the United

¹ Tyng-Ruey Chuang et al., “Openness,” *Internet Policy Review* 11, no. 1 (2022): 1–9, <https://policyreview.info/glossary/openness>.

States and China have long wrestled with the question of which software and hardware products manufactured by companies in the other country each government is willing to allow within its borders, even as U.S. and Chinese tech companies have pushed aggressively to spread their services worldwide.

In addition to encompassing multiple, competing priorities, countries' approaches and appetites for maintaining a globally open internet change over time. For instance, in October 2023 the United States withdrew its support at the World Trade Organization for the digital trade negotiation objective of protecting cross-border data flows—a goal it had previously championed.² By way of explanation for this reversal, the Office of the U.S. Trade Representative stated obliquely: “Many countries, including the United States, are examining their approaches to data and source code, and the impact of trade rules in these areas. In order to provide enough policy space for those debates to unfold, the United States has removed its support for proposals that might prejudice or hinder those domestic policy considerations.”³

This chapter considers two different types of policymaking that have a tendency to inhibit the global openness of the internet: bans on foreign online content and services and requirements for data localization. These two categories of policies have direct implications for the global openness of the internet: the blocking of foreign content and services makes it harder for users to access the full range of online information, and data localization inhibits the ease with which organizations can move data around the world. In sum, both types of measures diminish the global openness of the internet—even if there may be good reasons that the governments implementing such policies would wish to do so. Indeed, the primary purpose of this chapter is to understand what those reasons are, how deeply they are rooted in genuine concerns about cybersecurity and cyber resilience, and what other factors are involved in the conception, passage, and implementation of such policies. Each category has been researched and written about in depth in many other places. The objective of this chapter is not to summarize or repeat that extensive body of work, but rather to consider the security motivations behind these decisions and, more specifically, to examine the extent to which policymakers believe that these approaches will achieve cyber risk resilience.

The chapter argues that while blocking foreign online content and services and requiring data localization are not always viewed as increasing resilience against general categories of cyber risks, such as cybercrime, many governments do view them as helping increase resilience against very specific, high-probability risks, such as cyberespionage campaigns conducted by specific adversaries. Governments' willingness to embrace these policies, therefore, often depends on calculations for how likely certain cyber risks are to occur. Where policymakers choose to adopt policies that limit global openness, it is often because of a specific concern about the risks posed by particular international actors—for instance, tech firms based in foreign countries—rather than a belief that doing so will broadly improve the resilience of their cyber infrastructure in the face of most cyberattacks, outages, and other disruptions. This argument helps explain why countries' positions on global openness and the supporting policies sometimes shift as a result of changing cyberthreat models and perceptions of whether outside (and particularly international) online influences will be positive or negative from a national security standpoint as well as an economic one.

² Meredith Broadbent, “USTR Upends U.S. Negotiating Position on Cross-Border Data Flows,” Center for Strategic and International Studies (CSIS), December 12, 2023, <https://www.csis.org/analysis/ustr-upends-us-negotiating-position-cross-border-data-flows>.

³ “USTR Statement on WTO E-Commerce Negotiations,” Office of the U.S. Trade Representative, Press Release, October 24, 2023, <https://ustr.gov/about-us/policy-offices/press-office/press-releases/2023/october/ustr-statement-wto-e-commerce-negotiations>.

Bans on Foreign Content and Services

Countries have been filtering and blocking online content for several decades, and programmers and advocates have been developing tools and techniques to circumvent those restrictions for almost as long. China, as a steadfast proponent of the information isolationism approach described in chapter 1, is widely regarded as having highly sophisticated technical infrastructure for filtering online content and blocking online services like Facebook that are based overseas. Yet many other countries have also implemented policies in recent years to block content and services they deem to be undesirable. For instance, beginning in June 2020, the Indian Ministry of Electronics and Information Technology decided to block the Chinese-owned video service TikTok along with dozens of other Chinese apps following a confrontation between the Indian and Chinese militaries. While several other governments have banned TikTok from being used on government-issued devices, the Indian ban was unusual in that it also applied to general, nongovernment internet users. India cast its decision as being motivated primarily by cybersecurity concerns relating to a particular international actor: China. The Indian government accused the Chinese apps it had blocked of “stealing and surreptitiously transmitting users’ data in an unauthorised manner to servers outside India.”⁴ These types of decisions to block outside services run by a foreign company for fear of its ties to its domestic government may, in fact, provide security benefits to countries even as they limit the global openness of the internet by imposing regional restrictions on what types of data and services can operate online.

Whether these bans improve nations’ cyber risk resilience is a more complicated question. Decreasing the diversity of online services available to users is not an obvious means of improving resilience because greater diversity generally provides a wider range of options if a service were to malfunction or go offline. Thus, the crucial question is the following: resilience against which types of cyber risk? Potentially, a ban of a foreign service motivated by fears of cyberespionage or data harvesting could improve resilience to that particular threat from a specific country, even if it has minimal—or negative—impacts on the nation’s ability to recover from or withstand other types of cyberthreats (for instance, outages of similar services).

Not all bans on foreign services and websites are motivated by concerns about espionage and data harvesting, however. For instance, in 2023 the government of Brazil temporarily banned the Dubai-based encrypted messaging service Telegram across the entire country due to fears that neo-Nazi groups were using the platform to incite violent attacks on schools.⁵ Previously, the Brazilian government imposed similar measures on WhatsApp, owned by Facebook’s parent company Meta.⁶ Such blocks are not unique to Brazil. Russia also attempted to block Telegram, though it largely failed in its implementation.⁷ In many cases where the targets of bans on foreign online services are end-to-end encrypted messaging services, the motivation for these policies is law enforcement’s inability to monitor communications on the platform or collect information from the service providers rather than a fear of foreign interception or cyberespionage. For instance,

⁴ Hannah Ellis-Petersen, “India Bans TikTok after Himalayan Border Clash with Chinese Troops,” *Guardian*, June 29, 2020, <https://www.theguardian.com/world/2020/jun/29/india-bans-tiktok-after-himalayan-border-clash-with-chinese-troops>.

⁵ André Spigariol, “Brazilian Judge Bans Telegram App amid School Attacks Inquiry,” *New York Times*, April 26, 2023, <https://www.nytimes.com/2023/04/26/briefing/brazil-telegram-ban.html>.

⁶ Vinod Sreeharsha, “WhatsApp Blocked in Brazil as Judge Seeks Data,” *New York Times*, May 2, 2016, <https://www.nytimes.com/2016/05/03/technology/judge-seeking-data-shuts-down-whatsapp-in-brazil.html>.

⁷ Neil MacFarquhar, “Russia Tried to Shut Down Telegram. Websites Were Collateral Damage,” *New York Times*, April 18, 2018, <https://www.nytimes.com/2018/04/18/world/europe/russia-telegram-shutdown.html>.

the Brazilian ban on Telegram followed the company's refusal to turn over data about neo-Nazi activity on the platform that CEO Pavel Durov claimed the company did not have access to.⁸

In these cases, cybersecurity is again a central motivator for governments' decisions to block foreign-run services, but the cybersecurity concerns motivating these decisions are primarily centered on whether law enforcement and intelligence agencies will be able to investigate domestic cyberthreats, crimes, and conspiracies. By contrast with bans that are intended to block foreign cyberespionage efforts, bans on encrypted messaging services have the potential to actually expose more users to espionage and other interception risks by limiting the availability of strong encryption protections that can limit espionage efforts. In these cases, as well, policies that limit the global openness of the internet by blocking foreign-run encrypted messaging services can offer governments certain forms of resilience against activities facilitated by these apps, while still leaving them susceptible to other threats that these apps simultaneously help protect against. This tension between different security goals speaks to the importance of balancing different cyber risks and determining which one to prioritize in cybersecurity policymaking. It also helps explain why governments might alter some of their policies in seemingly contradictory ways as their ideas about the relative likelihood and severity of different cyberthreats change.

Another major category of bans of foreign online services covers those motivated by a government's desire to prevent a domestic audience from being exposed to particular types of content or information. In 2023, for instance, Somalia banned both TikTok and Telegram, citing concerns about disinformation and the dissemination of "horrific images."⁹ In cases like this one, where a ban is motivated by a desire to control or eliminate specific types of political or sensitive content, cyber risk resilience is typically not a major factor driving governments away from global openness of their internet infrastructure. Instead, fears about domestic stability and political unrest often motivate these efforts to restrict online content. And while such measures might offer a degree of protection or resilience against certain types of political unrest or instability, they offer little protection against cyberthreats and are often undermined by tools intended to help users circumvent such blocks, including virtual private networks (VPNs) and proxy servers. These circumvention tools and services are themselves sometimes blocked in further pursuit of information control, but even then users are still often able to find alternative workarounds. In early 2024, for instance, Russia attempted to ban VPNs but struggled to implement and enforce the ban.¹⁰

One final set of foreign online service bans worth considering are those motivated by economic concerns surrounding issues of competition and development. While these motivations are distinct from the domestic and international security fears that underlie many of the previously discussed policies, economic and security concerns are often difficult to completely disentangle in such bans. For instance, in 2023 Indonesia blocked TikTok Shop, which is TikTok's e-commerce platform, when the Ministry of Trade announced new rules that required separation of e-commerce and social media platforms in order to "protect local merchants, prevent algorithms from dominating

⁸ "Telegram CEO Claims Complying with Brazil Order Impossible," Associated Press, April 27, 2023, <https://apnews.com/article/telegram-brazil-block-d0af428c25c40c11658e44f31e8116bf>.

⁹ "Somalia Bans TikTok, Telegram and 1XBet over 'Horrific' Content, Misinformation," Reuters, August 21, 2023, <https://www.reuters.com/world/africa/somalia-bans-tiktok-telegram-1xbet-over-horrific-content-misinformation-2023-08-21>.

¹⁰ Patrick Tucker, "VPN Ban Is Kremlin's Latest Effort to Quash Dissent," Defense One, March 1, 2024, <https://www.defenseone.com/technology/2024/03/vpn-ban-kremlins-latest-effort-quash-dissent/394635>.

the market and stop the use of personal data for business purposes.”¹¹ Indonesia’s list of reasons for blocking services like TikTok Shop hint at a range of different factors contributing to the government’s decision: a desire to support local businesses, a concern about unfair competition from algorithm-driven retailers, and a privacy concern about how social media data might be used and repurposed for e-commerce.

This speaks to the complexity of the factors underlying online bans and the challenges of trying to draw out all the different motivations a government might have for implementing them. Even if policymakers truly do believe there will be cybersecurity and cyber risk resilience benefits to a certain policy, they may also want to restrict access to certain types of information or promote domestic industry stakeholders above foreign competitors. For instance, even outside of Somalia, which explicitly referenced violent content as a motivating factor for its ban of TikTok, debates over blocking TikTok in other countries often touch on concerns about misleading, false, or potentially incendiary information, in addition to risks of espionage and data collection.¹² Moreover, proposed solutions, such as requiring the sale of TikTok to a domestically based company, can also hint at possible economic drivers for these policies.¹³ The U.S. government first proposed this measure in 2023 and is now operationalizing a version of it.

Taken together, a complex and varied set of motivations underlie government policies that pull online ecosystems further from ideas of global openness and free global exchange of online services and information. While many of these policies are passed in the name of improving cybersecurity and data protection, they can protect countries against very different sorts of cyber risks, ranging from overseas cyberespionage to an inability to investigate domestic threats and crimes or concerns about political unrest and instability. Such measures can be seen as promoting cyber risk resilience insofar as the governments implementing them believe that the threats these policies protect against are the most important, or most imminent, ones they face. However, these policies often involve trade-offs that make users more susceptible to certain types of risks, even as they may protect those users’ governments from other threats. Genuine security concerns motivate many of the content- and service-blocking policies that inhibit the global openness of the internet, but these concerns are often competing with other security threats and resilience risks that might actually be allayed by a more open internet.

Requirements for Data Localization

Another set of policies that directly limit the global openness of the internet are data localization policies that require online services and companies to store data locally within a particular country. Such measures can be disadvantageous for large cloud providers, which might prefer to build fewer data centers and shift data between them depending on demand. Compliance with these policies is also often difficult for smaller online companies, which must invest in data storage infrastructure in every country to which they wish to expand. Still, there are many reasons governments may resist giving online service providers the flexibility to

¹¹ John Yoon and Muktita Suhartono, “TikTok Forced to Close Shopping Feature in Its Second-Largest Market,” *New York Times*, October 5, 2023, <https://www.nytimes.com/2023/10/05/business/tiktok-shop-indonesia-ecommerce-ban.html>.

¹² Caitlin Chin-Rothmann, “Banning TikTok Will Not Solve U.S. Online Disinformation Problems,” CSIS, March 8, 2023, <https://www.csis.org/analysis/banning-tiktok-will-not-solve-us-online-disinformation-problems>.

¹³ Chang Che, “China Says It Will ‘Firmly Oppose’ Forced Sale of TikTok,” *New York Times*, March 23, 2023, <https://www.nytimes.com/2023/03/23/business/china-forced-sale-tiktok.html>.

maintain physical infrastructure in fewer locations and move data freely across borders. Like bans on end-to-end encrypted messaging services, data localization policies are often aimed at facilitating law-enforcement investigations and ensuring that governments have access to the data that their citizens are storing online. Yet, like online service bans, these policies are also motivated by a combination of different factors. For instance, some governments cite concerns about having their data intercepted or surveilled if it is stored in foreign countries. Critics have suggested that there may be economic motives as well, since these laws favor companies that are willing and able to invest in building local data storage facilities.

China's Cybersecurity Law, passed in 2017, was one of the most influential policy measures to enshrine a data localization requirement. It requires that many companies either store Chinese "personal data" or "important data" on servers in China or undergo a risk assessment performed by the Chinese government before the data can be transferred outside the country. Over time, the Chinese government has revised these rules, at times tightening them but at other times attempting to limit the requirements imposed on companies—particularly Chinese companies with global operations.¹⁴ Martin Chorzempa and Samm Sacks cite China's "slowing growth and declining foreign investment" as possible reasons the government might be willing to relax some of its proposed data localization requirements. Their analysis highlights the complex range of motivations that underlie these measures and the interplay between security and economic concerns as well as the associated trade-offs. Other countries have also eased their data localization laws at times in order to encourage industry development and investment. In 2019, for instance, Indonesia rolled back some of its data localization requirements to allow foreign companies to store and process more data abroad.¹⁵

Data localization laws have become increasingly popular since the passage of China's 2017 law, to the point that one McKinsey report in 2022 estimated that 75% of all countries "have implemented some level of data localization rules."¹⁶ For instance, Vietnam requires foreign companies to store Vietnamese users' data domestically, as does Malaysia.¹⁷ The European Union General Data Protection Regulation imposes some requirements on the transfer of personal data outside the EU. In 2023, Meta was fined 1.2 billion euros for violating those requirements and ordered to stop transferring its European users' data to U.S. servers.¹⁸ To a large extent, these measures appear to be motivated by a mixture of security and privacy concerns. The security concerns stem from fears that law-enforcement investigations could be stymied by overseas storage of data that would place people's information beyond the reach of domestic law-enforcement entities. The privacy concerns largely relate to fears that other countries' laws may not sufficiently protect users' data from resale, interception by law enforcement or intelligence agencies, or other threats.

¹⁴ Martin Chorzempa and Samm Sacks, "China's New Rules on Data Flows Could Signal a Shift Away from Security toward Growth," Peterson Institute for International Economics, October 3, 2023, <https://www.piie.com/blogs/realtime-economics/chinas-new-rules-data-flows-could-signal-shift-away-security-toward-growth>.

¹⁵ Tabita Diela, "Indonesia Relaxes Data Storage Rules to Allow Placement Abroad," Reuters, October 24, 2019, <https://www.reuters.com/article/idUSKBN1X31P4>.

¹⁶ Satyajit Parekh et al., "Localization of Data Privacy Regulations Creates Competitive Opportunities," McKinsey & Company, June 30, 2022, <https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/localization-of-data-privacy-regulations-creates-competitive-opportunities>.

¹⁷ James Andrew Lewis, "Cloud Computing in Southeast Asia and Digital Competition with China," CSIS, August 7, 2023, <https://www.csis.org/analysis/cloud-computing-southeast-asia-and-digital-competition-china>.

¹⁸ Adam Satariano, "Meta Fined \$1.3 Billion for Violating EU Data Privacy Rules," *New York Times*, May 22, 2023, <https://www.nytimes.com/2023/05/22/business/meta-facebook-eu-privacy-fine.html>.

While genuine security fears may partly drive these policies, and the policies may provide some resilience against risks like stymied criminal investigations and foreign exploitation of data, they also undermine countries' resilience to other types of risks, namely those posed by outages. By mandating that data be stored on only specific, geographically clustered sets of servers, governments actually provide themselves and their internet users less resilience to threats like natural disasters, power outages, or cyberattacks that might take those servers offline with no overseas backup of their data. As with the policy measures focused on banning online content and services, data localization requirements make clear trade-offs between different types of cyber risk resilience. Governments are forced to prioritize and choose which risks they feel are most pressing when designing and implementing cybersecurity policies.

Conclusion

Countries pursue cybersecurity policies that diverge from globally open approaches for a variety of reasons, ranging from fears about cyberespionage to concerns about local law-enforcement efficacy, economic development, and political stability. In many cases, governments seem to believe that these approaches will help them more effectively achieve certain types of cyber risk resilience against very specific threats or concerns. But as is often the case with policies that limit global openness, there are inherent trade-offs in those decisions. The security and resilience benefits provided by a policy designed to block the free flow of online information across borders could be outweighed by a set of countervailing security and resilience weaknesses introduced by those same policies.

Long-Term Trade-Offs in Cybersecurity Policy Decisions

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NOTE: The views expressed in this chapter are those of the author and do not reflect the official policy or position of the National Defense University, the U.S. Department of Defense, or the U.S. government.

EXECUTIVE SUMMARY

This chapter argues that national cybersecurity policy decisions often entail complex trade-offs between multiple risks and opportunities without full awareness of the long-term consequences for national security and other core national interests.

MAIN ARGUMENT

Cybersecurity policy decisions have significant but inadequately understood long-term effects on core national interests. While policymakers often focus on balancing the short-term costs, benefits, and feasibility of particular approaches to addressing acute current cybersecurity risks, longer-term consequences of these decisions are not always adequately considered. Drawing on the alternative “open” and “closed” models of cybersecurity policy that have been discussed throughout this report, this chapter examines how cyber policy decisions often entail complex trade-offs with long-term repercussions for core areas of national interest, including national security, economic growth, political stability, civil liberties, and geopolitical relationships. While policymakers attempt to balance some of these complexities, policy outcomes are often influenced by limited information, constrained resources and decision processes, international interdependencies, and pressure to act quickly in response to current perceived threats. This chapter provides a roadmap to enhance strategic foresight in crafting durable cybersecurity policies in the long-term national interest.

POLICY IMPLICATIONS

- Cybersecurity policy decisions often require balancing across several short- and long-term risks and opportunities with significant repercussions for national security, economic growth, and other core national interests. To address this, governments should incorporate long-term strategic impact assessments into cybersecurity policymaking processes.
- While adequately addressing current serious cyber risks is critical, policymakers should also avoid overly reactive policy responses to single high-profile threats or incidents without adequate consideration of trade-offs, implementation feasibility, and long-term effects.
- Cybersecurity policy development processes should include sustained engagement with technical, private-sector, civil-society, and other stakeholder communities to reduce information gaps and unintended consequences. This should allow ongoing feedback and adjustment as policies are adopted and implemented.
- Cybersecurity policymakers should use a combination of strategic foresight tools, stakeholder feedback mechanisms, and international collaboration to improve their understanding of possible long-term future risks and build cybersecurity policy solutions that ensure maximal adaptability to different future developments.

In 2012 when the Russian Federation adopted its first law requiring some categories of internet content to be blocked, there was debate both inside the country and abroad as to what the measure portended for the future of the internet within Russia. The law ostensibly only aimed to protect children online. But detractors saw it as an incremental move toward restricting a rapidly developing blogosphere and casting the national approach to internet policy closer to that of China's heavy censorship regime—in other words, the emergence of what chapter 1 of this report describes as “information isolationism.” The law's passage came in the immediate aftermath of the country's largest mass protest events in the post-Soviet period, and the move could be seen as an attempt to respond to a perceived threat to regime stability by curtailing previously freewheeling online discourse.¹

Over the next several years, as the Russian government formulated and implemented additional internet-related laws, some stakeholders interested in maintaining a more open online environment participated in government consultations in attempts to steer the policy in more desirable directions. But they also came to refer to the Duma as the “crazy printer” for its rapid production of new internet laws.² In retrospect, it is easy to see that the precipitous shift in Russia's domestic approach to internet policy aligned with the government's known threat perceptions around “information security” and the potential effects of information flows on domestic political instability that had been articulated in intergovernmental proposals and national strategic documents since the 1990s. Concerns about these risks had, until that point, been balanced with other interests and risks, however, including the effects of new policies on public opinion, the costs of or inability to fully implement measures, and the economic benefits of rapid internet development and potential contribution to diversification and modernization of the economy.

In recent decades, as the internet and digital technologies have become increasingly interwoven into all areas of society, governments around the world have become aware of and sought to mitigate new associated risks. While there is, by now, some degree of shared international awareness of the existence of cyberspace-related security threats, there is far less agreement as to what those threats are or how best to address them. This is only made more complicated by the constantly changing nature of the digital substrate that cuts across and affects many different areas of governance and different stakeholders. Old threats morph and change, while new threats emerge from unexpected arenas, and specific threats can also be context-contingent or have different perceived salience levels for different actors. Thus, as governments have developed laws, policies, and regulations aiming to protect against cyber risks, there has often been substantial variation in what or whom they seek to protect, what security risks they aim to reduce, and, perhaps most critically, how they balance multiple such challenges.

As recounted in this report, the primary ends and means of national cybersecurity policies often differ widely, even as there are some overlapping objectives. While most countries depend to some degree on global flows of data and of digital goods and services, for example, some limit these flows in efforts to protect intellectual property, safeguard political stability, support technological self-sufficiency, and reduce potential security vulnerabilities of critical infrastructure or supply chains. Likewise, while it is common for governments to have an interest in some forms of data

¹ “Russia Internet Blacklist Law Takes Effect,” BBC News, November 1, 2012, <http://www.bbc.com/news/technology-20096274>.

² “From ‘Protecting Children’ to ‘Discrediting the Army’: A Brief History of 10 Years of Russian Internet Censorship,” Meduza, November 5, 2022, <https://meduza.io/en/feature/2022/11/06/from-protecting-children-to-discrediting-the-army>; and Michele A. Berdy, “Of Ear Noodles and Maniacal Printers,” *Moscow Times*, March 17, 2023, <https://www.themoscowtimes.com/2023/03/17/of-ear-noodles-and-maniacal-printers-a80521>.

access for law enforcement and national security, countries vary in the extent to which such uses are constrained by rule of law and civil liberties or privacy protections. There also are substantial differences across countries in how and to what extent different societal stakeholders are consulted and their interests accounted for in the policymaking process.

Overall, some countries' approaches to cybersecurity are more "open" to global flows of data and digital goods and services, with government access and use of data constrained by the rule of law and with substantial mechanisms for private-sector and civil-society engagement with the policymaking process—generally aligning with what is described earlier in this report as the "cybersecurity approach." Other, more isolationist or "closed" national approaches are more resistant to cross-border free flows of data and digital goods and services, with fewer controls on government access to private data and little or no engagement with nongovernmental stakeholders in policymaking—aligning closer with the "information security approach." Approaches also differ in more granular ways. Significant models today focus primarily on market access and economic growth, individual rights and privacy, or information security and political stability.³ But these models and alternatives also continue to evolve with ongoing changes in the technological landscape and the emergence of new challenges.

A number of different factors can affect a country's choice of cyber policies, which are often responsive to the relative salience of particular risks but can also be shaped by awareness of available solutions and a state's capacity to implement them, perceived needs to balance between risks or stakeholder interests, international pressures or cooperation, and the constraints of the national legal and political system and the policy decision process. Cyber policies are often an accumulation of incremental decisions, furthermore, lacking substantial foresight about future rapidly shifting technological developments and seeking to address immediate challenges with more limited consideration of the long-term and strategic effects of different approaches.

This chapter aims to illustrate why long-term effects and trade-offs might warrant further consideration as countries prepare to make consequential short-term cybersecurity policy decisions. While the short-term costs, benefits, and feasibility of particular approaches to addressing cybersecurity risks will always be critical considerations for policymakers, the chapter suggests that longer-term effects on critical infrastructure, economic growth, cybersecurity outcomes, individual rights, and relationships with other countries in the international community might not always be immediately evident to decision-makers. Cybersecurity policy decisions can have substantial effects on the development and security of a country's digital infrastructure and technological capacities, or affect what kinds of economic opportunities will be cultivated through or supported by the country's technology sector. They can also shape the internet ecosystems, freedom of information, or privacy that are available to a country's citizens. Cybersecurity policy decisions can influence what community of countries a nation will draw closer to globally through trade, economic, or political alignments, and they might have repercussions for potential future rivalries or conflict. Recent historical examples demonstrate how decisions drawing a country incrementally closer to open or closed approaches on some factors of cyber policy have had long-term implications. These, in turn, hold lessons for how to address current challenges in cybersecurity decision-making even despite uncertainties about future trajectories.

³ Anu Bradford, *Digital Empires: The Global Battle to Regulate Technology* (New York: Oxford University Press, 2023); and Nicholas D. Wright, ed., *Artificial Intelligence, China, Russia, and the Global Order: Technological, Political, Global, and Creative Perspectives* (Maxwell AFB: Air University Press, 2019).

The remainder of the chapter is divided into four sections. The first section examines in more detail open versus closed approaches to national cybersecurity, underscoring the extent to which many countries fail to neatly align with one or another approach. The second provides an overview of factors and dynamics often influencing these countries' choices, examining why longer-term consequences might not always be the most prominent considerations. The third section, in turn, examines some potential longer-term consequences of adopting more open or closed cyber policies. It considers how these might create substantive trade-offs that are often not adequately understood at the initial point of decision. The chapter concludes with a discussion of how best to extrapolate and learn lessons from such prior experiences in light of current uncertainties about subsequent developments in a constantly changing global digital environment.

Open, Closed, and Other Variations

This report has shown that while most states today understand that they face some risks and threats associated with cybersecurity, they often perceive threats and the relative salience of specific risks quite differently, leading to very different conceptualizations of cybersecurity, strategies for how to pursue it, and consequent wide variation in specific policies and regulatory approaches adopted. We have characterized several broad models that involve a more open or closed approach to cybersecurity. These idealized orientations describe some essential aspects and prominent themes of how major cyber powers approach cybersecurity today, but they do not capture the often contingent decisions and circumstances through which the countries reached their current orientations. Nor do they necessarily characterize the full range of current and future variation as countries globally grapple with appropriate policies to address complex and rapidly changing technologies and circumstances.

Most countries do not fall neatly into clearly distinct and coherent ideological camps along the lines of open versus closed approaches or even following the more specific strategic models exemplified by the approaches of the United States (market-driven), European Union (digital rights), Russia (information isolationism), and China (information isolationism plus) described by Elaine Korzak in chapter 1. But even for these most prominent archetypes, it is important to recognize the extent to which their evolution has not always been a straight line but at times has involved deep conflicts between stakeholders, limitations due to internal processes and resource constraints, and inconsistent but ongoing consideration of different risks and equities to be balanced. Sometimes decision processes themselves are short-circuited by lack of or limited available solutions or capacity to implement them, or prioritization has abruptly changed due to lessons taken from world events or other sudden changes in context. Examples of complex balancing and a lack of one clear and consistent ideological approach can be seen in many of the cybersecurity policy decisions even of states considered exemplars of particular approaches.

These dynamics can be clearly seen in Russia's long-winding path toward its current highly closed approach to information security. While the country adopted its first legal framework for the government's System for Operative Investigative Activities (SORM) surveillance system to support law enforcement and intelligence in 1995 and adopted its first national strategic Doctrine of Information Security in 2000, it also rolled back a first attempt at exerting more centralized government control over national critical internet infrastructure in 1999 in response to pressure from civil-society and private-sector stakeholders, emphasized goals of internet and technology

sector development as sources of economic modernization through the early 2010s, and for years avoided overt internet censorship or the blocking of international online platforms or services, in some cases developing its own alternatives that could successfully compete for users domestically and in nearby countries.⁴ Ultimately, Russia's pivot toward overt censorship came as a response to changing threat perceptions driven by increasing national internet and social media penetration and international and domestic threat demonstration events. The Kremlin perceived the Arab Spring and Russia's Bolotnaya protest movement as showcasing existential risks to regime stability resulting from international information flows. Thus, even Russia's subsequent rapid shift in cybersecurity risk prioritization followed a long period of experimentation and balancing across multiple risks and objectives.⁵

For many countries, the process of determining what cyber risks to prioritize and how to address them is complex and conflictual. Smaller, less developed, middle regime-type, or not clearly aligned countries often must balance not only between different risks and ideological approaches but also between the demands or pressures of erstwhile partners or adversaries, varying normative expectations, and the potential impacts of different sets of international relationships and economic commitments. They are also often constrained by limited information, limited economic means, or modest technological capabilities; are not always aware of the larger repercussions of or linkages across policy decisions; or only consider cybersecurity risk through the prisms of prominent regional or international approaches or those of their closest partners.

We see some of these dynamics clearly when examining the broader evolution of internet censorship and surveillance in the 2000s and early 2010s. While some states like China rapidly developed advanced capabilities, others known for their offline control over expression or association were surprisingly late to adopt any form of online censorship or surveillance due in part to a lack of technological skill or state capacity. As affordable products and services became available on global networks, this online-offline asymmetry began to subside in some places. Middle regime-type countries were constrained in different ways, however, often wrestling with the balance between reputational and economic costs associated with reduced global digital connectivity or overt violations of democratic internet norms, on the one hand, and risk of regime-destabilizing internet-enabled mass protest movements, on the other. This challenging risk balance in turn fueled greater experimentation to mitigate perceived risks without resorting to more obvious restrictions that could scare away investors or prompt companies to exit. At least initially, such approaches balanced continued global digital connectivity and internet sector development with more subtle forms of closedness in attempts to manage information flows and discourse while maintaining the economic benefits of digital openness. However, the tension between these goals often increases as the relative economic and social importance of online activity grows.

A similar wrestling with conflicting pressures and threat perceptions, limited information, and constricted options can be seen in some of the policy choices countries have made since the late 2010s concerning how to situate themselves in relation to digital infrastructure and emerging areas of U.S.-China competition. Some developing countries have faced conflicting pressures—for

⁴ "Europe's Great Exception: Why Local Firms Dominate the Russian Internet," *Economist*, May 19, 2012, <https://www.economist.com/business/2012/05/19/europes-great-exception>; Andrei Soldatov and Irina Borogan, *The Red Web: The Struggle Between Russia's Digital Dictators and the New Online Revolutionaries* (New York: Public Affairs, 2015); and Jaclyn A. Kerr, "Authoritarian Management of (Cyber-) Society: Internet Regulation and the New Political Protest Movements" (PhD diss., Georgetown University, 2016), chap. 4.

⁵ Jaclyn A. Kerr, "Russia's Critical Juncture: The Ukraine War and the Battle for the Soul of the Web," *SAIS Review of International Affairs* 42, no. 2 (2022): 63–84.

example, on whether to strike technology development deals with China (such as participating in the Belt and Road Initiative and Digital Silk Road programs or purchasing Huawei Safe City or 5G technologies) or instead to adopt more expensive versions of the same technologies, or even to forgo these technologies altogether. In some cases, states have decided to rely on possibly less advanced homegrown alternatives.⁶ Some U.S. partners and allies have opted to participate in the U.S.-promulgated Clean Network Initiative, abstaining from the inclusion of Huawei technologies into critical networks (whether for reasons of diplomatic relations or national security).⁷ While some of these decision processes have involved high-level diplomatic meetings and careful balancing of global and national concerns, other decisions have been made iteratively in local contexts. Local public safety and governance imperatives concerning crime, traffic, or energy management, for example, have driven many city governments in developing countries to adopt affordable smart (or “safe”) city, facial recognition, or other sensor technologies without detailed attention to trade-offs touching on local-level data governance, let alone national-level security, economic, or diplomatic prerogatives.⁸

On top of this inherent complexity of national cybersecurity risk balancing, the international digital environment is constantly continuing to evolve, leading to the emergence of new risks, threat perceptions, and regulatory possibilities. Rapid changes in the digital technology environment and global context have often precipitated abrupt shifts in threat perceptions, sometimes leading to new incoherencies in a national cybersecurity approach and new risk-balancing dilemmas. We see such dynamics in the abrupt attention to potential risks and harms associated with end-to-end encryption in the United Kingdom following the 2011 London riots in which participants heavily used encrypted BlackBerry messaging.⁹ We see it also in the sudden attention to risks associated with social media platforms and online information flows in the United States following evidence of Russian online influence campaigns targeting the 2016 U.S. election or prominent incidents involving domestic hate crimes linked to online extremist discussions.¹⁰ Similarly, the emergence of international competition over artificial intelligence (AI) technology development has prompted a gradual weakening of the United States’ long-standing commitment to support an international norm of free flows of data across borders. Some data is increasingly viewed as a critical asset in the training of AI capabilities and hence integral to international competition.¹¹

Sometimes sudden shifts in perceived risk salience have also led to poorly considered attempts at rapid policy solutions, demonstrating the perils of low-information decision-making without adequate attention to risk trade-offs or second-order repercussions. This can result, for example, in the implementation of highly restrictive forms of regulation, such as using draconian performance requirements coupled with potentially devastating *ex post* liability, or creating extreme

⁶ David Sacks, “China’s Huawei Is Winning the 5G Race. Here’s What the United States Should Do To Respond,” Council on Foreign Relations (CFR), March 29, 2021, <https://www.cfr.org/blog/china-huawei-5g>.

⁷ David P. Fidler, “The Clean Network Program: Digital Age Echoes of the ‘Long Telegram’?” CFR, October 5, 2020, <https://www.cfr.org/blog/clean-network-program-digital-age-echoes-long-telegram>.

⁸ Beth Kerley, “Smart Cities and Democratic Vulnerabilities,” National Endowment for Democracy, 2022, https://www.ned.org/wp-content/uploads/2022/12/Beth-Kerley_Smart-Cities-and-Democratic-Vulnerabilities.pdf; and Robert Muggah, “‘Smart’ Cities Are Surveilled Cities,” *Foreign Policy*, April 17, 2021, <https://foreignpolicy.com/2021/04/17/smart-cities-surveillance-privacy-digital-threats-internet-of-things-5g>.

⁹ Josh Halliday, “London Riots: BlackBerry to Help Police Probe Messenger Looting ‘Role,’” *Guardian*, August 8, 2011, <https://www.theguardian.com/uk/2011/aug/08/london-riots-blackberry-messenger-looting>.

¹⁰ For further discussion, see Jaclyn A. Kerr, “Concept Misalignment and Cyberspace Instability: Lessons from Cyber-Enabled Disinformation,” in *Cyberspace and Instability*, ed. Robert Chesney, Max Smeets, and James Shires (Edinburgh: Edinburgh University Press, 2023), chap. 3.

¹¹ David Lawder, “U.S. Drops Digital Trade Demands at WTO to Allow Room for Stronger Tech Regulation,” Reuters, October 25, 2023, <https://www.reuters.com/world/us/us-drops-digital-trade-demands-wto-allow-room-stronger-tech-regulation-2023-10-25>.

intermediary liability for regulatory targets. Such approaches can prompt protective actions by private-sector targets that lead to far-reaching chilling effects on either economic engagement or discourse. They also can reduce the possible mechanisms through which civil society or the private sector can share views and engage in the policy process. The failed 2012 U.S. Stop Online Piracy Act (SOPA) and PROTECT IP Act (PIPA) are examples of such challenges in which legislation written to maximize protection against one emergent threat did not adequately account for potential far-reaching effects. The bills alarmed civil liberties activists as well as internet platforms and led to large-scale protests—including by some of the United States’ leading technology companies.¹²

The initial effort to update the Wassenaar Arrangement export controls to address contemporary cybersecurity issues is another clear instance where a rush to address one set of risks created new problems. The voluntary international export controls framework aims at controlling the spread of conventional arms and commercial items that also have military applications by coordinating national export control measures. In December 2013, additions following the Arab Spring sought to reduce the availability of digital network intrusion and surveillance technologies to repressive governments.¹³ The controls triggered an outcry from the cybersecurity industry and research community, which argued that their breadth would make digital technologies less secure by rendering many of the daily practices and international collaborations of cybersecurity professionals subject to export-control licensing requirements.¹⁴ The rules were ultimately successfully renegotiated with relevant exemptions in 2017 based on critical input from key experts.¹⁵

Developing countries can face even more complex trade-offs as they formulate cybersecurity policies for their quickly digitizing economies and societies. In Vietnam, expanding cybersecurity challenges—including cybercrime and state-backed cyberattacks—have prompted rapid adoption of multiple new measures. Vietnam’s new cybersecurity law (Decree 147), which went into force in December 2024, has faced criticism for its provision that internet platforms remove all posted “illegal content” within 24 hours, putting extreme pressure on internet intermediary companies with potential for incentivizing platform self-censorship and chilling effects on online discourse.¹⁶ While such provisions might suggest an overall prioritization of information security and political stability over economic development of the internet sector and the traditional U.S. or EU approaches to cybersecurity, Vietnam has also been credited with making swift improvements in its national commitment to cybersecurity measured by the International Telecommunications Union Global Cybersecurity Index. These efforts are particularly focused on protecting against cyber aggression by China even as some parts of Vietnam’s cybersecurity provisions have been

¹² David A. Fahrenthold, “SOPA Protests Shut Down Web Sites,” *Washington Post*, January 18, 2012, https://www.washingtonpost.com/politics/2012/01/17/gIQA4WYl6P_story.html.

¹³ Russell Brandom, “The International Rules That Have the Security World on Alert,” *Verge*, May 22, 2015, <https://www.theverge.com/2015/5/22/8644185/wassenaar-arrangement-export-zero-day-research>.

¹⁴ Russell Brandom, “Google Says Controversial Export Proposal Would Make the World ‘Less Secure,’” *Verge*, July 20, 2015, <https://www.theverge.com/2015/7/20/9005351/google-wassenaar-arrangement-proposal-comments>.

¹⁵ Shaun Waterman, “The Wassenaar Arrangement’s Latest Language Is Making Security Researchers Very Happy,” *CyberScoop*, December 20, 2017, <https://cyberscoop.com/wassenaar-arrangement-cybersecurity-katie-moussouris>; and Garrett Hinck, “Wassenaar Export Controls on Surveillance Tools: New Exemptions for Vulnerability Research,” *Lawfare*, January 5, 2018, <https://www.lawfaremedia.org/article/wassenaar-export-controls-surveillance-tools-new-exemptions-vulnerability-research>.

¹⁶ The new law also includes a data localization measure requiring companies to store user data and requires that all social media users register their accounts with their real names, with this data and user information to be provided to the government upon request. State media quoted Le Quang Tu Do, director of the Department of Radio, Television and Electronic Information in the Ministry of Information and Communications, stating that Directive 147 aims to “regulate behaviour in order to maintain social order, national security, and national sovereignty in cyberspace.” “Critics Decry Vietnam’s ‘Draconian’ New Internet Law,” *Agence France-Presse*, December 22, 2024, available at <https://www.theguardian.com/world/2024/dec/23/vietnam-identity-verification-internet-law-reactions-decree-147-censorship>.

criticized by digital rights advocates for their resemblance to China's 2017 Cybersecurity Law. Ultimately, countries seeking to foster competitive digital economies face complex trade-offs as they balance between protecting against digital crime and international cyber aggression, on the one hand, and avoiding onerous regulatory burdens for their technology sectors, on the other.¹⁷

Overall, when considering specific national decisions around how to conceptualize cybersecurity threats and risks and what strategies, policies, and regulations to adopt to mitigate risks, it is important to recognize the extent to which these decision processes can be messier than imagined. Decisions often involve incomplete awareness of the threat environment, inadequate information and technical understanding, and limited consideration of policy interdependencies, implementation challenges, or countervailing risks. Decision-makers likewise face rapidly developing technologies and global events, sometimes prompting swift adaptations with early and imperfect information. Constrained and iterative decisions can result in policies that lack coherent vision for long-term effects or interactions between issues. The next section takes a closer look at how constraints on cybersecurity decision-making processes can create gaps in understanding around trade-offs and long-term national strategic considerations.

Factors Influencing Cybersecurity Policy Decisions

An ideal cybersecurity policy process would carefully consider all risks and potential security mitigation approaches. It would gather information from all relevant stakeholders about their threat perceptions, interests, and capabilities, including a clear accounting of potential repercussions of different possible regulatory actions. After delineating the costs and benefits of different types of solutions, such a decision process would balance risks and interests to arrive at an optimal approach. In the real world, such centralized rational decision-making under conditions of perfect information is the stuff of fantasy. Governments often make cybersecurity policy decisions without equal awareness of different threats, risks, and solutions. Many factors beyond information limitations further constrain decision processes, going well beyond just the need to balance distinct risks or account for different stakeholder views. These can include inadequacies of information, imperfect rationality of processes, capacity limitations, and international interdependencies. Even when policymakers have acquired good information about these factors, resource constraints and institutional factors may limit policy options.

The modeling of cybersecurity policy decisions as thoughtful trade-off calculations by unitary, fully informed, rational decision-makers is misleading for reasons that relate to the actual messiness of many policy decision processes—especially concerning rapidly developing new areas of technology and policy. For one thing, the decisions often are not made by one centralized authority but involve a wide array of actors and institutions, crossing different sectors and stakeholder groups, sometimes physically dispersed or through processes spread iteratively across time. For example, accumulating municipal government decisions have influenced policies

¹⁷ Vietnam has sought to address major challenges from cybercrime and from Chinese cyberattacks. Sustained efforts to strengthen laws, institutions, and public awareness have been positively assessed by international cybersecurity metrics, with the Global Cybersecurity Index, for example, showing the country rapidly improving from 100th to 25th out of 194 countries between 2017 and 2020. Vietnam's National Cybersecurity Strategy (ratified in 2022) set a goal of maintaining a national ranking of at least 25th to 30th in this index. Robert Lemos, "Ransomware, Junk Bank Accounts: Cyber Threats Proliferate in Vietnam," Dark Reading, April 2, 2024, <https://www.darkreading.com/cyber-risk/ransomware-junk-bank-accounts-cyberthreats-proliferates-in-vietnam>; and Viet Dung Trinh, "Vietnam's Securitisation of Cybersecurity Under Growing Chinese Cyber Threats," Australian Institute of International Affairs, April 9, 2025, <https://www.internationalaffairs.org.au/australianoutlook/vietnams-securitisation-of-cybersecurity-under-growing-chinese-cyber-threats>.

around smart cities, facial recognition technologies, and 5G, while EU member and U.S. state laws have pioneered new approaches to data localization and digital privacy, Internet of Things (IoT) security, online hate and extremism, social media platforms, and other cybersecurity-related policies.¹⁸ In this way, some issues are framed at local levels before there is wider agreement or even recognition of the challenges.

Informational and cognitive constraints may also affect cybersecurity policy decision-making. When circumstances force policymakers to make decisions with limited information—about relevant risks, mitigations, and costs and benefits, and dealing with a technical subject matter of which they have limited understanding—they might frequently rely on simplifying heuristics or other intuitive reasoning mechanisms.¹⁹ Decision-makers tend to have greater awareness of more immediate and concrete risks than abstract, distributed, or far-off ones, for example, which can lead to significant focus on recent, overtly noteworthy, or sudden shock events, the experiences of most visible or near-peer countries, or overt and frequently occurring events rather than those that are highly impactful but lower frequency, or that might fly under the radar with serious cumulative consequences. This might lead some states to pay greater attention to high-frequency cybercrime than rarer but potentially devastating state-backed cyberattacks involving critical infrastructure or digital espionage. It also might make some policymakers less likely to address ongoing but less visible cyber harms such as those faced by highly distributed stakeholders with limited influence. Similarly, decision-makers might be inclined to accept conceptualizations of cybersecurity or information security adopted by partner countries, or to consider policy frameworks and approaches that they have observed implemented by economically or politically similar states.

Cognitive biases can lead to complacency dynamics that overlook or downplay vulnerabilities that are not overtly obvious or have yet to be clearly exploited. But when surprising threat demonstration events do occur, they can also lead to distorted reasoning and overreaction, limiting the consideration of other risks and priorities. Arguably such dynamics were evident in some early reactions to cyber-enabled information and influence operations and election interference in the United States and EU. Likewise, Russia's increasingly precipitous shift toward an information isolationism model following domestic protests showed a growing unwillingness to engage with stakeholder communities, even while policies enacted had enormous negative long-term economic consequences for its once thriving tech sector. While Russian internet and cybersecurity companies were regionally and globally competitive through the early 2010s, the country's subsequent isolationist approaches led to market losses and the end of international research collaborations.²⁰

¹⁸ See, for example, Julia Edinger, "Smart Cities' Evolve Beyond Devices to Become More Local," *Government Technology*, October 16, 2024, <https://www.govtech.com/smart-cities/smart-cities-evolve-beyond-devices-to-become-more-local>; Dani Anguiano, "Montana Becomes First U.S. State to Ban TikTok," *Guardian*, May 17, 2023, <https://www.theguardian.com/us-news/2023/may/17/tiktok-ban-montana>; Alison Smale, "Merkel Backs Plan to Keep European Data in Europe," *New York Times*, February 16, 2014, <https://www.nytimes.com/2014/02/17/world/europe/merkel-backs-plan-to-keep-european-data-in-europe.html>; Daisuke Wakabayashi, "California Passes Sweeping Law to Protect Online Privacy," *New York Times*, June 28, 2018, <https://www.nytimes.com/2018/06/28/technology/california-online-privacy-law.html>; and Adam Satariano and Christopher F. Schuetz, "Where Online Hate Speech Can Bring the Police to Your Door," *New York Times*, September 23, 2022, <https://www.nytimes.com/2022/09/23/technology/germany-internet-speech-arrest.html>.

¹⁹ Herbert Lin and Jaclyn Kerr, "On Cyber-Enabled Information Warfare and Information Operations," in *Oxford Handbook of Cyber Security*, ed. Paul Cornish (Oxford: Oxford University Press, 2022), 251–72; and Amos Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science*, September 1974, 271–1974.

²⁰ See, for example, Ravie Lakshmanan, "U.S. Bans Kaspersky Software, Citing National Security Risks," *Hacker News*, June 21, 2024, <https://thehackernews.com/2024/06/us-bans-kaspersky-software-citing.html>; and Peter Mironenko, "Yandex: The End of an Era for Russia's Most Innovative Firm," *Bell*, February 12, 2024, <https://en.thebell.io/yandex-the-end-of-an-era-for-russias-most-innovative-firm>.

State cybersecurity risk mitigations can also be significantly limited by the country's internal resource and institutional constraints. National economic, human capital, organizational, and governance capacities can either enable or restrict a state's ability to effectively implement particular policies or regulatory approaches. A less developed country with a limited technical workforce might not be able to emulate a policy that requires extensive technical manpower, for example, while a weak state or one with highly distributed or federated authorities might struggle to rapidly implement a significant change in the central government's regulatory reach.²¹ A country's legal and political system can also impose internal constraints on decision processes, such as when democratic or multistakeholder decision processes introduce friction. These internal dynamics can affect how responsive policies are to different risks and stakeholders. In 2023, for example, when the U.S. Environmental Protection Agency issued a memorandum requiring states to include assessments of cybersecurity in their mandatory surveys of public water systems, the rule was withdrawn following lawsuits by several state attorneys general and groups representing rural water utilities concerned about added costs. Likewise, repeated U.S. attempts to pass federal data breach notification legislation have failed in Congress, despite the adoption of federal notification requirements for some industry sectors (such as finance and health) as well as state-level laws in all 50 states.²²

While such internal political processes can make it more challenging for regulators to take action to reduce known risks, they can also slow the adoption of approaches that have not received adequate technical oversight. When stakeholder protests prevented the passage of the United States' SOPA and PIPA legislation in 2012, they also drew lawmakers' attention to underlying technical and economic problems with the specific approach and its potential negative impacts on business. On the other hand, deliberately analogous protest efforts by civil society and internet organizations in Russia in 2013 failed to stop a very similarly designed law from passage partly because these groups, even though consulted, had far less ability to project influence within the political system.²³

The degree of state capacity and the nature of government relations with different stakeholders can shape not only awareness of risks but the availability of critical information for determining or implementing effective policy approaches to address these risks. While many policymakers lack adequate understanding of the risks faced or defensive actions available to private-sector enterprises, such as internet intermediaries or critical infrastructure companies, they also rely on these actors to address information asymmetries and enact "delegated governance" as regulatory targets.²⁴ Various collaborative organizations such as Information Sharing and Analysis Centers and public-private partnerships often play important roles in bridging this gap, fostering

²¹ Most states attempting to implement internet content control measures in the 2000s and 2010s lacked sufficient manpower or technical knowledge to implement China's initial labor-intensive approach, for example, but many more could afford to purchase preprogrammed network filtration and deep-packet inspection devices when these became available from Blue Coat and similar companies on global markets. See Morgan Marquis-Boire et al., "Planet Blue Coat: Mapping Global Censorship and Surveillance Tools," Citizen Lab, January 15, 2013, <https://citizenlab.ca/2013/01/planet-blue-coat-mapping-global-censorship-and-surveillance-tools>.

²² Stephen Titcombe, "Data Breach Laws by U.S. State," TermsFeed, February 16, 2025, <https://www.termsfeed.com/blog/us-state-data-breach-laws>.

²³ Kevin Rothrock, "Orphaned in U.S., SOPA Finds Home in Russia," Global Voices, June 14, 2013, <https://globalvoices.org/2013/06/14/orphaned-in-us-sopa-finds-home-in-russia>; and Maryam Omid, "Blackout: Why Russian Internet Sites Are Going Dark over Anti-Piracy Laws," *Calvert Journal*, August 1, 2013, <http://calvertjournal.com/articles/show/1273/blackout-why-russian-internet-sites-are-going-dark-over-antipiracy-laws>.

²⁴ Laura DeNardis, *The Global War for Internet Governance* (New Haven: Yale University Press, 2014).

sector- or industry-specific cybersecurity information sharing, regulation, and enforcement.²⁵ At best, mutually beneficial public-private partnerships and other stakeholder interactions can enable more effective implementation of cybersecurity policy. But inadequate consultation, distrust and limited information sharing, or overly punitive regulatory approaches can also limit the effectiveness of implementation; stifle the independence of the private sector, civil society, or media; and potentially obscure additional risks or alternative solutions.

Individual state cybersecurity policy decision processes are likewise influenced by regional and international interdependencies, being affected by the actions and policy decisions of other states as well as international organizations, markets, and transnational networks. These can involve uncoordinated dynamics where one state's behavior inadvertently affects that of another or deliberate coordination where one state actively seeks to affect another state's policy or its implementation. Without any deliberate coordination, for instance, decision-makers often observe and learn from policies adopted by other states—particularly when they see a state as similar or like-minded or believe there is evidence of the policy proving advantageous. This can lead to clusters of similar policies across regions, sets of partners, or states of comparable development levels. Such uncoordinated diffusion can occur also in which the actions of one or several states effectively reduce the costs of similar action by additional states, hence driving increased preference by later adopters.²⁶ This could manifest in a reduction in the reputational costs related to “internet freedom” norm violations as a larger number of countries adopt some form of information isolationist policies in reaction to political stability or data security concerns, for example, or decreased costs of implementing organization-level cybersecurity best practices as enterprise security products and services become affordable through global markets.

Deliberate coordination between states can also drive policy diffusion, whether through collaboration or coercive pressure, in bilateral relationships or within international organizations. This can lead to the agreed adoption of specific shared regulatory frameworks, technical standards, or doctrinal approaches—sometimes enabling greater cross-jurisdictional legal, market, or technical interaction and interoperability. The EU's gradual movement toward increasingly harmonized cybersecurity policies across member states through various pieces of legislation demonstrates the possible roles of regional organizations and common markets in policy coordination.²⁷ There is further evidence of several of these diffusion mechanisms in the global uses of the National Institute of Standards and Technology (NIST) Cybersecurity Framework,²⁸ international technical assistance and capacity-building programs, the dispersion of information security doctrines and collaboration across states of the Shanghai Cooperation Organisation, or

²⁵ For example, in the U.S. financial sector the Financial Services Sector Coordinating Council and Financial Services Government Coordinating Council play vital roles in bringing together private-sector financial institutions and government agencies to protect the cybersecurity of financial services infrastructure. In the U.S. energy sector, the Federal Energy Regulatory Commission collaborates with the North American Electric Reliability Corporation to develop and monitor Critical Infrastructure Protection reliability standards for the bulk electric system and power grid. See “Protecting Critical Financial Infrastructure,” Financial Services Sector Coordinating Council, <https://fsscc.org/about-fsscc-13>; and Jeff Darrington, “FERC and NERC: Cyber Security Monitoring for the Energy Sector,” Graylog, March 3, 2025, <https://graylog.org/post/ferc-and-nerc>.

²⁶ Beth A. Simmons and Zachary Elkins, “On Waves, Clusters, and Diffusion: A Conceptual Framework,” *Annals of the American Academy of Political and Social Science*, no. 598 (2005): 33–51, <https://ssrn.com/abstract=1518452>.

²⁷ Grace O'Neill, “CyberNext BRU: Harmonizing Cybersecurity Regulations in the EU Single Market,” Center for Cybersecurity Policy and Law, February 25, 2025, <https://www.centerforcybersecuritypolicy.org/insights-and-research/cybernext-bru-harmonizing-cybersecurity-regulations-in-the-eu-single-market>; and “A Security Symphony: Harmonizing Cybersecurity Regulation,” Aspen Digital, May 2024, <https://www.aspendigital.org/report/security-symphony>.

²⁸ Amy Mahn, “NIST's International Cybersecurity and Privacy Engagement Update—Mexico City, RSA Conference, and More,” National Institute of Standards and Technology, Cybersecurity Insights, June 12, 2024, <https://www.nist.gov/blogs/cybersecurity-insights/nists-international-cybersecurity-and-privacy-engagement-update-mexico>.

the spread of data protection policies modeled on the General Data Protection Regulation (GDPR) beyond the EU.²⁹ Specific legal or technical solutions can also diffuse transnationally, such as through networks connecting similar agencies and experts across governments or through regional market mechanisms. Such dynamics can be seen in the emulation of Russia's lawful intercept law and its implementation across intelligence agencies in different former Soviet state governments as well as in the spread of network surveillance technologies built to the SORM standard across parts of the former Soviet Union region in the 2000s and 2010s.³⁰

In sum, cybersecurity policy decision processes are often distributed across actors, space, and time; face informational, cognitive, and resource constraints; and are affected by interdependencies and other pressures. The resulting decisions may seldom resemble rational decisions by a unitary actor balancing all current and future risks. While pressure from stakeholders might help correct serious missteps or overreach in the moment (particularly in contexts that allow for more such private-sector and civil-society engagement), it is challenging for countries to gain adequate awareness of potential longer-term trade-offs and consequences adhering to current pressing cybersecurity policy decisions. The next section turns to examine this challenge.

Long-Term Consequences and Trade-Offs

Even well-considered cybersecurity policy decisions often involve known trade-offs, with decision-makers choosing to minimize one risk while not addressing or even increasing vulnerability to another form of risk. But since these decision processes are in fact often constrained and imperfect and the technology and global context keep changing, some trade-offs embedded in policy decisions will not always be obvious. Others, even if known, can develop in unexpected ways and lead to unforeseen consequences or necessitate later policy reversals. These dynamics are only magnified the more limited a country's access to relevant information and the more constrained its decision-making process. But the consequences of one-sided policies that fail to account for trade-offs can be significant.

There are a number of different types of trade-offs that can occur in cybersecurity policy decision processes. These can involve, for example, decisions concerning how to prioritize different families of risk such as those related to national security, economic growth, law enforcement, political stability, and societal rights. Trade-offs can involve decisions about at which levels of society to prioritize risk mitigation—those that affect the whole state or government, its private sector and economy or civil society, or individual citizens. They can also involve systems of international relationships—decisions about which partnerships, pressures, or opportunities to prioritize in framing and addressing national cybersecurity risks. They can involve balancing between different time horizons such as short-term, clear threat perceptions versus longer-term, less precise risk considerations due to greater ambiguities around future developments. Typically trade-offs relate to an assumed or actual zero-sum relationship between maximizing protection against one type of risk and another risk form, though the extent of these incommensurabilities might also relate to policy design and implementation. Hence, balancing between multiple risks might involve

²⁹ See Dan Simmons, "17 Countries with GDPR-Like Data Privacy Laws," *Comforte Blog*, January 13, 2022, <https://insights.comforte.com/countries-with-gdpr-like-data-privacy-laws>.

³⁰ Jaclyn A. Kerr, "Information, Security, and Authoritarian Stability: Internet Policy Diffusion and Coordination in the Former Soviet Region," *International Journal of Communication* 12 (2018): 3814–34, <https://ijoc.org/index.php/ijoc/article/viewFile/8542/2460>.

necessary compromises in which the protection against one risk is realized partly through lessening protection against another; however, it also can involve innovative solutions whereby the actual need for such trade-offs is reduced through a novel policy approach that better handles both risks.

State decisions over open versus closed cybersecurity policy directions as discussed in this report often do entail significant trade-offs. Sometimes these have been carefully considered and weighed by decision-makers, though in other instances the nature or extent of the effects have not been as evident or accounted for in the moment of decision. Considering global developments in cybersecurity policy, we can see examples of both deliberate and inadvertent trade-offs as well as various attempts to balance between priorities in each of the policy areas we have examined. We also see cases where states have subsequently changed or adjusted their approaches in response to growing understanding or changing priorities.

Cross-Border Flows of Data and Information

When making decisions concerning the cross-border flows of data and information, state cybersecurity decision-makers often consider the effects of such flows on sovereignty, political stability, and national security. Such choices, however, can also have far-reaching effects on the vibrancy of a country's information ecosystem, national economic growth and tech sector development, the protection of civil liberties, or the resilience of critical information resources during conflict. Some prominent countries adopt largely closed or open cybersecurity policies on these issues, either favoring or opposing internet censorship, the blocking of foreign digital apps and internet intermediaries, and requirements for data localization. Yet many countries fall in the middle, embracing a mix of open and closed measures. In other cases, countries have altered these policies as their awareness of trade-offs changes.

Both more open and closed countries have changed their approaches at times to account for unanticipated vulnerabilities or harms. In recent years, as discussed in chapter 3, some general shift toward greater data localization and limits on certain foreign platforms or content has emerged, even in broadly open countries. While many countries initially embraced free flows of data and information across borders, the rise of digital espionage, online extremism, and disinformation as perceived threats and the increasing awareness of data as a resource for security-relevant competition in advanced AI development have led to shifting understandings of some norms and trade-offs. We see this in recent efforts to force the sale or blockage of TikTok in the United States or in European laws such as Germany's 2017 Network Enforcement Act, which has implications for online discourse and intermediary liability.³¹ Likewise, countries favoring information security do not always take fully closed approaches to data flows, balancing stability and security threat perceptions with concerns over economic or political consequences. This is evident, for example, in Russia's failure to block Telegram or YouTube at the start of its full-scale war against Ukraine (even as it blocked other popular social media applications). The decision was apparently at least in

³¹ Ann E. Marimow and Cristiano Lima-Strong, "Supreme Court Upholds TikTok Ban-or-Sale Law Slated to Start Sunday," *Washington Post*, January 17, 2025, <https://www.washingtonpost.com/politics/2025/01/17/tiktok-ban-supreme-court-decision-ruling/>; and Diana Lee, "Germany's NetzDG and the Threat to Online Free Speech," Yale Law School, Media Freedom and Information Access Clinic, October 10, 2017, <https://law.yale.edu/mfia/case-disclosed/germanys-netzdg-and-threat-online-free-speech>.

part due to widespread reliance on these technologies among government officials and a desire to permit their continued use to spread pro-war content to large audiences.³²

Countries have long adopted data localization requirements with a variety of motives. Some aim to ease data access by government and law enforcement, while others focus on protecting data from foreign espionage or fostering local investment. Primary goals can vary from national security and governability to privacy and economic growth. These policy decisions have not always clearly accounted for all trade-offs. This can be seen in cases where prominent national data localization rules were subsequently relaxed—at least in part for economic reasons—such as in China, Indonesia, and India.³³ Likewise, countries might fail to consider the greater physical vulnerability of storing critical data resources in a smaller number of geographically co-located facilities, particularly under conditions of military or hybrid conflict or during a natural disaster. Ukraine rapidly changed its approach to government data in response to Russia's full-scale invasion and war against the country, for example. With the help of international private-sector partners, it quickly moved critical data resources to cloud-based platforms to increase national digital resilience.³⁴

Openness or Closedness to the Import of Foreign Digital Products

National cybersecurity policy decisions around openness or closedness to the import of foreign digital products similarly reckon with a number of trade-offs related to both economic growth and key areas of national security. Some of these trade-offs have not always been immediately apparent. While many states have traditionally approached technology import policies primarily through the prism of economic development or competitiveness, changes in the global geopolitical and technological environment have drawn attention to often significant national security implications. These include risks such as the incorporation of vulnerabilities into critical infrastructure or supply chains increasing their susceptibility to espionage or cyberattack, vendor lock-in or debt-trap relations, or a lack of resilience due to dependence on a single major supplier for key technology inputs.

Understanding the issue through liberal economic logic, many countries initially adopted the open approach of embracing overseas suppliers and global markets for digital goods as an optimal path to ensure access to cutting-edge or affordable technologies. This approach enabled their own goods to compete in global markets while nourishing domestic technological resilience in part through vendor diversity. Countries opting for more closed policies also sometimes did so for economic reasons, prioritizing the development of homegrown technologies and import substitution as paths to the development of the domestic tech sector. But awareness of the potential

³² This included content spread by nationalists and amateur war correspondents, for example. As of April 2026, Telegram and YouTube have both finally been blocked nationally in Russia as authorities crack down further on the use of foreign messaging and video applications, though both remain widely accessed via virtual private networks. See, for example, "YouTube Blocks Russian Parliament Channel, Drawing Ire from Officials," Reuters, April 11, 2022, <https://www.reuters.com/world/europe/youtube-blocks-russian-parliament-channel-drawing-ire-officials-2022-04-09/>; Victor Davidoff, "Reading the Tea Leaves of Russia's Pro-War 'Z-Universe,'" *Moscow Times*, October 14, 2022, <https://www.themoscowtimes.com/2022/10/13/reading-the-tea-leaves-of-russias-pro-war-z-universe-a79078>; Alexey Strelnikov, "YouTube, WhatsApp Blocked in Russia," DW, February 12, 2026, <https://www.dw.com/en/youtube-whatsapp-blocked-in-russia/a-75940102>; and Stanisław Kaleta, "Moscow 'Fully Blocks' Telegram App amid Information Crackdown," TVP World, April 11, 2026, <https://tvpworld.com/92576745/moscow-fully-blocks-telegram-app-in-crackdown>.

³³ Arindrajit Basu, "The Retreat of the Data Localization Brigade: India, Indonesia and Vietnam," *Diplomat*, January 10, 2020, <https://thediplomat.com/2020/01/the-retreat-of-the-data-localization-brigade-india-indonesia-and-vietnam/>; and Martin Chorzempa, "China's New Rules on Data Flows Could Signal a Shift Away from Security Toward Growth," Peterson Institute for International Economics, October 3, 2023, <https://www.piie.com/blogs/realtime-economics/2023/chinas-new-rules-data-flows-could-signal-shift-away-security-toward>.

³⁴ Ryan White, "How the Cloud Saved Ukraine's Data from Russian Attacks," C4ISRNet, June 22, 2022, <https://www.c4isrnet.com/2022/06/22/how-the-cloud-saved-ukraines-data-from-russian-attacks>.

national security risks associated with a maximally open approach has grown in recent years, as states have observed supply chain problems, rising geopolitical tensions and conflict, and more serious and frequent cyber incidents. Now economic considerations must increasingly be balanced with those around national security risks.

As a result, a number of countries have shifted toward more closed approaches in recent years, though sometimes with mixed results and uncertain trajectories. Russia, for example, has attempted to increase reliance on domestically developed digital technologies as its geopolitical relations with the West have soured. Yet these efforts have progressed more slowly than hoped, intensifying reliance on Chinese imports in the face of Western sanctions.³⁵ To optimize for desired results, high-tech import substitution policies must also have conditions in place for workforce development, whether by training domestic workers or attracting and retaining the most skilled workers globally. These goals can sometimes be in tension, however, with populist or nationalist efforts at import substitution or global competition driven efforts where leading competitors are major sources of international tech talent.

The United States has also faced challenges in adapting its policies to balance between economic and security considerations with regard to supply chains and digital product imports. Its launch of the Clean Networks Initiative in 2020 sought to address a perceived increased threat from China not only by curbing inclusion of China's Huawei 5G technologies into domestic digital networks but by discouraging their use in the critical networks of partners and allies. But this initiative also faced criticism for failing to address the need of many partners for an equally affordable alternative technology.³⁶ While it was late to develop a competitive 5G alternative, the United States has also undertaken a number of policies aimed at fostering development of industrial capacity and technologies for more secure digital supply chains. The 2022 CHIPS and Science Act, for example, aimed to rebuild domestic semiconductor chip manufacturing capacity and invest in high-tech workforce development, while the International Technology Security and Innovation Fund sought to support and incentivize development of trusted technology production capacities in partner countries.³⁷ In an effort to balance between fully open or closed approaches, the United States has promoted the idea of building "trusted ecosystems" of secure products and supply chains in which technologies meet shared cybersecurity standards and are interoperable, and where markets exist at sufficient scale to allow competition between multiple vendors while ensuring ecosystem participants can trust the "whole stack."³⁸ This approach nonetheless leaves many countries on the fence, either choosing to engage across ecosystems or being already deeply intertwined with several.

The EU and its 27 member states have also wrestled with balancing economic and security risk considerations around the import of digital goods. Since 2020, the EU has sought ways to

³⁵ Jaclyn Kerr, "Russia's Asymmetric Enablers," in *Assessing Russian Plans for Military Regeneration: Modernization and Reconstitution Challenges for Moscow's War Machine* (London: Royal Institute of International Affairs, 2024), 36–43, <https://www.chathamhouse.org/2024/07/assessing-russian-plans-military-regeneration>.

³⁶ See, for example, Robert K. Knake, "What's Wrong with the Clean Network Initiative? China Can't Join It," CFR, September 30, 2020, <https://www.cfr.org/blog/whats-wrong-clean-network-initiative-china-cant-join-it>; and Axel de Vernou, "The U.S. Must Offer a Substantive 5G Alternative to China's Huawei," *National Interest*, May 17, 2024, <https://nationalinterest.org/blog/techland/us-must-offer-substantive-5g-alternative-chinas-huawei-211072>.

³⁷ Justin Badlam et al., "The CHIPS and Science Act: Here's What's in It," McKinsey & Company, October 4, 2022, <https://www.mckinsey.com/industries/public-sector/our-insights/the-chips-and-science-act-heres-whats-in-it>; and "The U.S. Department of State International Technology Security and Innovation Fund," U.S. Department of State, <https://www.state.gov/the-u-s-department-of-state-international-technology-security-and-innovation-fund>.

³⁸ Antony J. Blinken, "Technology and the Transformation of U.S. Foreign Policy" (speech at Moscone Center, San Francisco, May 6, 2024), <https://2021-2025.state.gov/technology-and-the-transformation-of-u-s-foreign-policy>.

gradually reduce reliance on “high-risk vendors” such as Huawei and ZTE in critical networks and IT supply chains, first through voluntary measures that balance consideration of security risks with concerns about costs and economic impacts. The economic concerns have come particularly from telecommunications providers (some of which had already purchased Huawei products or were concerned about the costs of alternative vendors) as well as from more resource-constrained member states and those with strong trade ties to China. The EU launched 5G Toolbox and Supply Chain Security Toolbox assessment tools to support determination of vendor risk levels while restricting funding to infrastructure projects involving high-risk vendors.³⁹ However, despite significantly rising cyberthreats, adoption rates of recommended restrictions have been relatively low.⁴⁰ In January 2026 the European Commission proposed a new package of amendments to the EU’s Cybersecurity Act and NIS2 Directive that will, if enacted, make aspects of the two toolboxes binding, particularly increasing coordinated protections across member states for networks and supply chains in eighteen critical sectors.⁴¹

Some cybersecurity policies affect a country’s openness to foreign digital technologies even though they are not explicitly designed with such an outcome in mind. This can be the case with vendor cyber certification requirements, for example. These requirements focus on ensuring strong cybersecurity practices across IT vendors but also tend to limit the market to established incumbents and those that can afford to pay for the certification process, thus potentially deterring or restricting market entry by international competitors. In the United States the rollout of the Cybersecurity Maturity Model Certification 2.0 began in 2025, requiring mandatory certification by defense contractors to demonstrate a sufficient level of cybersecurity practices to handle sensitive information.⁴² This program in turn has led to adjustments in third countries. For example, the Canadian Program for Cyber Security Certification, which began in 2025, was adopted to enable Canadian companies to compete for cross-border defense contracts. Yet it remains to be seen how the new requirements will ultimately affect the participation of vendors from other countries in U.S. defense contracts going forward.⁴³

Trends at the industry or enterprise level can also be affected by national cybersecurity policies in ways that might have unforeseen follow-on effects on overall market openness. For example, the recent trend toward consolidation of IT services with a limited number of vendors has been adopted by cybersecurity leaders of many organizations as a way to reduce costs and increase security—especially by relying on only most-trusted vendors. In the short term, such an approach might be a cost-effective response to mitigate against increased cybersecurity threats and manage new regulatory burdens. In the long term, however, it can create exclusionary dynamics, limiting competition between vendors, and can also make it harder for organizations to change vendors because of switching costs. Such consolidation can eventually lead to undesirable economic and security consequences associated with vendor lock-in and risk monocultures. It is important

³⁹ Laurens Cerulus, “Europe’s Huawei Plan Explained,” *Politico*, January 29, 2020, <https://www.politico.eu/article/europe-eu-huawei-5g-china-cybersecurity-toolbox-explained>.

⁴⁰ China also had recently restricted access to its market by European telecommunications competitors, Sweden’s Ericsson and Finland’s Nokia, raising further geopolitical and economic concerns about a lack of reciprocity. See Egle Markeviciute, “EU’s Plans to Reduce Dependency on High-Risk Vendors,” *EU Tech Loop*, December 10, 2025, <https://eutechloop.com/eus-plans-to-reduce>.

⁴¹ “EU Cybersecurity Act Overhaul Targets Rising Threats to Critical Infrastructure,” *Innovation News Network*, January 21, 2026, <https://www.innovationnewsnetwork.com/eu-cybersecurity-act-overhaul-targets-rising-threats/65673>.

⁴² Heather Wishart-Smith, “Cybersecurity Compliance: The Costs, Risks and Race to Certification,” *Forbes*, July 7, 2025, <https://www.forbes.com/sites/heatherwishartsmith/2025/07/07/cybersecurity-compliance-the-costs-risks-and-race-to-certification>.

⁴³ Hunter Fox and Wendy J. Wagner, “Cyber Security Certification Is Coming: What Canadian Defence Suppliers Need to Know,” *Gowling WLG*, April 25, 2025, <https://gowlingwlg.com/en/insights-resources/articles/2025/cyber-security-certification-is-coming>.

that policymakers do their best to account for the possibility of such inadvertent second-order consequences when considering long-term trade-offs.

Balancing Data Access for Law Enforcement with Legal Constraint

The balances between law-enforcement data access and legal constraints, on the one hand, and between more centralized policy approaches and greater inclusion of private-sector and civil-society stakeholder interests, on the other, can be particularly fraught in a rapidly evolving digital threat environment. Countries often lean toward closed approaches to address overt and daunting challenges to public safety or national security. They aim to confront a growing menace of cybercrime or counter imminent terrorist or extremist threats. But neglecting rule-of-law constraints and stakeholder engagement can cause these efforts to backfire, fueling distrust in government among citizens and enterprises, undermining other areas of security, or diminishing decision-makers' access to important contextual information to guide subsequent policy. Such approaches are also not always fully effective. Extreme open approaches that minimize law-enforcement access or emphasize stakeholder interests over the mitigation of cybersecurity risk can create significant challenges too, however.

Getting the balance right between law-enforcement data access and legal constraints can be challenging. On the one hand, public awareness of government data access programs related to law enforcement or national security can fuel societal perceptions of state surveillance and concerns about heightened scrutiny of online speech or activity. This can prompt increased individual use of anonymity, encryption, and circumvention tools. It can also drive digital communication platforms to offer technologies and terms of service facilitating greater user anonymity or making government data access more challenging. Further efforts to block or restrict all uses of technologies such as Tor or virtual private networks or to require encryption backdoors or real-name registration of social media accounts are not always successful in improving law enforcement or national security. They can also increase distrust and, in some cases, do disproportionate harm to vulnerable populations. On the other hand, lack of adequate legal data access options for law enforcement or national security can exacerbate serious societal risks from cybercrime, terrorism, illicit markets, or other significant security threats.⁴⁴

While easing law-enforcement data access to assist in addressing clear cases of public safety or terrorism might be seen as a reasonable solution to these problems in some settings, a perception of overreach can trigger backlash, diminishing trust and reducing the private sector's willingness to engage in information sharing and other forms of voluntary collaboration with the government. Where collaborative relationships cannot be established, even to address severe security risks, this can lead law enforcement and national security organizations to favor exploitation of digital vulnerabilities to access data considered critical to their enforcement or threat mitigation activities. But this can in turn decrease digital ecosystem cybersecurity, even while facilitating certain law enforcement or national security efforts.

⁴⁴ Mike Isaac and Kevin Roose, "Disinformation Spreads on WhatsApp Ahead of Brazilian Election," *New York Times*, October 19, 2018, <https://www.nytimes.com/2018/10/19/technology/whatsapp-brazil-presidential-election.html>; Gerry Shih, "Inside the Vast Digital Campaign by Hindu Nationalists to Inflamm India," *Washington Post*, September 26, 2023, <https://www.washingtonpost.com/world/2023/09/26/hindu-nationalist-social-media-hate-campaign/>; and Mercy Muendo, "Kenya Targets WhatsApp Administrators in Its Fight Against Hate Speech," *Conversation*, August 31, 2017, <https://theconversation.com/kenya-targets-whatsapp-administrators-in-its-fight-against-hate-speech-82767>.

Centralized Policy Approaches vs. the Greater Inclusion of Stakeholder Interests

The challenges around data access policies demonstrate the importance and difficulty of getting an appropriate balance between centralized policy approaches and those that involve greater inclusion of stakeholder interests. Ultimately a mix of both is usually necessary to achieve desired ends and avoid inadvertent outcomes. As national cybersecurity challenges have grown, many countries have moved toward more centralized national (or supranational in the case of the EU) policies and have come to see some forms of required measures as necessary to achieve greater security against severe threats. But with the adoption of more compulsory regulatory approaches, as opposed to voluntary frameworks, it becomes especially crucial that decision-makers consider the full spectrum of potential effects on relevant actors—not only those effects that are intended and desirable—and retain mechanisms for feedback and learning as policies are adopted and revised.

Stakeholder engagement can be an important feedback mechanism for determining cybersecurity policies that do not create unforeseen costs or risks. Approaches that ignore the perspectives of private-sector or civil-society stakeholders run the risk of creating regulatory burdens that damage economic development while failing to achieve security objectives. Successful stakeholder relationship building, on the other hand, can enhance national cybersecurity while protecting vital economic and societal interests. Healthy engagement can encourage the adoption of policies that effectively balance cybersecurity with other societal needs such as long-term economic growth and protection of civil liberties. Such efforts may also position key actors in a country's digital ecosystem to participate more effectively in regional or international cybersecurity capacity-building programs and multistakeholder engagements, in turn fostering a positive feedback loop with further development of cybersecurity best practices and strategic implementation.

Receptivity to feedback and willingness to update policies responsively to stakeholder needs can be critical in finding appropriate balances that both improve cybersecurity and protect core national interests. As it sought to increase government data access across digital communications in the 2010s, Russia initially faced challenges enforcing its SORM data-storage and -sharing requirements, which demanded that all internet service providers purchase and install relevant equipment on their networks—a requirement that was prohibitive for some of the country's many smaller providers. This and later efforts focused on implementing Russia's information isolationist approach to cybersecurity often were pushed through legislatively with little adaptation to stakeholder interests, including broadly impactful economic or societal considerations. The country's initial effort in 2018 to block the highly popular encrypted communication platform Telegram for noncompliance with a law requiring government access to all encryption keys led not only to widespread and costly cascading technical failures across the Russian internet but also to mass street protests.⁴⁵

Allowing for greater input from private-sector and civil-society stakeholders can, by contrast, help prevent distrust and build mutually beneficial public-private partnerships and data-sharing arrangements. Such reciprocal relationships can be particularly important to the preparedness of national critical industries and infrastructure for high-level cyberthreats, facilitating crucial mechanisms of cyber incident and threat information sharing. In the United States in 2020, for example, the National Security Agency launched its Cybersecurity Collaboration Center to build stronger cybersecurity partnerships with private industry by enabling mutually beneficial

⁴⁵ "Russia Internet Freedom: Mass Rally in Moscow against Telegram Ban," BBC, April 30, 2018, <https://www.bbc.com/news/world-europe-43948618>.

bidirectional information sharing—particularly around critical industries of the defense industrial base sector.⁴⁶ Stakeholder engagement can help not only in building vital cybersecurity information-sharing mechanisms across potentially vulnerable industries that are important to national security and the economy. By assuring input from technical experts, it can also help avoid adopting policies that inadvertently lead to greater digital ecosystem insecurity or instability.

Furthermore, engagement can be critical to selecting cybersecurity policies that are good for the long-term health and competitiveness of a country's economy—particularly the tech sector. Hearing from private-sector stakeholders allows policymakers to better understand possible economic and business impacts when weighing different cybersecurity policy options. Both IT vendors and their business customers can be affected by regulatory compliance costs, for example, with vendors passing some costs on to their customers, possibly with impacts across multiple sectors. Such burdens are unlikely to affect all stakeholders equally. Depending on how regulations are designed, for example, they might have a disproportionate impact on smaller businesses that are less able to absorb additional overhead costs. Especially in smaller or less developed countries, regulations might affect a country's ability to attract or retain foreign firms or investments. Companies could exit or be deterred from entering smaller markets where there is not enough return on investment to cover the market-unique product requirements or regulatory costs. In the absence of a thoughtful policy engagement process, even well-intended cybersecurity policies run the risk of ultimately damaging the competitiveness of a country's tech sector or reducing the market's access to affordable cutting-edge technologies.

A number of commonly considered cybersecurity regulation options create new costs for private-sector stakeholders as a necessary step in reducing risks. These policies are adopted in efforts to ensure greater cybersecurity throughout the industry or ecosystem. But even as they help reduce important security risks, they can have long-term economic effects, such as creating tech sector winners and losers, reducing vendor diversity, affecting market openness or access to rapid innovation, or altering overall growth. Regulation elements such as pre-release testing requirements, secure-by-design standards, licensing or certification, software vendor liability, or insurance mandates, for example, are among those adopted in various forms in efforts to shift incentives, create new forms of accountability, and improve cybersecurity. These each create new compliance costs that can in turn alter market dynamics. They also are increasingly common or widely considered components of more aggressive cybersecurity regulations that seek to mitigate serious and growing risks.

Pre-release testing and secure-by-design requirements, for example, have played important roles in cybersecurity regulations of IoT devices in California and the UK and of products with digital components under the EU's Cyber Resilience Act.⁴⁷ Cybersecurity licensing, certification, or labeling systems have also been adopted by a number of countries and subnational jurisdictions or sectors, including in Malaysia, Singapore, and Ghana, as well as sector-specific requirements

⁴⁶ Christopher Bing, "Amid Big Hacks, U.S. Spy Agency Touts Collaboration Center with Private Industry," Reuters, June 22, 2021, <https://www.reuters.com/technology/amid-big-hacks-us-spy-agency-touts-collaboration-center-with-private-industry-2021-06-22>.

⁴⁷ "California Enacts First IoT Cybersecurity Law," Keller and Heckman LLP, October 17, 2018, <https://www.khlaw.com/insights/california-enacts-first-iot-cybersecurity-law>; Mathew J. Schwartz, "Secure by Design: UK Enforces IoT Device Cybersecurity Rules," BankInfo Security, April 29, 2024, <https://www.bankinfosecurity.com/secure-by-design-uk-enforces-iot-device-cybersecurity-rules-a-24964>; and Muktak Kanaskar et al., "EU Cyber Resilience Act: Key Steps, Compliance Challenges, and Practical Guidance," Cyber Defense Magazine, December 13, 2025, <https://www.cyberdefensemagazine.com/eu-cyber-resilience-act-key-steps-compliance-challenges-and-practical-guidance>.

in New York State and a shared voluntary framework across the EU.⁴⁸ The EU's Product Liability Directive, which takes effect in December 2026, updates prior liability law to include digital products such as software, digital services, and AI. Vendors will be held strictly liable for damages caused by defective software or failure to provide software updates. Similar national policies have been under recent consideration in the UK and United States.⁴⁹ Even as these policies are designed to reduce important security risks, they may also have long-term effects on the economy such as creating tech sector winners and losers, reducing vendor diversity, affecting market openness or access to rapid innovation, or altering overall growth. This makes it all the more important that the relevant stakeholder perspectives be accounted for in consideration of trade-offs so as to ensure an appropriate balance of long-term security, economic, and other risks and avoid unnecessary or inadvertent harms.

It also is critical that regulators routinely assess the effects and effectiveness of such policies once they are adopted and retain flexibility to adjust their approach in order to achieve the correct balance of effects. Since 2019, South Korea's Personal Information Protection Act has mandated cyber liability insurance for all companies managing personal data. The policy was recently updated, however, to target only larger companies but impose more significant fines for negligence. This approach was tailored to better handle the costs of large data breaches and address gaps observed in the policy's initial uptake. While smaller companies had been observed to not be acquiring the required insurance, larger ones were taking the path of minimal compliance by not purchasing large enough policies to cover actual payouts in case of breaches. This "insurance gap" was prominently demonstrated by the large 2025 Coupang data breach. Policymakers updated their approach in order to improve both compliance and effectiveness.⁵⁰

International Cooperation and Engagement

Ultimately, a country's more open or closed approach to cybersecurity policy may be deeply influenced by its international engagement with other countries and the nature of its cooperative

⁴⁸ Recent national-level policies in Malaysia (2024) and Singapore (2022) require that cybersecurity vendors obtain licenses to operate as cybersecurity service providers. Since 2020, Ghana has taken a unique approach of requiring licenses to practice for all individual cybersecurity professionals. Following poor early compliance, a drive to enforce legal penalties against service providers or individuals practicing without the required license or accreditation was announced in January 2026. Since 2025, New York's Department of Financial Services has required regulated financial entities to submit annual cybersecurity compliance certifications. The EU adopted its first EU-wide voluntary certification framework in February 2025, allowing for greater standardization. As opposed to earlier country-specific programs, the EU Cybersecurity Certification Framework provides a common system for certifying the cybersecurity of ICT products across all EU members. See Robert Lemos, "Licensed to Bill? Nations Mandate Certification and Licensure of Cybersecurity Pros," Dark Reading, April 22, 2024, <https://www.darkreading.com/cyber-risk/licensed-to-bill-nations-mandate-certification-licensure-of-cybersecurity-pros>; Alexander Chipman Koty, "How to Get a Cybersecurity License in Singapore," ASEAN Briefing, June 21, 2022, <https://www.aseanbriefing.com/news/how-to-get-a-cybersecurity-license-in-singapore>; Isaac Kofi Tsoenamawu, "Cyber Security Authority to Enforce Licensing Crackdown Starting January 31," *High Street Journal*, January 24, 2026, <https://thehighstreetjournal.com/cyber-security-authoritytoensurelicensing>; Meghan O'Connor, "New York Cybersecurity Regulation Requires Submission of Compliance Certification or Acknowledgement of Noncompliance Next Week," Quarles and Brady LLP, April 9, 2025, <https://www.quarles.com/newsroom/publications/new-york-cybersecurity-regulation-requires-submission-of-compliance-certification-or-acknowledgement-of-noncompliance-next-week>; and European Commission, "EU Cybersecurity Certification Framework," January 20, 2026, <https://digital-strategy.ec.europa.eu/en/policies/cybersecurity-certification-framework>.

⁴⁹ See, for example, Lisa M. Baird et al., "The New EU Product Liability Directive: Implications for Software, Digital Products, and Cybersecurity," Reed Smith Client Alerts, July 2, 2025, <https://www.reedsmith.com/articles/eu-product-liability-directive-software-digital-products-cybersecurity>; Alexander Martin, "Software Companies Must Be Held Liable for British Economic Security, Say MPs," Recorded Future News, Record, November 24, 2025, <https://therecord.media/software-companies-liable-britain-security>; and Eric Geller, "The Struggle for Software Liability: Inside a 'Very, Very, Very Hard Problem,'" Recorded Future News, Record, October 22, 2024, <https://therecord.media/cybersecurity-software-liability-standards-white-house-struggle>.

⁵⁰ See, for example, "South Korea Amends Privacy Law to Authorize Fines of Up to 10% of Total Revenue," Hunton Andrews Kurth LLP, Privacy and Cybersecurity Law Blog, February 13, 2026, <https://www.hunton.com/privacy-and-cybersecurity-law-blog/south-korea-amends-privacy-law-to-authorize-fines-of-up-to-10-of-total-revenue>; Jun Ji-hye, "Firms Stick to Bare-Minimum Data Breach Insurance Despite Major Leaks," *Korea Times*, December 8, 2025, <https://www.koreatimes.co.kr/business/companies/20251208/firms-stick-to-bare-minimum-data-breach-insurance-despite-major-leaks>; and Kang Woo-ryang, "Coupang's 1 Billion Won Insurance Falls Short after 33.7 Million Data Breach," *Chosun Daily*, December 8, 2025, <https://www.chosun.com/english/market-money-en/2025/12/08/KXCKOF2BYJEFHJQJYDZ16MQZSE>.

arrangements around the development of the technology sector. Decisions over these forms of cooperation and engagement can involve important trade-offs concerning the long-term risks and benefits of different international relationships and interdependencies. This has been apparent in choices by many countries concerning which strategies, laws, or regulatory frameworks to emulate; which capacity-building partnerships to join; and what forms of economic interdependence to welcome in development of their national digital ecosystems and critical information infrastructures. As with the other cybersecurity policy decisions, however, the long-term effects and consequences of these relationship choices are not always initially apparent.

Threat perceptions about specific actors might change and alter understandings of the geoeconomic and security costs and benefits of pursuing particular policies or relationships, for example. Balancing across these considerations can lead to hedging or pivots concerning what cybersecurity strategies and collaborative relationships to pursue, or around what technological supply chain and trade interdependencies to further cultivate. South Korea's cyber strategy has shifted notably in recent years, for example. While through most of the 2010s Seoul took a primarily defensive approach, with cyberthreat perceptions focused largely on North Korea, this began to change as South Korea found itself increasingly targeted by Russian and Chinese cyber aggression. When Russia's APT-28 targeted the 2018 Pyeongchang Olympics, South Korea collaborated closely with the United States and UK to disrupt and attribute the attack, leading the country's subsequent 2019 National Cybersecurity Strategy to adopt a more proactive forward defense approach similar to that of its partners.⁵¹

Shifting geopolitics, such as increased tensions around great-power competition after a period of relative unipolarity or changes in the regional security environment, can create new challenges—particularly for countries seeking to maximize their own prosperity and security while navigating relationships with multiple global or regional powers. Geopolitical competition over the development and export of digital technologies intensifies this complexity. Today's digital technologies are foundational to telecommunications, critical infrastructure, and commerce. They also generate new security risks as well as cross-sector and international technical and economic interdependencies, deepening linkages across some areas of policy that were previously handled more autonomously. This is particularly apparent in areas of economic development and security. For countries not clearly aligned, facing economic or political pressure from powerful actors, or weighing difficult-to-compare economic and security benefits associated with different partnerships, the long-term risks and benefits of different possible collaborations can be ambiguous or involve perilous balancing efforts.

Ambivalence, uncertainty, and balancing are evident in the many mixed approaches that countries have taken in their technological imports and partnerships. Vietnam, for example, is one of many countries that has adjusted its approach to supply chains for critical technology in response to growing security and geopolitical concerns—but balanced also with pressing economic considerations. This has involved multiple pivots. Like many countries in Southeast Asia, through the 2010s Vietnam incorporated technology from the Chinese vendor Huawei into its domestic networks. These products were long viewed by many developing and middle-income countries as offering affordable and relatively high-quality solutions to their technological sector

⁵¹ "Cyber Attacks at the 2018 PyeongChang Winter Olympics," RSI Security, February 14, 2018, <https://blog.rsisecurity.com/cyber-attacks-at-the-2018-pyeongchang-winter-olympics>; and Kim Arin, "Seoul's Spy Agency Accuses China of Major Cyber Attacks," *Korea Herald*, January 24, 2024, <https://www.koreaherald.com/article/3312349>.

needs at a time when equivalent products on the market were prohibitively expensive. Later these technologies came to be seen as potential sources of insecurity by some countries, however, due both to potential direct cybersecurity risks and critical supply chain interdependencies and to emerging geopolitical concerns and pressures from other partners. Thus, as Vietnamese threat perceptions about Chinese cyberattacks and regional maritime power projection increased—also facing diplomatic pressure from the United States’s Clean Network Initiative campaign—Vietnam chose to opt out of most future uses of Huawei beginning in 2019–20, instead turning to domestically developed or internationally sourced Viettel 5G and 4G technologies.⁵² Vietnam never took an official stance against Huawei, however, and by 2025 it had struck new arrangements with both Huawei and ZTE to help develop its 5G sector.⁵³

Many countries still hedge in their choices between different major digital ecosystems and cybersecurity policy approaches, whether for clear strategic reasons or as a result of multiple disaggregated decision processes. Even as the reach of China’s Digital Silk Road widens, bringing with it support for digital infrastructure, including 5G, safe city, and facial recognition technologies, U.S. companies often lead in cloud computing, software, AI, or satellite services in the same markets. This is the case in a number of critical regional developing economies, such as Indonesia, Brazil, and Kenya.⁵⁴ At the same time, we see broad emulation of the United States’ NIST Cybersecurity Framework and standards and of laws based on the EU’s GDPR privacy protections, with some countries favoring close relations with China for development investment but with the United States or EU for security or governance partnerships.⁵⁵ While some of these mixed approaches have accrued through gradual iteration and reaction, in other cases they mark countries’ deliberate efforts to maintain “strategic autonomy” or hedge against future uncertainties through balancing multiple relationships and a “mixed stack” of diverse vendors in their digital ecosystems.

Changes in a country’s internal and external political environments can also expose prior approaches to cybersecurity and cyber partnerships as outdated and result in rapid pivots in policy. In the Caucasus, for example, Georgia and Armenia both could be moving toward new forms of cyber policy and cooperation. Since the 2008 Russia-Georgia war in which Georgia was targeted by Russian cyberattacks, Georgia had received Western cyber resilience assistance, building close collaborative relationships with the UK, EU, and United States. But the growing political alignment of the Georgian government with Russia has led to a pause in much of this collaboration.⁵⁶ At the same time, possibly disillusioned with Russian security guarantees in the wake of the 2020

⁵² See, for example, Viet Dung Trinh, “Vietnam’s Securitisation of Cybersecurity Under Growing Chinese Cyber Threats,” Australian Institute of International Affairs, April 9, 2025, <https://www.internationalaffairs.org.au/australianoutlook/vietnams-securitisation-of-cybersecurity-under-growing-chinese-cyber-threats>.

⁵³ Minh Tran, “Vietnam’s Cautious Acceptance of Chinese 5G Technology: Recent Agreements with Huawei and ZTE Reflect Economic Pragmatism Rather Than a Shift in Strategic Trust,” *Diplomat*, December 4, 2025, <https://thediplomat.com/2025/12/vietnams-cautious-acceptance-of-chinese-5g-technology>.

⁵⁴ Vivek Chilukuri and Ruby Scanlon, “New Report Outlines Strategy to Compete with China’s Digital Silk Road in Kenya,” Center for New American Security (CNAS), July 22, 2025, <https://www.cnas.org/press/press-release/new-report-outlines-strategy-to-compete-with-chinas-digital-silk-road-in-kenya>; Vivek Chilukuri and Ruby Scanlon, “Countering the Digital Silk Road: Indonesia,” CNAS, March 20, 2025, <https://www.cnas.org/publications/reports/countering-the-digital-silk-road-indonesia>; and Ruby Scanlon and Bill Drexel, “Countering the Digital Silk Road: Brazil,” CNAS, April 24, 2025, <https://www.cnas.org/publications/reports/countering-the-digital-silk-road-brazil>.

⁵⁵ Evan Williams, “China’s Digital Silk Road Taking Its Shot at the Global Stage,” East Asia Forum, May 9, 2024, <https://eastasiaforum.org/2024/05/09/chinas-digital-silk-road-taking-its-shot-at-the-global-stage>; Mahn, “NIST’s International Cybersecurity and Privacy Engagement Update”; and Simmons, “17 Countries with GDPR-Like Data Privacy Laws.”

⁵⁶ Mamuka Kirkitadze, “From Ally to Outcast, What’s Next for Georgia’s Cyber Future?” Binding Hook, December 24, 2024, <https://bindinghook.com/articles-binding-edge/from-ally-to-outcast-whats-next-for-georgias-cyber-future>; and “UK-Georgia Cyber Partnership Launch Georgian Language Cybercrime Report,” RUSI, April 22, 2024, <https://www.rusi.org/news-and-comment/rusi-news/uk-georgia-cyber-partnership-launch-georgian-language-cybercrime-report>.

Nagorno-Karabakh war, Armenia has recently sought closer cyber collaboration with Western countries, including the EU, United States, and UK.⁵⁷ Depending on how these collaborative relationships develop—pulling the two Caucasus countries closer to, respectively, more open or closed digital ecosystems and security concepts—they could have significant repercussions for the long-term economic modernization and cybersecurity of both countries and the region.

As the many examples discussed in this section demonstrate, cybersecurity policy decisions can have significant implications for a country's future economic, political, and social development. Unfortunately, even the more predictable trade-offs inherent in particular decisions are often not well understood by decision-makers at the time. Significant uncertainties in the future trajectory of world affairs, combined with the rapid development of technology, exacerbate these challenges. The conclusion to this chapter examines potential best practices countries may consider to improve understanding of known trade-offs and to limit the consequences of future uncertainties.

Conclusion: Reflecting on Future Trade-Offs in Current Decisions

Strategic foresight is a challenge in most important areas of state policymaking, but it is particularly daunting in cybersecurity. A policy arena concerned with rapidly changing ecosystems of digital technologies, cybersecurity also has critical connections to complex areas of both domestic and international affairs. This makes it particularly difficult—but all the more crucial—for policymakers to gain adequate awareness of the possible long-term trade-offs and consequences related to their current decisions. Though some uncertainties are inevitable, policymakers should take all possible steps to improve understanding. Some unforeseen consequences can be prevented by deliberately addressing limitations in how the decision-making process integrates and evaluates available information about known risks and stakeholder equities. But cybersecurity risk assessment and policy processes should also address the limitations on strategic foresight directly. Policymakers can do this by deliberately integrating consideration of the impacts of alternative future paths and by ensuring processes of stakeholder feedback that permit course corrections if or when policies cease to be well suited to a changing threat environment.

Any attempt to ensure full consideration of trade-offs inherent in cybersecurity policy decisions must address key limitations in the decision process that inhibit awareness of the multiple risks being balanced. This should include efforts to improve decision-maker understanding of the full array of known risks, threats, and trade-offs bearing on any given cybersecurity policy decision as well as key policy linkages and interdependencies across issue areas. Policymakers at all levels should also be aware of the different constraints that affect their range of policy options, the availability of particular risk mitigation solutions, and the potential effectiveness of different approaches. These can be affected, for example, by state capacity and regulatory reach, economic capability or workforce potential, or access to certain technologies or foreign assistance alternatives. Policymakers should also be as aware as possible of the complex international interdependencies and relationships that might affect or be influenced by their decisions.

⁵⁷ "OSCE Supports Armenia's Efforts to Combat Cybercrime," OSCE, March 19, 2025, <https://www.osce.org/node/587822>; Christopher Crowson, "Armenia Demonstrates Increased Resilience in Cybersecurity," Civilnet, May 26, 2025, <https://www.civilnet.am/en/news/953123/armenia-demonstrates-increased-resilience-in-cybersecurity>; Ani Avetisyan, "European Union, United States Woo Armenia with Economic Assistance Package: Support Is Designed to Encourage Armenia's 'Resilience,'" Eurasianet, April 8, 2024, <https://eurasianet.org/european-union-united-states-woo-armenia-with-economic-assistance-package>; and "Armenia Advances in Cybersecurity," Cyber4am, November 27, 2024.

National strategy development and updating processes can help address these informational challenges by facilitating knowledge sharing between the different expert and stakeholder communities—both domestic and international—thereby building common awareness of the varying threat perceptions, issue areas, interests, gatekeepers, and decision points iteratively involved in the development of a country’s cybersecurity strategy, policy, and regulations. These processes can be designed to help illuminate and mitigate the influence of unforgiving heuristics and biases in decision-making. But such efforts must also seek to account for the potential longer-term effects of policy choices beyond those attributed to known trade-offs. In other words, they must seek to understand possible subsequent changes in the threat environment in order to build risk mitigation solutions that are most likely to endure successfully despite future uncertainties. Decision-makers must not only focus on problems of the moment but build processes that also encourage consideration of likely areas of change over the medium and long term. This includes some areas of known rapid change or high uncertainty—in both the technology environment and the global context—that could lead to transformations in the threat environment.

Current ongoing changes in technology and the global context are likely to have transformative effects on the risks and opportunities that must be accounted for in cybersecurity policy. The IoT, sensors, and automation continue to erode the digital-physical distinction, creating new risks associated with data confidentiality, integrity, and availability; physical harm; and critical infrastructure. Ongoing rapid advances in AI research and applications will intersect with each of these areas, exacerbating challenges even as they create enormous opportunities. In this context, cybercrime and adversarial cyberoperations will continue to play persistent roles in defining the risks states face, both as threats to public safety and in the context of hybrid or above-threshold interstate conflict. The cyber dimension of warfare will likely also continue to grow as 21st-century conflicts are increasingly defined by autonomous systems and robotics, electronic warfare, space and counterspace technologies, and data-based battlefield analytics. States will also face new types of cyber-enabled risks as ongoing advances in digital technology ecosystems increasingly interact with a panoply of other information-centric and dual-use emerging technologies.

Alongside these rapid technological developments, ongoing and potential future changes in the global context further complicate the long-term risks and threats that cybersecurity policymakers should consider. These could include the potential cybersecurity ramifications of major geopolitical, economic, or societal developments, including ongoing trends or surprise events. Escalating great-power economic or military competition, for example, could exacerbate the fragmentation of digital networks and supply chains, further facilitating the development of several major digital ecosystems or changing the dynamics of regional digital integration. Advanced AI’s demand for energy and data could likewise make data centers and data-collection infrastructure new areas of international collaboration or cyber vulnerability.

In each of these areas of potential long-term change relevant to cyber risks, there exist some dynamics of greater certainty, but also significant unknowns or areas of expert debate. While the exact contours may vary, it is possible to project with some degree of certainty that increased digitization of real-world systems will continue, as will increasing integration of AI. But there is still considerable uncertainty about when exactly to expect certain technical breakthroughs, such as in AI or quantum computing, and their exact repercussions for digital security. We also cannot be sure exactly how far current trends toward separate digital ecosystems will continue.

Yet even though cybersecurity risk assessments cannot know how some of these consequential future developments will unfold, they can use strategic foresight tools, stakeholder feedback mechanisms, and international collaboration to improve their understanding of possible future risks and build solutions that ensure maximal adaptability to different developments. Scenario planning exercises can be used, for example, to project alternative futures based on known trends and sources of uncertainty.⁵⁸ Testing the likely effects of different cybersecurity policy options against several hypothetical future scenarios can provide insights into which proposals promise greatest resiliency against future unknowns. States can also maintain the adaptability of their cybersecurity policies to address new emerging challenges if they build fair, inclusive processes and trusting consultative relationships with civil-society and private-sector stakeholders. This will foster feedback mechanisms of information sharing that can facilitate monitoring of the effectiveness or inadvertent consequences of existing policies and serve as early warnings of emerging areas of friction or novel risks and threats. International cybersecurity cooperation also has an important role to play in the development of longer-range thinking. Capacity-building programs, for example, can encourage decision-makers to engage in multistakeholder settings and cultivate strategic foresight skills.

Cybersecurity risks will continue to rapidly evolve, and there is seldom a one-size-fits-all solution to any of the myriad policy challenges involved in balancing and mitigating cyber risks. Solutions for particular states and challenges will often involve balancing between various policy directions. But states can improve their odds of achieving their desired outcomes without adverse consequences if they approach cybersecurity policy decisions with as full an understanding as possible of the multiple risks and interests implicated. While some long-term consequences of these decisions will remain necessarily uncertain due to the many unknowns in the future development of digital technology and the global context, states can improve the long-term viability of their policy approaches through the integration of strategic foresight tools and stakeholder and partner engagement into their cybersecurity policy processes.

⁵⁸ Lawrence Wilkinson, "How to Build Scenarios," *Wired*, November 1, 1995, <https://www.wired.com/1995/11/how-to-build-scenarios>.

Conclusion

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The preceding chapters described key features of alternative approaches to cybersecurity governance. These approaches diverge along several fault lines, including their international orientations toward the security implications of information technologies, their alignment with the maturation trajectories of emerging economies, the motivations behind approaches that restrict or welcome digital openness, and the linkages between policy choices informed by exogenous circumstances and their long-term (and not always immediately obvious) consequences for resilience, innovation, private-sector implementation and economic outcomes, and other important values. States face genuine and difficult trade-offs when designing cybersecurity policy, a challenge compounded by—as Jaclyn Kerr documents in chapter 4—policymakers’ finite and incomplete knowledge and foresight about risks and policy outcomes.

No approach to cybersecurity governance is without trade-offs, but openness to global markets for digital goods and services has notable economic and security advantages and carries fewer costs compared with what this report refers to as closed or information security approaches. Openness enables domestic innovation ecosystems to participate in global value chains in support of economic development objectives. It expands the range of cost-effective digital technologies available to domestic businesses and consumers. And it positions governments to extract better signals about risks and mitigation strategies by positioning them to engage in diversified public-private partnerships and exchanges with other like-minded governments. Closed approaches risk limiting access to the global marketplace for digital products and services, constraining choices about productivity-enhancing capabilities and cybersecurity services.

Choices about cybersecurity governance are also choices about the governance of other digital and emerging technology domains, particularly artificial intelligence (AI). AI is poised to become the most consequential technology of the coming decades—reshaping economies, restructuring labor markets, transforming military capabilities, and reordering the terms of great-power competition. AI systems are digital systems and therefore subject to the same broad regulatory and policy requirements as other digital systems, including for cybersecurity. At the same time, key aspects of what make AI systems unique, compared with other digital technologies, will introduce new attack vectors for adversaries. AI systems must be resilient against these attacks—a topic that will be explored further in the second report in this series. Enterprises could also be reluctant to embrace AI if they do not trust it or are uncertain about regulatory expectations. Cybersecurity resilience and AI diffusion goals are both served best by more open approaches to cybersecurity governance, characterized by the free flow of data, multi-stakeholder engagement, and international collaboration.

Markets, Rights, and Isolationism

In chapter 1, Elaine Korzak identifies a fundamental divide between states that conceptualize cybersecurity narrowly—focused on protecting the confidentiality, integrity, and availability of digital systems—and those that embrace a broader information security paradigm concerned with controlling information flows as a tool of political stabilization. This divide is not merely semantic; it reflects the state’s conceptualization and prioritization of what values or interests it is trying to protect. For the cybersecurity camp (led by a broad grouping of market economies), cybersecurity resilience is concerned primarily with safeguarding individuals, private- and public-sector entities, and critical infrastructure. For the information security camp (led by

Russia and China), the paramount goal of policy is to protect the incumbent political order from threatening influences.

Most states, however, do not fall neatly into one or the other camp; they exist on a spectrum between the two camps. As Anthony Adams documents in chapter 2, cybersecurity capacity building is an increasingly important focus of development cooperation. Cyber-related development cooperation with advanced democracies in East Asia, Europe, and North America tends to reflect the cybersecurity camp's emphasis on enhancing the resilience of critical infrastructure and the broader economy against threats to the confidentiality, integrity, and availability of digital infrastructure, whereas development cooperation with the information security camp is more likely to involve provision of infrastructure with domestic surveillance capabilities.

This distinction between the two camps also helps anchor the four national archetypes that inform this report's analyses. The United States pursues a market-driven approach that treats private-sector innovation as both the engine of cybersecurity and the primary mechanism for incentivizing cybersecurity resilience. The European Union's digital rights approach emphasizes individual citizens and deploys proactive regulation—including instruments like the NIS2 Directive and the Cyber Resilience Act—to establish minimum security floors for any company (foreign or domestic) that wants to deliver products or services in Europe's single market. Both approaches also reflect "global openness," as described by Josephine Wolff in chapter 3. Russia's information isolationism model, on the other hand, couples maximalist state control over the domestic information space with active decoupling from foreign technology providers. And China's advanced information isolationism approach goes even further, extending comprehensive state authority into cyberspace while leveraging a competitive domestic technology sector as an instrument of both security and geopolitical influence.

As Kerr documents in chapter 4, the policy priorities and preferred policy instruments associated with each of the archetypes shape the development of its technology sector, the nature of its international partnerships, the openness of its information ecosystem, and its long-run capacity to adapt to a rapidly evolving threat environment. The security and economic stakes of getting this choice right—or at least getting it less wrong—are therefore substantial.

The Compounding Costs of Information Isolationism

Countries that adopt an information isolationism approach often do so for reasons that appear logical but without sufficient consideration for the long-term costs. Wolff's analysis in chapter 3 shows that policy interventions associated with information isolationist approaches to cybersecurity—content blocking, foreign service bans, and data localization requirements—are not always mere pretexts for censorship or protectionism (even if they look large as factors). When governments mandate data localization, for example, they are often motivated by legitimate concerns about the accessibility of data for domestic law enforcement, the risks of foreign intelligence agencies exploiting records stored overseas, and increasingly appropriation of national data by foreign companies to train AI models. These are real vulnerabilities, and the policy responses to them reflect real threat assessments. India's ban of dozens of Chinese applications following the 2020 Galwan Valley confrontation sought to address a genuine and specific threat: the risk that user data collected by those applications could be harvested by an adversarial state.

Indeed, Russia's information isolationism model has an intelligible internal logic. If the primary threat a government perceives is the destabilizing potential of cross-border information flows—as Russian strategic doctrine has held since at least the 1990s—then building the technical and legal infrastructure to control those flows is a rational response, even if it undermines resilience against other threats or hampers innovation and economic development. Kerr's reconstruction of Russia's incremental march toward internet censorship shows that each individual step, from the 2012 children's protection law to the eventual blocking of major social media platforms, was responsive to a specific perceived threat.

However, the systemic costs of these approaches accumulate. Russia's pivot toward information isolationism, Kerr argues, came with severe long-term economic costs to its technology sector. Russian internet and cybersecurity firms that had been globally competitive through the early 2010s lost market access and international research partnerships as isolationist policies deepened. Import substitution programs designed to replace foreign technology with domestic alternatives progressed far more slowly than hoped, ultimately increasing Russia's dependence on foreign technology (often from China) rather than achieving the self-sufficiency the strategy sought. Meanwhile, policies driven by nationalist and populist pressures made it harder to attract and retain the international technical talent on which competitive technology development depends. Russia's information isolationism embedded stagnation into the country's long-term technological trajectory.

China's advanced information isolationism model presents a more complex case, because it is backed by a genuinely competitive domestic technology sector that Russia lacks. China has mandated data localization, restricted foreign digital services, and asserted comprehensive control over the national information space while still sustaining remarkable rates of technological development and economic growth. Whether China can sustain innovation and growth over the long term within its existing governance model is an open question. Even so, China's ability to pursue advanced information isolationist policies and grow its economy and digital innovation ecosystem reflects a unique combination of advantages that no other country possesses—a world-class innovation ecosystem, a large tax base, industrial policy capacity, a massive domestic market that can simultaneously attract foreign firms but nurture and sustain viable domestic alternatives to foreign platforms to eventually replace them, and a technologically enabled authoritarian political system that makes surveillance and repression possible to achieve at scale. Many countries have one or more of these attributes, but only China possesses them all. For the vast majority of states, therefore, mimicking China's policies would produce Russia's outcome: isolation without the compensating domestic technological depth.

The Economic and Security Case for Openness

The argument for more open cybersecurity governance rests on the claim that it better sustains the conditions for both economic growth and genuine security. Openness sustains vendor diversity and competitive markets in ways that closed approaches systematically undermine. As Kerr notes in chapter 4, policies that require domestic data storage, restrict foreign digital services, or impose certification regimes with high compliance costs tend to favor established incumbents over new entrants and reduce the menu of digital products and services available to businesses and governments—potentially denying them access to innovative, cost-effective solutions available

in the global marketplace. Private- and public-sector enterprises forgo productivity gains that may come from using better digital solutions and have fewer options for achieving cybersecurity resilience goals.

Over time, risk may become concentrated in the products of a limited number of vendors—a technological monoculture that is itself a significant security risk. In biological systems, a monoculture emerges from a lack of biological diversity; the hazard is that a single disease or other vulnerability has the potential to devastate the entire ecosystem if the organisms that make up the ecosystem are all vulnerable to it. For digital ecosystems, similarly, the availability of more options for critical products and services means more options for reducing systemic risks and less vulnerability to single points of systemic failure.

Open approaches are also more compatible with the kind of adaptive, stakeholder-engaged policymaking that Kerr identifies in chapter 4 as essential for navigating an environment of rapid technological change. The core failure mode she diagnoses in closed approaches is not merely that they produce bad initial policies, but that they foreclose the feedback mechanisms that would allow those policies to be corrected. Russia’s attempt to block Telegram—which produced cascading technical failures across the Russian internet, resulting in mass street protests—is a paradigmatic example: a policy pushed through with minimal stakeholder engagement, implemented without adequate technical understanding of its consequences, and ultimately ineffective even on its own terms. The EU’s iterative development of its cybersecurity regulatory framework, however imperfect, reflects a fundamentally different model—one in which industry, civil society, and member state governments are engaged in the policy process in ways that surface the costs and unintended consequences of proposed regulations before they are locked in. The result is not necessarily faster policymaking, but it is more durable and effective policymaking.

The policy frameworks governing cybersecurity and digital openness also have far-reaching implications for how transformational technologies such as AI will be incorporated into a country’s digital ecosystem. A country that adopts data localization requirements aligned with China’s Cybersecurity Law, incorporates Huawei infrastructure into its digital backbone, or accepts Chinese safe city and facial recognition technologies through the Digital Silk Road is not just making a cybersecurity choice; it is embedding itself in a digital ecosystem whose AI governance norms, surveillance architecture, and data access regimes will shape how AI diffuses within its borders for years to come. Kerr’s observation that many of these choices are made “iteratively in local contexts”—driven by municipal governments seeking affordable smart city solutions rather than by national-level strategic deliberation—is especially troubling in this light. The long-term AI governance consequences of adopting particular surveillance or urban infrastructure technologies are unlikely to be salient to local officials facing immediate public safety or service delivery pressures; yet those consequences may prove very difficult to reverse once the infrastructure is embedded.

Kerr’s account of Russia’s desiccated digital innovation ecosystem also foreshadows how a closed, information isolationism approach to digital governance could constrain a country’s AI future. Russia’s cybersecurity and internet companies, globally competitive in the early 2010s, lost market access and research partnerships as isolationist policies deepened. Applied to AI, this dynamic will be even more damaging: AI development is more dependent than previous technology generations on international research collaboration, access to model ecosystems maintained by globally distributed communities, and cross-border data flows, all of which restrictive cybersecurity

regimes systematically impede. Russia's import substitution programs have already struggled to replace foreign software and hardware; attempting to indigenize AI capabilities faces even steeper barriers. Russia's trajectory is a cautionary example of a closed governance philosophy creating structural ceilings for AI development based on limited domestic capabilities.

Implications

The countries best positioned to navigate the cybersecurity challenges of the coming decades will not be those that have most successfully insulated themselves from the global digital ecosystem. They will be those that have built open, competitive, and well-governed digital environments capable of attracting investment, generating innovation, and sustaining the international partnerships on which both economic growth and genuine security ultimately rest. The case for open cybersecurity governance is, at its core, the case for treating security and prosperity not as competing values to be traded off against each other but as mutually reinforcing conditions that only open approaches can sustain over time.

Kerr's prescriptions for improving strategic foresight in cybersecurity policymaking apply with particular urgency to AI. She argues that policymakers typically focus on immediate, concrete threats rather than abstract or long-term risks; that decision processes are fragmented across actors and levels of government; and that the feedback mechanisms needed to correct course are often absent or underpowered in closed models. Each of these pathologies is likely to be especially acute in the AI domain, where the pace of technological change means that governance frameworks designed for today's AI capabilities will be tested by systems that are qualitatively more powerful within years rather than decades.

The scenario planning exercises Kerr recommends as a tool for improving foresight are particularly apt here. Policymakers who stress-test their cybersecurity frameworks against a future in which AI dramatically lowers the cost of cyberattacks, enables new forms of disinformation, or makes surveillance architecture qualitatively more powerful will quickly discover that frameworks designed around today's threat environment may be badly misaligned with tomorrow's. The countries best positioned to navigate that future will be those whose cybersecurity governance frameworks are adaptive rather than rigid—that is, open models with strong stakeholder engagement mechanisms—and not those that have locked in specific controls and restrictions that cannot easily be revised as the environment changes.

It is difficult to overstate the importance of stakeholder engagement in the course of developing cyber policies and laws. The preponderance of expertise and experience in what cyber risks face organizations, which practical measures for reducing and managing those risks work and do not work, how much money they cost to implement, and what the impacts are resides outside government in the private sector or in quasi-governmental critical infrastructure organizations. The same is true when it comes to the innovation ecosystems for digital products, including for cybersecurity. Of course, governments have expertise of their own, and they should view this multistakeholder community of nongovernmental experts as a force multiplier for helping the government craft practical, cost-effective cyber resilience strategies aligned with national priorities and values.

States will continue to face pressure to restrict information flows in the name of political stability, to localize data in the name of law-enforcement access, and to exclude foreign vendors in

the name of supply chain security. In each of these domains there are genuine security arguments to be made for restriction. But as Korzak, Adams, Wolff, and Kerr collectively demonstrate, those arguments must be weighed against the long-term costs: the erosion of vendor diversity, the loss of international research and commercial partnerships, the chilling effects on innovation, and the foreclosure of the stakeholder feedback mechanisms that allow policy to adapt as threats evolve—all of which weaken cybersecurity resilience.

With the AI era well underway, countries that recognize the deep interconnection between cybersecurity governance and broader digital technology advancements and that approach both with adequate attention to long-term trade-offs will be best positioned to navigate the AI diffusion contest. They will maintain the openness to cross-border data flows, international talent, and global platform access that AI development requires, while building the regulatory capacity to govern AI's risks and distribute its benefits equitably. They will engage private-sector and civil-society stakeholders in developing AI governance frameworks, recognizing that the information asymmetries between government and industry are even more severe in AI than in earlier digital domains. And they will invest in the capacity building and international collaboration that allow countries at lower levels of technological development to participate in the global AI ecosystem rather than being absorbed into whichever closed digital system happens to be most affordable or most aggressively promoted in their region. A country that is more resilient against cybersecurity threats has more autonomy and agency to chart its own digital development and innovation journey.

The cybersecurity governance choices being made today will affect countries' technological development for years and even decades to come. Choosing and implementing open approaches will allow countries to be best positioned to secure the associated innovational gains and better integrate with global technology systems and leaders. Conversely, those that choose the short-term gains of isolationist approaches are far more likely to miss out on the benefits or rapid technological advances. In sum, today's cybersecurity governance decisions are setting the terms on which AI will diffuse across the global economy. The window for course correction is open, but it will not remain open indefinitely. The structural advantages that open approaches offer—in adaptability, in economic dynamism, and in the sustenance of the international collaborations on which AI progress depends—make the case for open cybersecurity governance not merely as a matter of principle but as a strategic imperative for any country that intends to be a meaningful participant in the AI-defined economy of the coming decades.



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